

More than One Way to Skin a Cat: Preserving Verified Modes in RLVR

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Abstract

A mathematical reasoning model can become more accurate while losing alternative correct solutions. To measure this loss, we introduce **MODEBENCH**, a benchmark of five executable domains that identifies different correct outputs for one problem. We then show that **ReplayMaxRL**, which combines sampling-aware reinforcement learning with a memory of previously verified outputs, improves correctness while preserving more valid alternatives. We find gains from replay with both **GRPO** and **MaxRL**, across three and two model scales respectively, and on a harder graph-coloring task. These results separate two requirements often treated as one: discovering successful behaviors and preserving them.

1 Correct answers can conceal the loss of alternatives

Reinforcement learning with verifiable rewards (RLVR) trains from executable correctness feedback [Guo et al., 2025], rewarding familiar and new correct solutions alike. In mathematical search, losing alternatives narrows the verified candidate set available from repeated sampling [Brown et al., 2024, Snell et al., 2024]. Figure 1 shows this failure: correctness rises while sampled alternatives disappear.

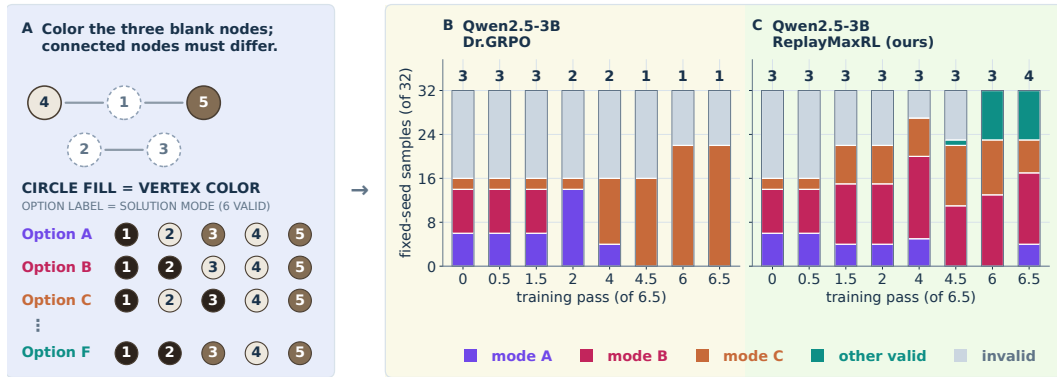


Figure 1: **Correctness can rise while alternatives disappear.** One graph admits six valid colorings. Bars pool four eight-sample draws from matched Qwen2.5-3B runs. At pass 6.5, Dr.GRPO produces 22/32 correct from one mode; ReplayMaxRL produces 32/32 across four.

The loss also appears across our datasets. Averaged over five domains, eight passes of Dr.GRPO reduce the initial number of verified alternatives beyond the first correct output in eight samples by 99.4% at Qwen2.5-0.5B, 71.6% at Falcon3-1B, and 41.5% at Qwen2.5-3B (Appendix Figure 10).

18 In an idealized model, our analysis shows that additional parameters can preserve more sampled
 19 alternatives at matched accuracy when they strengthen updates shared across correct outputs (Ap-
 20 pendix E.3, Theorem E.5); size alone provides no guarantee. We therefore ask how training can retain
 21 alternatives after they become rare.

22 2 ModeBench makes output identity executable

23 To distinguish a different solution from a different string, MODEBENCH assigns each accepted output
 24 a canonical key; each distinct key defines a verified *mode*. The validator returns both correctness
 25 and identity (Figure 2): computation routes in Countdown and MathIR, tested program behaviors in
 26 Python, and valid configurations in Graph and PantryPlan. Cosmetic aliases merge; distinct routes
 27 can share an answer (key rules: Appendix B.1).

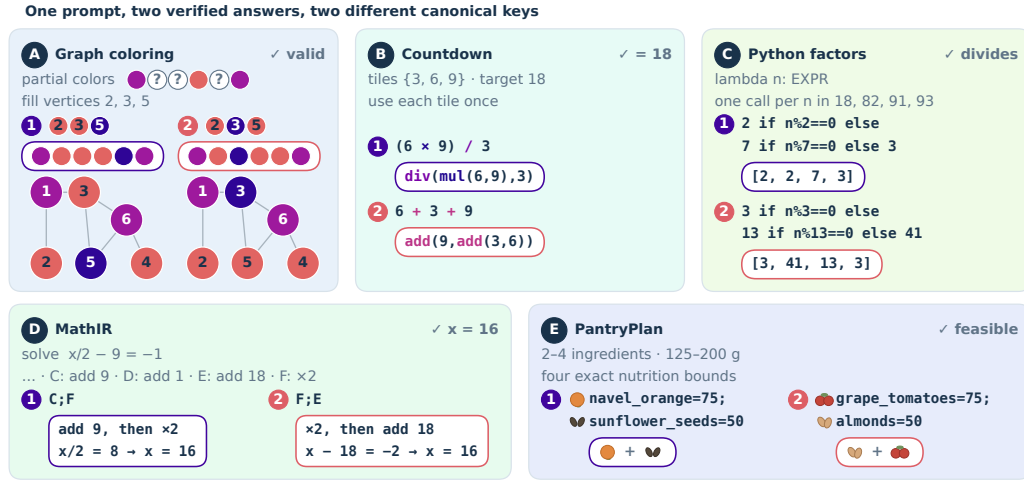


Figure 2: **One prompt, different verified outputs.** Keys retain a color vector, normalized computation tree, tested return vector, algebraic state trajectory, or ingredient support. Cosmetic aliases merge; these keys identify task-defined outputs rather than free-form reasoning.

28 3 ReplayMaxRL: learn from new samples and remember successes

29 Output identity also makes preservation actionable. A verified mode missing from fresh rollouts
 30 supplies no direct on-policy signal. ReplayMaxRL pairs fresh learning with a memory of earlier
 31 successes (Figure 3).

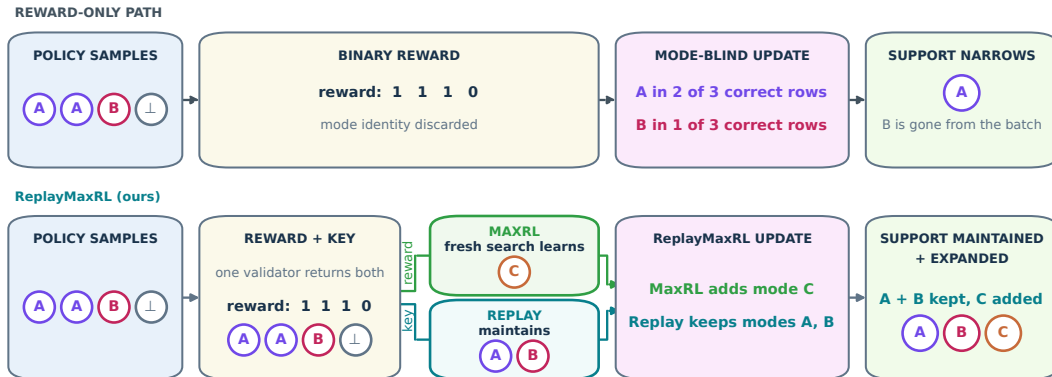


Figure 3: **Discovery and preservation use the same verified samples.** MaxRL learns from fresh rewards; replay retains exemplars by key and revisits rare successes. Memory cannot protect a mode before it is discovered.

MaxRL trains policy π_θ for success within k samples across sampling budgets [Tajwar et al., 2026]. Each training prompt x keeps one policy-generated exemplar e_c per admitted key in bank $\mathcal{B}_x$. With $|e_c|$ its token length, recurrent replay minimizes

$$L_{\text{replay}}(x) = -\frac{1}{|\mathcal{B}_x|} \sum_{c \in \mathcal{B}_x} \frac{\log \pi_\theta(e_c | x)}{|e_c|}. \quad (1)$$

Equal averaging of per-token scores makes replay independent of discovery frequency. Replacing MaxRL with Dr.GRPO [Liu et al., 2025], which debiases GRPO [Shao et al., 2024], yields our ReplayDr.GRPO. Appendix K gives the full update. Appendix E proves categorical collapse and replay retention (Theorems E.3 and E.12).

Related work. Prior work studies diversity collapse [Cui et al., 2025, Dang et al., 2025], sampling-aware objectives, and success replay [Zhang et al., 2025]. Our focus is execution-defined identity for measuring and retaining verified alternatives. Appendix D gives the full discussion.

4 Does memory preserve useful output support?

We test whether remembering training successes benefits new problems. `pass@8` is the chance that eight samples contain any correct response; `distinct@8` is their average number of distinct verified modes. Reporting both separates success from the alternatives available for selection, although mode counts remain coupled to correctness (definitions: Appendix A).

Five seeds are registered per comparison for eight passes; each of 128 held-out prompts per domain receives four eight-sample draws. Controls execute the replay code with its gradient zeroed. Figure 4 compares Dr.GRPO with replay and alternatives targeting reward normalization, on-policy diversity (UCPO; Lochab et al., 2026), success replay (RLEP-Dr), and semantic entropy. Protocols and uncertainty are in Appendices B and C.

A ReplayDr.GRPO (ours) – Dr.GRPO across three model scales

	$\Delta \text{ pass@8}$						$\Delta \text{ distinct@8}$					
Qwen 0.5B	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34	+2.11	+1.15	+0.40	+0.32	+1.02	+1.00
Falcon 1B	+0.15	+0.31 [†]	+0.98	+0.64	+0.12	+0.44[†]	+0.68	+0.42[†]	+1.11	+0.66	+0.94	+0.77[†]
Qwen 3B	+0.15	+0.14	+0.59	+0.41	+0.11	+0.28	+0.58	+0.33	+0.78	+0.43	+0.67	+0.56
	Graph	Countdown	Python	MathIR	Pantry	Average	Graph	Countdown	Python	MathIR	Pantry	Average

B Initial reference + alternatives – terminal Dr.GRPO at Qwen2.5-0.5B

	$\Delta \text{ pass@8}$						$\Delta \text{ distinct@8}$					
Before training	+0.13	-0.38	-0.17	-0.29	+0.38	-0.07	+0.24	-0.38	-0.17	-0.29	+1.90	+0.26
ReplayDr.GRPO (ours)	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34	+2.11	+1.15	+0.40	+0.32	+1.02	+1.00
MaxRL (no replay)	+0.21	+0.10	+0.00	-0.07	+0.01	+0.05	+0.35	+0.13	+0.00	-0.07	+0.01	+0.09
UCPO	-0.00	+0.13	+0.00	+0.03	+0.05	+0.04	-0.00	+0.18	+0.00	+0.03	+0.05	+0.05
RLEP-Dr	+0.10	+0.10	+0.00	+0.13	+0.00	+0.07	+0.15	+0.13	+0.00	+0.12	+0.00	+0.08
Semantic-MaxEnt	+0.00	+0.14	+0.00	-0.12	+0.20	+0.04	+0.00	+0.18	+0.00	-0.12	+1.12	+0.24
GRPO	+0.04	+0.16	+0.00	-0.05	+0.02	+0.03	+0.05	+0.17	+0.00	-0.05	+0.02	+0.04
	Graph	Countdown	Python	MathIR	Pantry	Average	Graph	Countdown	Python	MathIR	Pantry	Average

Figure 4: **Replay improves correctness and verified support across scales.** (A) ReplayDr.GRPO minus Dr.GRPO. (B) Qwen2.5-0.5B alternatives and the untrained reference minus Dr.GRPO. Daggers mark four admissible pairs; others use five. Average uses shared seeds. Left: probability differences. Right: key counts. Bold marks column maxima; uncertainty: Appendix F.

Memory improves the candidates available at evaluation. ReplayDr.GRPO improves both means in all 14 complete blocks and the remaining four-seed block; `distinct@8` rises in all 74 admissible pairs (source exclusion: Appendix F.1). On Qwen2.5-0.5B Countdown, `pass@8` rises from .48 to .67 and `distinct@8` from .49 to 1.64. The gain includes more computation routes alongside greater success. These held-out prompts never enter the bank: replay improves candidates for new problems. A matched weighting ablation favors uniform over fresh-frequency replay on Graph, Countdown, and PantryPlan; Python and MathIR remain inconclusive (Appendix F.4).

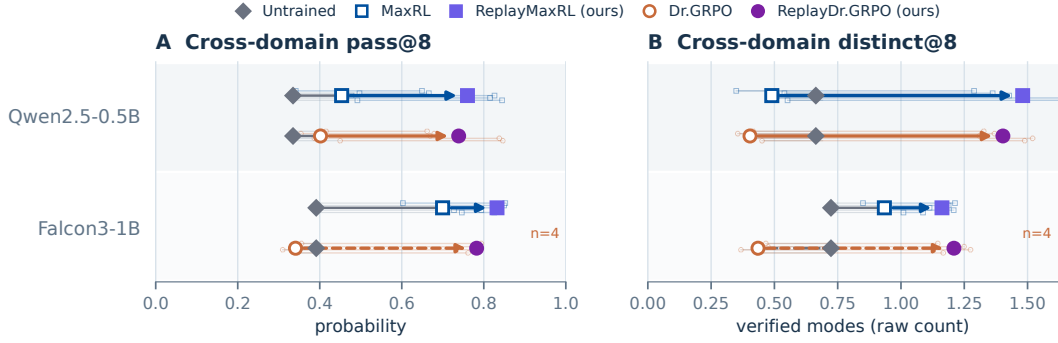


Figure 5: **Memory adds value beyond sampling-aware MaxRL.** Each track connects the untrained model, fresh-objective training, and its replay counterpart. Faint paths are seeds; bold paths average domains within seed. Falcon Dr.GRPO uses four common admissible seeds; other tracks use five. Both means improve at both scales.

59 **Stronger fresh learning still benefits from memory.** The first result leaves open whether a stronger
60 fresh objective would suffice. We therefore cross Dr.GRPO/MaxRL with replay absent/present
61 (Figure 5). Replay improves both cross-domain averages under both objectives; the Falcon Dr.GRPO
62 track is descriptive at four seeds. For ReplayMaxRL, both endpoints improve in all five Qwen
63 domains; Falcon correctness improves in five and breadth in four. The cross-domain averages are
64 post-hoc summaries; paired intervals are in Appendix C.2. Improving success across sampling
65 budgets does not remove the value of revisiting earlier successes: fresh learning still benefits from
66 persistent memory.

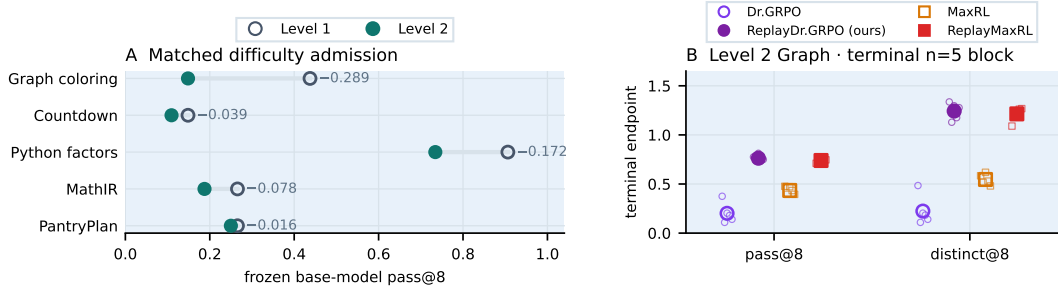


Figure 6: **Preservation matters on harder Graph.** (A) Frozen-model success across levels. (B) Completed Level-2 Graph factorial: small marks are five seeds, large marks their means. Every replay seed exceeds every non-replay seed on both endpoints.

67 **On harder problems, memory also changes the chance of success.** We next test Level 2, which
68 increases difficulty while preserving the validator, key rule, and valid-mode-count histogram (Ap-
69 pendix J). On harder Graph, Dr.GRPO/MaxRL reach .20/.43 pass@8, versus .76/.74 with replay.
70 The benefit extends to finding any correct candidate, giving memory a direct performance stake
71 (Figure 6). This complete five-seed result covers Graph; other Level-2 domains remain incomplete.

72 5 Conclusion and Limits

73 Remembering earlier successes complements fresh learning: replay improves held-out correctness
74 and sampled output support. Evaluation should therefore ask which verified alternatives remain
75 available, alongside what training learns to solve. These are small synthetic tasks, and replay can
76 revisit only discovered, retained modes. Individual mode survival (Appendix L) and size-dependent
77 update geometry remain unmeasured. The results do not establish that broader support causes
78 accuracy gains, or that verified modes have downstream utility. Open-ended mathematical reasoning
79 remains an important test.

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Supplementary Material

The supplement expands the four-page paper in reading order. Appendix A defines the measurements and explains the replay objective. Appendix B gives the domain and comparison designs; Appendix C preserves the detailed results behind the main figures. Related work appears in Appendix D, followed by the formal analysis in Appendix E and supporting comparisons in Appendix F. The remaining sections specify the semantic baseline, execution contracts, prompts, datasets, algorithm, mechanism checks, scale extensions, and reproducibility record. Appendix O states the limits and disclosures.

A Extended Definitions and Objective Rationale

Correctness and solution support are distinct properties of a policy. For a prompt x with reference specification s_x , the validator maps a response to either failure or a canonical key:

$$V(\cdot, s_x) : \mathcal{Y} \rightarrow \mathcal{C}_x \cup \{\perp\}.$$

Here $\mathcal{C}_x$ is the set of valid solution keys and $\perp$ denotes failure. The learner observes a key only for a response it generates; neither the full set $\mathcal{C}_x$ nor its size is revealed.

For $Y \sim \pi_\theta(\cdot | x)$, the policy induces both a total probability of verified success and a distribution over how that probability is allocated:

$$P_\theta^+(x) = \Pr[V(Y, s_x) \neq \perp], \quad p_\theta^+(c | x) = \Pr[V(Y, s_x) = c | V(Y, s_x) \neq \perp].$$

The conditional distribution is defined only when $P_\theta^+(x) > 0$; when $P_\theta^+(x) = 0$, all three sampling metrics below are zero. The first quantity measures correctness. The second describes the repertoire of verified behaviors conditional on being correct. We call concentration of p_θ^+ onto fewer keys *mode collapse*, particularly when P_θ^+ remains stable or increases.

A.1 Measuring success and support

For K independent samples $Y_{1:K}$, let

$$D_K = |\{V(Y_i, s_x) : i \leq K\} \setminus \{\perp\}|, \quad F_K = K^{-1} \sum_i \mathbf{1}[V(Y_i, s_x) \neq \perp].$$

ModeBench reports three averages with different units. $\text{distinct@}K = \mathbb{E}[D_K]$ is an expected count in $[0, K]$ of unique verified keys; $\text{pass@}K = \Pr[D_K > 0]$ and $\text{mean@}K = \mathbb{E}[F_K]$ are probabilities in $[0, 1]$. For every draw, $F_K \leq \mathbf{1}[D_K > 0] \leq D_K/K$; hence $\text{mean@}K \leq \text{pass@}K \leq \text{distinct@}K/K$. The inequality is valid, but equal numerical changes are not commensurate: .02 in $\text{pass@}8$ is two probability points, whereas .02 in $\text{distinct@}8$ is .02 expected verified keys.

For a fixed prompt, write $\mu_c(x) = \Pr[V(Y, s_x) = c]$. Independence gives

$$\text{pass@}K(x) = 1 - (1 - P_\theta^+(x))^K, \quad \text{distinct@}K(x) = \sum_{c \in \mathcal{C}_x} [1 - (1 - \mu_c(x))^K].$$

These identities apply before averaging over prompts; applying the nonlinear formula to an average success probability generally gives a different result.

Raw $\text{distinct@}K$ measures sampled coverage, not the total number of positive-probability modes. For an uninformed reference, suppose a finite space has N actions and n_c actions map to valid key c . Uniform independent action sampling then gives $\sum_c [1 - (1 - n_c/N)^K]$ expected verified keys. The simpler expression $m_x[1 - (1 - 1/N)^K]$ applies when the m_x valid actions have distinct keys, as in our reported finite-action references. Open-ended program and derivation spaces have no canonical nominal “uniform” generator.

The registered terminal co-primary endpoints are $\text{pass@}K$, the probability of finding any correct output, and raw $\text{distinct@}K$, the number of verified keys found. We report both separately for each model–domain–level block. The derived contrast $\text{distinct@}K - \text{pass@}K = \mathbb{E}[D_K - \mathbf{1}[D_K > 0]]$ is an interpretive decomposition: it estimates the number of verified modes beyond the first. It is computable without knowing the full valid set $\mathcal{C}_x$ and remains coupled to the number of successful draws. We use it as an interpretive decomposition rather than a fully success-matched diversity measure or a third primary endpoint. Because every headline evaluation fixes $K = 8$, reporting $\text{distinct@}8/8$ would add no information beyond a linear rescaling; we retain the raw count and label its units explicitly.

255 A.2 Why binary rewards amplify common modes

256 A binary reward identifies success but is indifferent among successful keys. GRPO and Dr.GRPO
 257 center that reward within each sampled group, assigning every correct response the same task credit
 258 regardless of its execution key. Key frequency is therefore the remaining asymmetry: a common
 259 correct mode appears in more updates, while an absent mode supplies no direct sampled signal.

260 A categorical reduction makes this feedback loop explicit. Let p_c denote the probability of correct
 261 mode c and $P = \sum_{c \in \mathcal{C}} p_c$ the total correct mass. Under Euclidean gradient ascent in independently
 262 trained softmax logits on expected binary reward, two correct modes c and d satisfy

$$\frac{d}{dt} \log \frac{p_c}{p_d} = (p_c - p_d)(1 - P).$$

263 Whenever incorrect mass remains, $p_c > p_d$ causes the log-odds gap to grow: binary reward magnifies
 264 an existing imbalance rather than correcting it. Finite on-policy groups add two sampling effects. An
 265 unsampled mode supplies no direct sampled score term, while an all-correct group has zero centered
 266 task advantage. The full derivation and the GRPO/Dr.GRPO equivalence appear in Appendix E,
 267 which states the assumptions and distinguishes this idealized mechanism from finite language-model
 268 training.

269 A.3 Why key-balanced replay targets both success and support

270 For a prompt x , the bank $\mathcal{B}_x$ contains one verified exemplar per discovered key. The replay term
 271 applies equal weight to those keys. Uniform sampling over keys prevents discovery frequency from
 272 determining replay frequency. To explain its relationship to the evaluation metric, we first analyze an
 273 equal-weight categorical surrogate, without length normalization. A stored response’s probability
 274 need not equal its key’s total probability. For $C = |\mathcal{B}_x|$ banked keys, write $p(c)$ for their probabilities
 275 and q_B for their total mass. Then

$$-\frac{1}{C} \sum_{c \in \mathcal{B}_x} \log p(c) = -\log q_B + \log C + \text{KL}(U_C \parallel p(c \mid c \in \mathcal{B}_x)). \quad (2)$$

276 The objective rewards greater banked verified mass, which can improve pass@K , and a more balanced
 277 conditional distribution, which is aligned with distinct@K . This decomposition does not guarantee
 278 monotonic changes in both terms under a combined policy update. Indeed banked-key coverage
 279 contributes $\sum_c [1 - (1 - p(c))^K]$ to expected distinct@K ; at fixed q_B and C , concavity makes it
 280 largest under uniform allocation, while $1 - (1 - q_B)^K$, the probability of any banked key, depends
 281 only on total mass. This deliberate objective–metric alignment motivates treating held-out pass@K
 282 as co-primary. Uniformity also maximizes the smallest conditional banked-key mass, making equal
 283 retention explicit. Banks contain only training-prompt responses; evaluation responses never enter
 284 training. Canonical keys may recur across different prompts.

285 B Benchmark Construction and Experimental Design

286 B.1 Executable domains and matched levels

287 A domain is mode-verifiable when the same execution both accepts a response and produces its
 288 canonical identity. Cosmetic aliases must merge; different executed outcomes must remain distinct.
 289 MODEBENCH instantiates this contract in five domain families. Each family is generated rather
 290 than fixed: we release two disjoint levels with the same validator and canonical-key definition, each
 291 containing 384 training prompts and a fixed 128-prompt evaluation split.

292 Figure 2 illustrates the five executable domains at Level 1:

- 293 • **Graph coloring** completes a partial three-color assignment. Execution checks every fixed color
 294 and edge; the full color vector is the key, yielding 6.27 valid modes per evaluation prompt on
 295 average across the fixed evaluation split of 128 prompts.
- 296 • **Countdown** builds an exact arithmetic expression from a fixed multiset of numbers. The key is
 297 its normalized executed syntax tree: commutative reorderings merge, while different computations
 298 remain distinct (4.52 modes per evaluation prompt).

- **Python factors** writes a restricted one-line function that is executed on frozen inputs. Its return vector is the key, so different source programs with identical behavior merge (229.44 modes).
- **MathIR** solves an equation by selecting executable algebra actions. The normalized state trajectory is the key, preserving different valid derivations of the same answer (5.00 modes).
- **PantryPlan** selects ingredients for a constrained plan; a trusted environment finds and validates exact feasible quantities. The selected ingredient support is the key, abstracting away quantity and order aliases (18.19 modes per evaluation prompt).

ModeBench targets task-defined executable output support. ReplayMaxRL stores one verified exemplar per observed task-defined key and samples uniformly over those keys. Appendices I, H, and J give the complete execution, alias, and split contracts.

Matched levels. Level 2 preserves each validator, key, and valid-mode-count histogram while adding structural difficulty: more hidden Graph vertices and Countdown operands, Python inputs above the base range, variable-on-both-sides MathIR equations, and tighter PantryPlan constraints. Frozen evaluation admits a level only when it is harder but solvable; Section B.6 gives the design.

B.2 Registered comparisons and evidence policy

Three experiments test replay retention and direct alternatives, cross fresh objective with memory, and transfer the factorial to a harder matched level. Appendix Table 8 reports comparator coverage.

Problem precheck: collapse without an intervention. We evaluate unmodified Dr.GRPO and GRPO at passes 0 and 8 across three models, five domains, and five seeds per method (150 runs). Their five-domain macro-average of extra modes falls by 98.2%–99.4% at Qwen2.5-0.5B and 41.5%–71.6% at larger scales (Appendix Figure 10). `pass@8` falls in 10 of the 30 domain–method–scale comparisons, while macro `distinct@8` ends above its pass-0 reference at Qwen2.5-3B. The precheck characterizes the failure pattern before intervention comparison.

B.3 Shared protocol and evidence policy

Runs use 384 training prompts, group size 16, temperature 1, top- $p = 1$, one PPO epoch, $\beta_{KL} = 0$, and eight passes (3,072 updates). Evaluation uses a disjoint 128-prompt split, a separate temperature-0 greedy trace, and four fixed $K = 8$ draws at temperature 1 and top- $p = 1$ at half-pass checkpoints. Samples are independent within a draw; draw $d \in \{0, 1, 2, 3\}$ uses the registered domain seed plus d . Paired methods share model revision, initialization, prompt order, optimizer, decoding, checkpoints, placement, and recovery rules. We average prompts and evaluation draws within each training seed before pairing. These are approximate, unadjusted intervals for training-seed variation conditional on the frozen evaluation prompts and seeds, not simultaneous bounds across comparisons.

The primary reporting unit is a five-seed model–domain–level terminal block. Only complete blocks receive paired 95% Student- t intervals; smaller admissible intersections report descriptive means and exact n without intervals. We do not substitute shallower checkpoints, impute final results, or pool models or levels. `pass@8` and raw `distinct@8` are co-primary; adjusted breadth is derived. Figure 4’s within-seed, equal-weighted domain Average is descriptive. Nominal exploratory sign tests use only the 14 complete block means and assume independent block signs; the two endpoint tests are Holm-adjusted (Appendix F.1). Appendix N gives the complete identities and fail-closed checks.

B.4 Experiment 1: retention and direct comparators

We pair Dr.GRPO with ReplayDr.GRPO on five Level-1 domains, three model scales, and five registered seeds: 75 planned pairs, of which 74 are source-admissible. Fourteen blocks contain five seeds; Falcon Countdown contains four (Appendix F.1). The methods share replay code and differ in whether its derivative is applied. Subsequent banks, schedules, and realized workloads depend on each policy. Fully matched Qwen2.5-0.5B blocks also provide direct alternatives: ordinary GRPO normalization; UCPO on-policy advantage redistribution [Lochab et al., 2026]; our frequency-weighted RLEP-Dr adaptation [Zhang et al., 2025]; memoryless binary MaxRL; and fixed Semantic-MaxEnt. Figure 4B compares them with ReplayDr.GRPO. The semantic estimator is defined only as a comparator in Appendix G.

B.5 Experiment 2: objective–memory factorial

The central 2×2 factorial crosses the fresh objective (Dr.GRPO or binary MaxRL) with verified replay (absent or present), yielding Dr.GRPO, ReplayDr.GRPO, MaxRL, and ReplayMaxRL. The two within-objective replay contrasts identify memory; their difference is the replay-by-objective interaction on the common admissible seed intersection. The MaxRL pair is registered for five seeds in all five Level-1 domains and three scales. Ten MaxRL–ReplayMaxRL blocks are complete at five seeds; the full four-arm intersection contains nine complete blocks and one Falcon Countdown block with $n = 4$. All use shared final pass-8 evaluations; smaller intersections are descriptive and report exact n without intervals.

B.6 Experiment 3: transfer to a harder matched level

ModeBench Level 2 preserves each domain’s validator, canonical key, split sizes, and split-wise valid-mode-count histogram while changing structure: four hidden Graph vertices, four Countdown operands, four Python cases including a value above 96, variable-on-both-sides MathIR equations, and tighter PantryPlan constraints. The disjoint Level-2 prompts receive the same four-method, five-seed Qwen2.5-0.5B factorial. We estimate ReplayDr.GRPO–Dr.GRPO, ReplayMaxRL–MaxRL, MaxRL–Dr.GRPO, and their interaction within level. Cross-level effect patterns compare the unpaired prompt sets.

C Extended Experimental Results

The three experiments build one argument: replay retains verified support across scale, remains useful after stronger fresh-rollout optimization, and survives the first controlled increase in task difficulty. The main figures show results completed by the September 4 analysis cutoff; every claim uses the displayed terminal seed intersection.

C.1 Experiment 1: canonical replay retains verified support across scale

Replay improves both success and verified support in every completed model–domain block. ReplayDr.GRPO has positive block-mean effects on pass@8 and raw distinct@8 in all 14 complete five-seed blocks. Both descriptive means are also positive in the four-seed Falcon Countdown block. Raw distinct@8 rises in all 74 admissible pairs. Appendix F.1 gives the source exclusion and nominal exploratory sign summaries. At Qwen2.5-0.5B, ReplayDr.GRPO also has the largest mean Δ pass@8 among the displayed alternatives in every domain.

On Countdown, replay raises correctness while preserving distinct routes to the same answer. At Qwen2.5-0.5B, pass@8 rises from .480 to .672, while distinct@8 more than triples from .487 to 1.637; adjusted breadth increases by +.959 (unadjusted 95% interval [+.863, +1.054]). Because Countdown keys encode computation trees, the increase includes more observed computation routes alongside more frequent success. It does not measure the survival of specific banked routes through training. The other domains sharpen the boundary of that claim. Python improves from .172 to .528 on pass@8, but adjusted breadth changes by only +.043 ([-.077, +.164]); MathIR improves from .512 to .796, with adjusted breadth +.030 ([+.001, +.060]). Thus the correctness benefit is broad, while the size and meaning of the support benefit depend on whether a key denotes a route, a functional behavior, or a valid configuration.

C.2 Experiment 2: replay adds value beyond MaxRL

Stronger fresh-rollout optimization does not eliminate the value of memory at either completed scale. Averaging the five domains within seed, ReplayMaxRL minus MaxRL is +.307 on pass@8 and +.991 raw distinct@8 at Qwen2.5-0.5B (unadjusted 95% intervals [+.208, +.406] and [+.815, +1.167]). At Falcon3-1B, the corresponding effects remain positive at +.133 and +.228 ([+.043, +.223], [+0.094, +.361]). Because MaxRL already rewards success across sampling budgets, the remaining increment tests the benefit of adding verified replay: a banked key absent from fresh rollouts can still contribute through its retained exemplar.

The gain follows the memory intervention, not a particular fresh objective. Both replay arrows move right on both cross-domain endpoints for both models (Figure 5). On the Dr.GRPO track,

replay adds (+.337, +.999) at Qwen2.5-0.5B. At Falcon3-1B, the four common admissible seeds 55–58 give descriptive effects (+.441699, +.773926) on pass@8/raw distinct@8, with no interval. The MaxRL track retains its five admissible seeds. The domain breakdown in Appendix Figure 13 sharpens the boundary: ReplayMaxRL improves both endpoints in every Qwen domain and Falcon pass@8 in every domain, while Falcon raw support improves in four of five. The paired tracks show that replay can add value under either fresh objective, including the domain-level exception; they do not directly measure new-key discovery or individual-mode survival.

C.3 Experiment 3: difficulty makes memory decisive

Replay raises the chance of finding a correct solution on harder Graph. The matched construction lowers frozen-model pass@8 from .438 at Level 1 to .148 at Level 2 while keeping the task solvable. After training, Dr.GRPO and MaxRL reach only .202 and .434, whereas ReplayDr.GRPO and ReplayMaxRL reach .763 and .739. Replay thus adds +.561 pass@8 to Dr.GRPO (unadjusted 95% interval [+.445, +.677]) and +.305 to MaxRL ([+.250, +.360]). This is not a single-seed effect: the weakest replay run scores .711, above the strongest non-replay run at .477. The replay intervention therefore improves the chance that repeated sampling finds a correct candidate. These joint endpoint gains do not establish that greater breadth causes greater correctness.

On the harder task, memory matters more than the choice of fresh-rollout objective. Without replay, replacing Dr.GRPO with MaxRL adds +.233 pass@8 and +.322 raw distinct@8. Adding replay is larger: its pass@8/raw-distinct@8 effects are (+.561, +1.019) for Dr.GRPO and (+.305, +.668) for MaxRL. Once replay is present, the methods nearly converge—.763 versus .739 on pass@8 and 1.243 versus 1.214 on raw distinct@8. Thus sampling-aware fresh updates help without memory, while adding replay improves both fresh objectives. This is a complete five-seed result for Graph; the other four Level-2 domains remain incomplete, so we do not claim cross-domain Level-2 generality.

D Related Work

Verifiable learning and output support. Repeated sampling and verifier-guided selection depend on the proposal distribution [Brown et al., 2024, Snell et al., 2024]. Prior work reports reduced sampled output diversity under RLHF or reasoning RL [Kirk et al., 2024, Dang et al., 2025], declining policy entropy [Cui et al., 2025], and reduced high-budget problem coverage in tested RLVR models [Yue et al., 2025]. These measures differ from within-prompt execution-key breadth. Semantic clustering infers when generations express the same answer [Farquhar et al., 2024]. MODEBENCH instead lets accepting execution define the equivalence relation, yielding an exact behavioral key.

Sampling-aware exploration. Outcome bonuses, entropy control, set objectives, distribution matching, and uniform-correct objectives encourage broader verified-output support [Song et al., 2025, Chen et al., 2025, Li et al., 2025, 2026a,b, Lochab et al., 2026]. MaxRL combines best@ k objectives across sampling budgets [Tajwar et al., 2026]. For our binary rewards, best@ k equals pass@ k ; ReplayMaxRL tests whether explicit mode memory adds support beyond this fresh-rollout objective.

Replay and persistent memory. Experience replay reuses past experience and can mitigate forgetting [Lin, 1992, Rolnick et al., 2019]; RLEP reuses successful reasoning trajectories [Zhang et al., 2025]. Our replay component instead stores one exemplar per canonical executed key and samples uniformly over keys, making equal retention an explicit target rather than depending on how often each success occurs. Novelty search rewards behavioral novelty [Lehman and Stanley, 2011], while MAP-Elites retains one elite per behavior-descriptor cell [Mouret and Clune, 2015]. Our replay bank uses the execution key as the descriptor inside a policy-gradient update.

E Mode Collapse and Verified Replay

Main-body link. This appendix supplies the formal claims invoked in Sections 3 and 4.

A binary verifier assigns the same reward to every correct mode, but an update can still amplify the most common one. We first explain that mechanism in a finite categorical model, then ask what changes when we alter the update geometry, add entropy, or replay earlier successes. Replay directly

protects represented modes; discovery, admission, and reliable neural optimization remain separate requirements.

How to read the proof chain. Begin with the three-correct-mode example in Section E.1. The group calculations reveal why a correctness gradient need not balance correct alternatives, and the next section turns the local imbalance into an asymptotic collapse theorem. Sections E.3–E.6 then vary geometry and entropy: they show both how collapse can slow and why it is not a universal consequence of rewarding correctness.

The second part asks what replay adds. Sections E.7 and E.8 separate a missing fresh signal from a persistent bank signal, then turn bounded replay loss into protected probabilities. The remaining sections connect that argument to complete responses, controlled stochastic updates, and actual bank admission. Finally, Section E.12 shows how much those bounds can certify and what finite observations can establish about a named mode. Each proof states its assumptions and attributes the literature tools it uses.

All worked numbers below are illustrative calculations in the stated models, not measurements from the training runs. Unless a new example is introduced, the examples reuse the same three correct modes and one incorrect category. Comparisons keep the optimized objective, decoding law, and time variable explicit, since changing any of these can change the conclusion.

Here G counts training rollouts, K counts evaluation draws, m, n count correct and incorrect categories, and k is the bank size. In the mean-flow results, t is continuous optimization time and τ is a reparameterized learning clock. The later stochastic and admission results use explicitly defined update or opportunity indices. None is a wall-clock or GPU-runtime prediction. “Collapse” refers to limiting concentration: finite softmax logits already give positive probabilities at every finite time. A uniform lower bound for all future times is the stronger retention property proved below.

E.1 Categorical reduction and the GRPO mean update

Fix one prompt. Let the finite response categories be $\mathcal{A} = \mathcal{C} \dot{\cup} \mathcal{N}$, where the $m \geq 2$ categories in $\mathcal{C}$ are correct execution modes and the $n \geq 1$ categories in $\mathcal{N}$ are incorrect. Let $p_a = \text{softmax}(z)_a$, $P = \sum_{c \in \mathcal{C}} p_c$, and $q_c = p_c/P$. Thus q is exactly the conditional correct-mode distribution $p_\theta^+(\cdot | x)$ defined in Appendix A, and P is the correct mass $P_\theta^+(x)$.

Running example: three correct alternatives. Take three correct modes and one incorrect category, with

$$p = (0.30, 0.15, 0.05, 0.50), \quad P = 0.50, \quad q = (0.60, 0.30, 0.10).$$

Half of all draws succeed, but among successful draws the three modes appear in a 6:3:1 ratio. Increasing P and balancing q are therefore separate goals. We reuse this hypothetical distribution below to distinguish the effects of the fresh objective, optimizer geometry, and replay.

Draw a group $A_1, \dots, A_G \stackrel{\text{iid}}{\sim} p$ with $G \geq 2$, write $R_i = \mathbf{1}[A_i \in \mathcal{C}]$, and let $M = \sum_i R_i$. Both GRPO variants can be written, at the on-policy point where the PPO ratio equals one, as

$$\hat{g}_G = \frac{1}{G} \sum_{i=1}^G w(M) \left(R_i - \frac{M}{G} \right) \nabla_z \log p_{A_i}. \quad (3)$$

For Dr.GRPO, $w(M) = 1$ after absorbing the shared positive length scale. For binary-reward GRPO, $w(M)$ is the reciprocal group reward standard deviation on mixed groups. We absorb the same fixed response-length normalizer in both variants; response-dependent row normalization is outside this reduction. In either case $0 < w(\ell) < \infty$ for $\ell \in \{1, \dots, G-1\}$. We define the entire update as zero on all-equal groups, without evaluating an undefined reciprocal standard deviation; likewise, $w(M)M(G-M)$ below is defined as zero when $M = 0$ or $M = G$. We take explicit reference KL to be zero, as in the experiments.

Lemma E.1 (A group baseline does not remove frequency amplification). *For $0 < P < 1$, the expected group update is*

$$\mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P, \quad c_G(P) = \frac{\mathbb{E}[w(M)M(G-M)]}{G^2 P(1-P)} > 0. \quad (4)$$

491 For Dr.GRPO, $c_G(P) = (G - 1)/G$. Hence reward centering and binary reward standardization
 492 change the speed, but not the expected direction, of the binary-reward policy gradient.

493 *Proof.* Condition on the complete binary reward vector, whose sum is M . The score identities give

$$\mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 1] = \nabla_z \log P, \quad \mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 0] = \nabla_z \log(1 - P).$$

494 There are M correct rows with centered reward $(G - M)/G$ and $G - M$ incorrect rows with centered
 495 reward $-M/G$. Therefore the conditional expectation of the unaveraged sum in (3) is

$$w(M) \frac{M(G - M)}{G} (\nabla_z \log P - \nabla_z \log(1 - P)) = w(M) \frac{M(G - M)}{GP(1 - P)} \nabla_z P.$$

496 The outer factor $1/G$ and expectation over $M \sim \text{Binomial}(G, P)$ yield (4). For $w \equiv 1$, $\mathbb{E}[M(G -$
 497 $M)] = G(G - 1)P(1 - P)$, proving the Dr.GRPO value. $\square$

498 The binomial identity also gives the denominator-free expression

$$c_G(P) = \frac{G - 1}{G} \sum_{j=0}^{G-2} \binom{G-2}{j} w(j+1) P^j (1 - P)^{G-2-j}.$$

499 The binomial weights sum to one, so c_G lies between $(G - 1) \min_j w(j+1)/G$ and $(G -$
 500 $1) \max_j w(j+1)/G$. This establishes the smooth, positive extension to $P = 0, 1$ used in the
 501 subsequent flow and convergence arguments.

502 The next lemma specializes the practical-estimator analysis of Tajwar et al. [2026, Section 4.3 and
 503 Appendix D, Theorem 5] to our categorical notation; we include the derivation for completeness.

504 **Lemma E.2** (Binary MaxRL has the same correctness-gradient direction). *Let MaxRL assign*
 505 $A_i^{\text{MaxRL}} = GR_i/M - 1$ *when* $M > 0$ *and zero when* $M = 0$, *as in Appendix K.1. At the on-policy*
 506 *point, its expected update is*

$$\mathbb{E}[\hat{g}_G^{\text{MaxRL}}] = c_G^{\text{MaxRL}}(P) \nabla_z P, \quad c_G^{\text{MaxRL}}(P) = \frac{1 - (1 - P)^{G-1}}{P} > 0, \quad (5)$$

507 *with continuous endpoint values* $c_G^{\text{MaxRL}}(0) = G - 1$ *and* $c_G^{\text{MaxRL}}(1) = 1$.

508 *Proof.* Condition on $M = m > 0$. Each successful row has advantage $(G - m)/m$ and each failed
 509 row has advantage -1 . Using the same conditional score identities as above, the expected unaveraged
 510 sum is

$$(G - m)(\nabla_z \log P - \nabla_z \log(1 - P)) = \frac{G - m}{P(1 - P)} \nabla_z P.$$

511 After the outer factor $1/G$ and averaging over $M \sim \text{Binomial}(G, P)$,

$$\mathbb{E}[(G - M)\mathbf{1}\{M > 0\}] = G(1 - P) - G\Pr(M = 0) = G[(1 - P) - (1 - P)^G].$$

512 Substitution gives Equation (5). Its endpoint values follow by continuity (or the finite geometric-series
 513 identity), which also gives the stated continuous extension at both boundaries. $\square$

514 The same finite-series identity gives the potential used below:

$$\Psi_{\text{MaxRL}}(P) := \int_0^P c_G^{\text{MaxRL}}(s) ds = \sum_{k=1}^{G-1} \frac{1 - (1 - P)^k}{k}. \quad (6)$$

515 In particular, $\Psi_{\text{MaxRL}}(1) = \sum_{k=1}^{G-1} 1/k$. For binary rewards, $\text{best}@k$ equals $\text{pass}@k$, so this is
 516 a harmonic mixture of sampling objectives for $k = 1, \dots, G - 1$. The upper limit is $G - 1$, not
 517 G : the implemented centered estimator drops the entire update on all-failure groups. Subtracting
 518 an unconditional score baseline even on those groups instead gives the order- G estimator. This
 519 distinction concerns the exact finite-group expectation at the on-policy point; it does not assert
 520 unbiasedness of subsequent clipped or adaptive-optimizer updates.

521 **What the group normalization changes.** For the running example with $G = 4$, $\nabla_z P =$
522 $(0.15, 0.075, 0.025, -0.25)$. The two lemmas give

$$\mathbb{E}[\hat{g}_4^{\text{Dr.GRPO}}] = 0.75 \nabla_z P,$$

$$\mathbb{E}[\hat{g}_4^{\text{MaxRL}}] = 1.75 \nabla_z P.$$

523 These are expectations over all possible groups, not updates from one particular sample. MaxRL
524 moves faster along the same direction at this state, while the most common correct mode receives six
525 times the logit increment of the rarest. The next section asks what repeated unequal increments do to
526 the relative probabilities of equally rewarded modes.

527 E.2 Winner-take-all collapse under GRPO, Dr.GRPO, and MaxRL

528 Prior work studies objective-induced outcome collapse and inverse-probability correction [Sinha
529 et al., 2026]; UCPO analyzes correctness-indifferent collapse through sampling bias and gradient
530 amplification [Lochab et al., 2026, Section 4]. Here we connect our group estimators to a standard cat-
531 egorical flow and then study persistent replay over execution-defined keys. Consider the infinitesimal
532 expected-update flow $\dot{z} = \mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P$. This is the clean mean-field limit of fresh on-policy
533 groups and the first-order PPO update. By Lemmas E.1 and E.2, it covers GRPO, Dr.GRPO, and
534 binary MaxRL with their respective positive functions c_G . It deliberately removes sampling noise;
535 the result below therefore identifies a structural instability rather than blaming finite batches.

536 The conditional correct-mode flow derived below is a standard frequency-dependent replicator system
537 with fitness $f_c(q) = q_c$ [Hofbauer and Sigmund, 1998]; the following theorem is not claimed as a new
538 replicator-dynamics result. Its role is to show that the expected GRPO, Dr.GRPO, and binary-MaxRL
539 updates reduce to that familiar winner-take-all flow under the paper’s categorical assumptions.

540 **A local view of the instability.** For $q = (0.60, 0.30, 0.10)$ in the running example, $\sum_c q_c^2 = 0.46$.
541 In the effective time τ defined in the proof, Equation (7) gives

$$\frac{dq}{d\tau} = (0.084, -0.048, -0.036), \quad \frac{d}{d\tau} \log \frac{q_1}{q_2} = 0.30.$$

542 The two less common modes lose their share of successful outputs even while correctness increases.
543 This calculation is local; the theorem establishes that the imbalance persists all the way to the
544 single-mode limit.

545 **Theorem E.3** (Standard replicator specialization for binary-reward RL). *Start from finite logits with*
546 *$0 < P(0) < 1$. If the conditional correct-mode distribution $q(0)$ has a unique largest coordinate c^* ,*
547 *then*

$$P(t) \longrightarrow 1, \quad q_{c^*}(t) \longrightarrow 1, \quad q_c(t) \longrightarrow 0 \quad (c \neq c^*).$$

548 *Thus these GRPO, Dr.GRPO, and binary MaxRL mean flows converge to a single correct execution*
549 *mode for generic initial conditions, even though every mode in $\mathcal{C}$ has exactly the same reward.*

550 *Proof.* For a correct mode c and incorrect category a ,

$$\dot{z}_c = c_G(P) p_c (1 - P), \quad \dot{z}_a = -c_G(P) p_a P.$$

551 Also $\dot{P} = c_G(P) \|\nabla_z P\|_2^2 > 0$. If P converged to an interior value, continuity and positivity of c_G ,
552 together with

$$\|\nabla_z P\|_2^2 \geq P^2 (1 - P)^2 \left(\frac{1}{m} + \frac{1}{n} \right),$$

553 would keep $\dot{P}$ bounded away from zero, a contradiction. Hence $P \rightarrow 1$.

554 Define the effective optimization time

$$\tau(t) = \int_0^t c_G(P(s)) P(s) (1 - P(s)) ds.$$

555 If $\tau(\infty) < \infty$, every correct and incorrect logit would have finite total variation, because the absolute
556 velocity of each is at most $\dot{\tau}$. All logits would then converge to finite values, contradicting $P \rightarrow 1$.
557 Therefore $\tau(t) \rightarrow \infty$, leaving infinite effective time for the correct-simplex dynamics below.

On the correct simplex, changing from t to τ gives the replicator flow

$$\frac{dq_c}{d\tau} = q_c \left(q_c - \sum_{d \in \mathcal{C}} q_d^2 \right), \quad \frac{d}{d\tau} \log \frac{q_c}{q_d} = q_c - q_d. \quad (7)$$

The initially unique maximum remains the unique maximum. For $c \neq c^*$, set $\xi_c = q_c/q_{c^*}$, so $\xi_c(0) < 1$ and ξ_c is nonincreasing. Since $q_{c^*} \geq 1/m$,

$$\frac{d \log \xi_c}{d\tau} = -q_{c^*}(1 - \xi_c) \leq -\frac{1 - \xi_c(0)}{m}.$$

Hence $\xi_c(\tau) \leq \xi_c(0) \exp[-(1 - \xi_c(0))\tau/m] \rightarrow 0$. Every competing mode vanishes relative to c^* ; normalization then yields $q_{c^*} \rightarrow 1$, establishing the claimed single-mode limit. $\square$

Corollary E.4 (Ties are nongeneric). *If exactly $k < m$ correct modes tie for the largest initial probability, symmetry preserves the tie and the flow converges to the uniform distribution on those k modes while eliminating the rest. Only the exactly uniform initialization preserves all m correct modes in the limit; a tie among $1 < k < m$ maxima yields partial collapse onto those k modes. Any perturbation creating a unique maximum gives single-mode collapse by Theorem E.3.*

An exact tie is a different initial condition. Starting instead from $q = (0.45, 0.45, 0.10)$ gives the limit $(0.50, 0.50, 0)$. Changing the initialization to $(0.451, 0.449, 0.10)$ makes the first mode the unique winner. Thus an exactly symmetric two-mode limit does not imply robustness to a small imbalance in the initial policy.

Finite groups also introduce sampling effects: a mode absent from a group has no sampled response of its own carrying a score term, all-correct groups have zero task advantage, and sampling noise can break exact ties. Softmax normalization and shared parameters can still change an unsampled mode's probability. These facts are not needed for the theorem.

E.3 How model parameterization changes collapse

The cross-scale losses raise a further question: can a larger model learn correctness while preserving more correct alternatives? The preceding theorem already allows every logit to vary independently, so representational capacity alone cannot answer this question. We give a conditional mechanism: directions that raise correct logits together can improve correctness with less amplification of differences among correct modes.

For a smooth parameterization $z = z(\theta)$, write $J_\theta = \partial z / \partial \theta$ and $K_\theta = J_\theta J_\theta^\top$. The chain rule extends Lemmas E.1 and E.2 to Euclidean parameter gradient flow:

$$\dot{z} = c_G(P) K_\theta \nabla_z P, \quad \frac{d}{dt} \log \frac{q_c}{q_d} = c_G(P) (\mathbf{e}_c - \mathbf{e}_d)^\top K_\theta \nabla_z P, \quad (8)$$

where $\mathbf{e}_c$ is a coordinate basis vector. Thus the logit-gradient geometry, not the number of columns of J_θ alone, governs mode amplification. The independent-logit theorem has $K_\theta = I$.

To isolate shared correctness learning, let $v = \mathbf{1}_C$ be the vector that is one on correct categories and zero on incorrect categories, and consider the constant geometry

$$K_\lambda = I + \lambda v v^\top, \quad \lambda \geq 0. \quad (9)$$

The added direction moves all correct logits equally, so it improves their mass without directly changing their ratios. For example, $z = u + \alpha \sum_{j=1}^s h_j v$, with trainable u and scalar h_j , realizes this geometry with $\lambda = s\alpha^2$. More such coordinates strengthen the shared update if α is fixed; normalizing $\alpha = 1/\sqrt{s}$ removes that dependence. These coordinates are redundant for expressivity and assume perfect correctness alignment. They describe an optimization mechanism, not a feature that an arbitrary larger language model is guaranteed to possess.

Theorem E.5 (Shared correctness learning preserves more sampled modes at matched accuracy). *Compare the flows (8) with $K_\theta = K_\lambda$, the same finite initial logits, and any of the three functions c_G above. Fix a target correct mass $P_* \in (P(0), 1)$. At the first time each flow reaches P_* , distinct θK*

is nondecreasing in λ for every integer $K \geq 1$, while $\text{pass}@K$ is identical. The increase is strict for $K \geq 2$ when $q(0)$ is nonuniform. In addition, every correct mode satisfies

$$q_c^{(\lambda)}(P_\star) \geq q_c(0) \exp\left(-\frac{L}{\lambda + 1/m + 1/n}\right), \quad L = \log \frac{P_\star}{1 - P_\star} - \log \frac{P(0)}{1 - P(0)}. \quad (10)$$

Here m, n count output categories, not model parameters. For every fixed finite λ , a unique initial maximum still implies eventual single-mode collapse as $t \rightarrow \infty$.

Proof. Let $r_a = p_a/(1 - P)$ for $a \in \mathcal{N}$, and write $S_q = \sum_c q_c^2$, $S_r = \sum_a r_a^2$. In the effective time $d\tau = c_G(P)P(1 - P) dt$, the logit velocities are $dz_c/d\tau = q_c + \lambda$ and $dz_a/d\tau = -r_a$. Hence

$$\frac{dq_c}{d\tau} = q_c(q_c - S_q), \quad \frac{dr_a}{d\tau} = -r_a(r_a - S_r), \quad \frac{d}{d\tau} \log \frac{P}{1 - P} = S_q + S_r + \lambda. \quad (11)$$

The paths $q(\tau), r(\tau)$ therefore do not depend on λ . Also $\dot{P} = c_G(P)P^2(1 - P)^2(S_q + S_r + \lambda) > 0$; the same interior-limit argument as in Theorem E.3 gives $P \rightarrow 1$. At $P_\star$, the effective time τ_λ uniquely solves

$$L = \lambda\tau_\lambda + \int_0^{\tau_\lambda} (S_q(u) + S_r(u)) du.$$

The right-hand side increases strictly with both its positive time argument and λ , so τ_λ decreases strictly with λ . Since $S_q \geq 1/m$ and $S_r \geq 1/n$, $\tau_\lambda \leq L/(\lambda + 1/m + 1/n)$. Integrating $d \log q_c/d\tau = q_c - S_q \geq -1$ over this bounded interval gives the probability floor in Equation (10).

At fixed $P_\star$, define $\mathcal{D}_K(q) = \sum_c [1 - (1 - P_\star q_c)^K]$, the expected number of correct modes in K independent draws. For $a(u) = KP_\star(1 - P_\star u)^{K-1}$, the replicator equation gives

$$\frac{d\mathcal{D}_K}{d\tau} = \frac{1}{2} \sum_{c,d} q_c q_d (q_c - q_d) (a(q_c) - a(q_d)) \leq 0.$$

The inequality holds because a is nonincreasing; it is strict for $K \geq 2$ and nonuniform q . Thus the smaller τ_λ gives larger $\text{distinct}@K$, whereas $\text{pass}@K = 1 - (1 - P_\star)^K$ is unchanged. Finally, $S_q + S_r + \lambda \leq 2 + \lambda$ and $P \rightarrow 1$ imply $\tau \rightarrow \infty$. The unchanged correct-mode replicator flow then gives eventual collapse under a unique initial maximum. $\square$

Same accuracy, different remaining breadth. Start from the running example and stop each idealized flow when P first reaches 0.90. Both policies then have $\text{pass}@8 = 1 - 10^{-8}$. Numerical integration of Equation (11) gives the following illustrative values; the probability floors are calculated separately from Equation (10) with $L = \log 9$.

λ	Conditional mass q_3	Guaranteed q_3 floor	$\text{distinct}@8$
0	0.05255	0.01924	2.125
4	0.08582	0.06623	2.374

The trajectory values are rounded approximations; the displayed floors are rounded down. The shared direction preserves more of the initially rare mode at the same correctness, but its mass still falls below its initial 0.10. A better finite-accuracy trajectory can therefore coexist with eventual loss, as the next corollary makes precise.

Corollary E.6 (Collapse rate as correctness approaches one). *Under Theorem E.5, fix finite $\lambda \geq 0$ and suppose the initial correct-mode maximum is unique. For every other correct mode c ,*

$$\lim_{P \uparrow 1} \frac{\log q_c^{(\lambda)}(P)}{\log(1 - P)} = \frac{1}{\lambda + 1 + 1/n}. \quad (12)$$

Thus $q_c^{(\lambda)}(P) = (1 - P)^{1/(\lambda + 1 + 1/n) + o(1)}$: larger fixed λ reduces the decay exponent, but every minority mode still vanishes. The exponent describes the limiting ratio of logarithms; finite-accuracy probabilities also depend on the initial distribution.

629 *Proof.* Theorem E.5 gives $q \rightarrow \mathbf{e}_{c^*}$, so $S_q \rightarrow 1$. The incorrect conditional flow in Equation (11)
630 preserves ordering and makes $r_{\min}$ nondecreasing. For $h = \log(r_{\max}/r_{\min})$,

$$h' = -(r_{\max} - r_{\min}) = -r_{\min}(e^h - 1) \leq -r_{\min}(0)h.$$

631 Hence $r \rightarrow (1/n, \dots, 1/n)$ and $S_r \rightarrow 1/n$, including trivially $n = 1$. Integrating $d \log q_c / d\tau =$
632 $q_c - S_q$ and the correctness-clock equation and taking time averages yields

$$\frac{\log q_c(\tau)}{\tau} \rightarrow -1, \quad \frac{\log(P(\tau)/(1 - P(\tau)))}{\tau} \rightarrow \lambda + 1 + 1/n.$$

633 Since $\log(1 - P) = \log P - \log(P/(1 - P))$ and $\log P \rightarrow 0$, their ratio proves Equation (12). $\square$

634 **A smaller exponent still permits extinction.** With one incorrect category, the minority-mode
635 exponents are $1/2$ at $\lambda = 0$ and $1/6$ at $\lambda = 4$. Both are positive, so both limits lose minority modes
636 as $P \uparrow 1$. These are asymptotic logarithmic slopes; they do not replace the finite-accuracy values in
637 the preceding table.

638 **Correctness alone does not imply collapse.** The independent-logit component matters. With
639 purely shared geometry $\mathbf{K} = \lambda \mathbf{v} \mathbf{v}^\top$, $\lambda > 0$, all correct logits move equally, so $q(t) = q(0)$ while
640 $\dot{P} = \lambda c_G(P) P^2 (1 - P)^2 > 0$ and $P \rightarrow 1$. For $\mathbf{K} = a \mathbf{I} + b \mathbf{v} \mathbf{v}^\top$ with fixed $a > 0$ and $b \geq 0$, time
641 rescaling instead recovers Theorem E.5 with $\lambda = b/a$. Thus persistent mode-specific amplification,
642 not increasing correctness by itself, drives this collapse result.

643 **Why parameter count alone is insufficient.** For an integer $h \geq 1$, replace each independent
644 logit by $z_a = h^{-\gamma} \sum_{j=1}^h \theta_{aj}$, initializing $\theta_{aj}(0) = h^{\gamma-1} z_a(0)$. Every added parameter affects the
645 logits, and every model represents the same categorical policies, but $\mathbf{K} = h^{1-2\gamma} \mathbf{I}$. Its policy follows
646 exactly the original collapse trajectory with time multiplied by $h^{1-2\gamma}$. Increasing the parameter
647 count therefore makes collapse faster, unchanged, or slower for $\gamma = 0, 1/2, 1$, respectively. For
648 example, using $h = 4$ multiplies the speed by 4, 1, or $1/4$ under those three normalizations, despite
649 quadrupling the parameter count in every case. A size claim therefore requires assumptions about
650 how additional parameters change the update geometry.

651 **What this explains and what remains untested.** Theorem E.5 connects a specified kind of shared
652 learning to the paper’s sampling metrics at matched accuracy. It also clarifies why a slower loss
653 can coexist with eventual collapse, leaving a role for replay. It does not identify the cause of the
654 empirical cross-scale pattern: those runs start from different pretrained distributions, use different
655 learning-rate schedules, and compare equal training passes rather than matched accuracy. Falcon also
656 changes model family. Neither the assumed geometry nor its dependence on size is measured here.
657 Establishing that link would require controlled within-family scaling and measurements of shared
658 versus mode-specific updates. The result retains prompt isolation, exact mean updates, and zero
659 reference KL; it does not cover shared-prompt interference, adaptive optimizer effects, or finite-step
660 stochastic training.

661 E.4 An exact optimizer comparison from information geometry

662 Changing the optimization metric lets us isolate correctness learning from changes among correct
663 alternatives. The next result shows both effects in separate equations for P and q .

664 Natural gradient on the categorical simplex uses the Fisher metric, also called the Shahshahani metric
665 in evolutionary dynamics [Harper, 2009, Sections 2.3 and 3.1]. Its gradient formula gives an exact
666 comparison with the Euclidean-logit collapse result in Theorem E.3. We derive the specialization
667 below so that the dependence on optimizer geometry is explicit.

668 **Theorem E.7** (Natural-gradient selection and replay). *Retain the finite categorical setup and start*
669 *from an interior distribution $p(0)$ with $0 < P(0) < 1$. Let $\Psi \in C^2([0, 1])$ satisfy $c(P) := \Psi'(P) > 0$*
670 *on $[0, 1]$. Choose a nonempty fixed bank $\mathcal{B} \subseteq \mathcal{C}$, positive weights w_b summing to one over $\mathcal{B}$, and*
671 *extend them by zero to a vector $\bar{w}$ on $\mathcal{A}$. Exact Fisher natural-gradient ascent of*

$$\mathcal{J}(p) = \Psi(P) + \rho \sum_{b \in \mathcal{B}} w_b \log p_b, \quad \rho \geq 0,$$

672 induces

$$\dot{p}_i = c(P)p_i(\mathbf{1}\{i \in \mathcal{C}\} - P) + \rho(\bar{w}_i - p_i), \quad (13)$$

$$\dot{P} = (1 - P)(c(P)P + \rho), \quad \dot{q}_c = \frac{\rho}{P}(\bar{w}_c - q_c). \quad (14)$$

673 For $\rho = 0$, $P(t) \rightarrow 1$ while $q(t) = q(0)$: correctness learning preserves all relative probabilities
 674 among correct modes. For $\rho > 0$, define $A(t) = \int_0^t \rho/P(s) ds$. Every correct mode satisfies

$$q_c(t) = \bar{w}_c + (q_c(0) - \bar{w}_c)e^{-A(t)}, \quad A(t) \geq \rho t. \quad (15)$$

675 Furthermore,

$$1 - P(t) \leq (1 - P(0))e^{-\rho t}, \quad p_b(t) \geq P(0) \min\{q_b(0), w_b\} > 0 \quad (b \in \mathcal{B}, t \geq 0). \quad (16)$$

676 Thus full-support replay sends the conditional distribution to its bank target, whereas an incomplete
 677 bank still loses unbanked correct modes in the limit.

678 *Proof.* For tangent vectors u, v with zero coordinate sum, the categorical Fisher metric is $g_p(u, v) =$
 679 $\sum_i u_i v_i / p_i$. The tangent vector

$$v_i = p_i \left(\partial_i \mathcal{J} - \sum_j p_j \partial_j \mathcal{J} \right)$$

680 has zero sum and satisfies $g_p(v, u) = \sum_i \partial_i \mathcal{J} u_i$ for every tangent u . It is therefore the natural
 681 gradient. Here

$$\partial_i \mathcal{J} = c(P)\mathbf{1}\{i \in \mathcal{C}\} + \rho \bar{w}_i / p_i, \quad \sum_i p_i \partial_i \mathcal{J} = c(P)P + \rho,$$

682 which proves Equation (13). Summing over correct categories gives the P equation. Differentiating
 683 $q_c = p_c/P$ then gives the q equation in (14); the correctness terms cancel.

684 The P equation makes P nondecreasing. If $\rho = 0$, q is constant; if $\rho > 0$, solving its linear
 685 equation gives Equation (15). Since $P \leq 1$, this also gives $A(t) \geq \rho t$. Incorrect coordinates satisfy
 686 $\dot{p}_i = -(c(P)P + \rho)p_i$ and remain positive at every finite time. Correct conditional coordinates
 687 remain positive by the explicit solution, and $P \geq P(0) > 0$. Because c is bounded on $[0, 1]$ and
 688 $\rho/P \leq \rho/P(0)$, these formulas extend the interior solution to every finite time.

689 For $\rho = 0$, P increases and cannot tend to a limit below one: at any such limit $(1 - P)c(P)P$
 690 would stay positive. For $\rho > 0$, $\dot{P} \geq \rho(1 - P)$ yields the correctness bound in Equation (16). Each
 691 banked $q_b(t)$ lies between $q_b(0)$ and w_b , so multiplication by $P(t) \geq P(0)$ proves the probability
 692 floor. Finally, $A(t) \rightarrow \infty$ gives $q_c(t) \rightarrow \bar{w}_c$, so unbanked correct modes have zero limiting mass. $\square$

693 **Worked example: the metric changes which modes grow.** At the running example's $P(0) = 0.5$,
 694 $q(0) = (0.6, 0.3, 0.1)$, and $G = 4$, Dr.GRPO has $c(P) = 3/4$. Without replay, the Fisher flow
 695 gives $\dot{P}(0) = 0.1875$ and $\dot{q}(0) = 0$: the model learns correctness while keeping its correct-mode
 696 proportions. With the uniform three-mode target $w = (1/3, 1/3, 1/3)$ and $\rho = 0.1$, the same initial
 697 distribution instead gives

$$\dot{P}(0) = 0.2375, \quad \dot{q}(0) \approx (-0.05333, 0.00667, 0.04667).$$

698 Replay now increases the rare third mode's share among correct responses. Equation (16) also
 699 guarantees $p_3(t) \geq 0.05$ for all time in this ideal flow. These are instantaneous derivatives and a
 700 trajectory bound, not the result of taking a discrete optimizer step.

701 The smooth potentials with positive derivatives in Lemmas E.1 and E.2 satisfy the premises. The
 702 Fisher-gradient formula is established information geometry; the result above is its elementary
 703 specialization to correctness selection plus replay toward a fixed target. It makes the scope of the
 704 collapse claim concrete: increasing correctness alone does not force a loss of conditional support
 705 under every optimization geometry. The theorem concerns the exact categorical Fisher flow. It gives
 706 no guarantee for approximate natural gradient, a shared neural Fisher matrix, PPO, or AdamW, and
 707 its time variable is an ideal optimization clock rather than elapsed training time.

708 E.5 Exact entropy and sampled surprisal have different guarantees

709 To interpret an entropy ablation, we must specify whether the update differentiates exact entropy or
 710 applies a bounded score to sampled responses. The two lemmas below make that distinction explicit.

711 The preceding collapse theorem has no entropy term. It does not imply collapse under exact Shannon-
 712 entropy regularization, which has a different vector field.

713 **Lemma E.8** (Exact categorical entropy prevents extinction). *Let $d = m + n$, $p = \text{softmax}(z)$, and
 714 $J(z) = \Psi(P(z)) + \beta H(p(z))$, where $\Psi \in C^2([0, 1])$, $\beta > 0$, and $0 \leq \Psi'(P) \leq M < \infty$. Under
 715 the exact independent-logit gradient flow $\dot{z} = \nabla_z J$, put*

$$\bar{z} = \frac{1}{d} \sum_i z_i(0), \quad \ell = \min\{\min_i z_i(0), \bar{z} - M/\beta\}.$$

716 For every $t \geq 0$ and every category i ,

$$p_i(t) \geq \frac{1}{d} \exp(d(\ell - \bar{z})) > 0.$$

717 If Ψ is also concave, the policy converges to the unique interior maximizer of $\Psi(P) + \beta H(p)$. In
 718 particular, for Dr.GRPO with $a = (G - 1)/G$, the limit is

$$p_i^* = \frac{\exp(a \mathbf{1}\{i \in \mathcal{C}\}/\beta)}{m \exp(a/\beta) + n}.$$

719 Thus the conditional correct-mode distribution is uniform. The full entropy objective also retains
 720 positive incorrect mass, so its limiting correctness is below one for fixed $\beta > 0$.

721 *Proof.* The gradient components are

$$\dot{z}_i = p_i [\Psi'(P)(\mathbf{1}\{i \in \mathcal{C}\} - P) + \beta(-\log p_i - H(p))].$$

722 Their sum is zero, so $\bar{z}$ is conserved. Jensen’s inequality and $H(p) \leq \log d$ give $-\log p_i - H(p) \geq$
 723 $\bar{z} - z_i$. Consequently, at any coordinate with $z_i = \ell$, its velocity is at least $p_i[-M + \beta(\bar{z} - \ell)] \geq 0$.
 724 The region $z_i \geq \ell$ for every i is therefore forward invariant. Conservation of the sum also gives
 725 $z_i \leq d\bar{z} - (d - 1)\ell$, proving the probability bound.

726 These bounds make the fixed-mean logit trajectory compact. Since $\dot{J} = \|\nabla_z J\|_2^2$ and J is bounded
 727 on that compact set, the gradient has a finite squared-norm integral. Bounded derivatives on the set
 728 then give $\nabla_z J \rightarrow 0$, so all limit points are stationary. Interior softmax stationarity is equivalent to
 729 simplex stationarity. Concavity of Ψ and strict concavity of βH give a unique maximizer; it is interior
 730 because the entropy gain from adding a missing coordinate dominates the bounded first-order change
 731 of Ψ . Hence the entire policy converges to that maximizer. For linear $\Psi(P) = aP$, the simplex
 732 first-order conditions give the stated Gibbs distribution. $\square$

733 **Worked example: exact full entropy preserves incorrect mass too.** For a separate hypothetical
 734 fixture with two correct categories and one incorrect category, take $G = 4$, $\Psi(P) = 3P/4$, and
 735 $\beta = 1/4$. The reward-to-entropy ratio is $a/\beta = 3$, so the lemma gives

$$p^* = \frac{(e^3, e^3, 1)}{2e^3 + 1} \approx (0.48786, 0.48786, 0.02429), \quad P^* \approx 0.97571.$$

736 Both correct modes survive equally, but the incorrect category also retains about 2.43% probability.
 737 This is the limit of the exact full-output entropy flow under the lemma’s assumptions; it is not a
 738 prediction for the historical sampled semantic score.

739 This result concerns exact entropy over the finite output categories in the reduction. Entropy over
 740 completion strings, conditional verified keys, and sampled token distributions are different objectives.
 741 It does not certify a clipped surrogate or an adaptive neural optimizer. The concavity condition holds
 742 for Dr.GRPO and the implemented binary-MaxRL potential.

743 **Lemma E.9** (Bounded sampled scores have no probability-independent boundary force). *Condition
 744 on the training history, draw $A_1, \dots, A_G$ independently from p , and let a_i be any detached, possibly
 745 group-dependent advantages with $|a_i| \leq B$. At the on-policy point define*

$$\hat{g} = \frac{1}{G} \sum_{i=1}^G a_i (e_{A_i} - p).$$

746 For every category b ,

$$|\mathbb{E}[\hat{g}_b]| \leq \mathbb{E}[\hat{g}_b] \leq 2Bp_b(1 - p_b).$$

747 On a group with no draw of b , exactly $\hat{g}_b = -p_b G^{-1} \sum_i a_i$; it is zero if the applied advantages
 748 sum to zero. By comparison, the verified-replay component for a fixed target w has logit velocity
 749 $\rho(w_b - p_b)$, which approaches $\rho w_b > 0$ for a banked category as $p_b \rightarrow 0$.

750 *Proof.* Use the triangle inequality and $|a_i| \leq B$. Each marginal draw satisfies $\mathbb{E}[\mathbf{1}\{A_i = b\} - p_b] =$
 751 $2p_b(1 - p_b)$, irrespective of the dependence of a_i on the group. On the absence event all indicator
 752 terms are zero. The replay formula follows by differentiating $-\sum_b w_b \log p_b$. $\square$

753 The fixed historical semantic advantage has $B = \eta = .10$, including its unit pseudocount, reserved
 754 unseen bucket, and clipped surprisal. The bound does not prove inevitable collapse or rule out
 755 asymptotic retention by a particular bounded estimator. Unsampled probabilities can also change
 756 through softmax normalization or shared parameters. Exact entropy is outside the uniformly bounded-
 757 score premise because its surprisal is unbounded near zero.

758 E.6 Exact conditional entropy and replay target the same optimum

759 Two regularizers can favor the same uniform target while assigning different penalties to rare modes.
 760 The following identities locate that difference.

761 The distinction between confidence penalties and label smoothing is a known choice of KL direction
 762 [Pereyra et al., 2017, Section 3.2]. Log-barrier regularization has also been studied in policy-gradient
 763 theory [Agarwal et al., 2021, Section 5.2]. Here we specialize those identities to a target supported
 764 on verified correct modes. The resulting comparison separates an objective’s maximizer from what
 765 its value certifies about a rare mode.

766 Let $u_c = 1/m$ on all $m \geq 2$ correct categories, retain $P > 0$ and $q_c = p_c/P$, and write $H(q) =$
 767 $-\sum_c q_c \log q_c$, with $0 \log 0 = 0$.

768 **Lemma E.10** (Full-support objective comparison). *For a bank containing every correct category*
 769 *with a uniform categorical replay target,*

$$H(q) = \log m - \text{KL}(q\|u), \quad (17)$$

$$R_u(p) := -\frac{1}{m} \sum_{c \in \mathcal{C}} \log p_c = \log m - \log P + \text{KL}(u\|q). \quad (18)$$

770 If Ψ is continuous and strictly increasing on $[0, 1]$, then for all $\beta, \rho > 0$, both objectives

$$\Psi(P) + \beta H(q) \quad \text{and} \quad \Psi(P) - \rho R_u(p)$$

771 have the unique maximizing categorical distribution $P = 1, q = u$. Maximization is over the closed
 772 simplex with $P > 0$; replay loss is $+\infty$ if a protected coordinate is zero. This identifies the optimum;
 773 convergence of conditional-entropy flow or stochastic updates is not asserted.

774 *Proof.* Expanding the KL divergences gives the first identity. Substitution of $p_c = Pq_c$ into R_u gives
 775 the second. In the first objective, $H(q) \leq \log m$, with equality only at $q = u$; after choosing this q ,
 776 strict increase of Ψ selects $P = 1$. The second objective is

$$\Psi(P) + \rho \log P - \rho \log m - \rho \text{KL}(u\|q).$$

777 Its terms depending on P increase strictly, while the last term is maximized only at $q = u$. At $P = 1$
 778 all incorrect probabilities are zero, so the resulting full categorical distribution is unique. $\square$

779 **Worked example: separate correctness from conditional spread.** For the running distribution
 780 $p = (0.30, 0.15, 0.05, 0.50)$, the uniform target on the three correct modes gives

$$\begin{aligned} H(q) &\approx \log 3 - 0.20067 \approx 0.89795, \\ R_u(p) &\approx \log 3 - \log 0.5 + 0.24052 \approx 2.03228. \end{aligned}$$

781 The subtracted 0.20067 is $\text{KL}(q\|u)$; the added 0.24052 is $\text{KL}(u\|q)$. Raising P from 0.5 to 0.9 while
 782 holding q fixed leaves $H(q)$ unchanged but reduces replay loss by $\log(0.9/0.5) \approx 0.58779$. Thus

783 replay’s loss measures both correct mass and allocation within it; conditional entropy measures only
784 the latter. The correctness potential $\Psi(P)$ supplies the first objective’s incentive to increase P .

785 Full-output entropy $H(p)$ is a different comparator. Its Gibbs optimum in Lemma E.8 follows the
786 standard entropy-conjugacy calculation [Geist et al., 2019, Section 2.2]. That optimum assigns
787 positive incorrect mass for fixed $\beta > 0$ under linear correctness reward. By contrast, Lemma E.10
788 shows that exact *conditional* entropy has no such loss of correctness at its global optimum. The
789 historical semantic score used in the experiments is not an exact gradient of either entropy objective.

790 Existing literature also gives discrete-time guarantees for an exact comparator. Under their finite-
791 bandit setup and step-size conditions, Mei et al. [2020, Lemma 13 and Theorem 5] prove positive
792 probability floors and geometric convergence for exact tabular entropy-regularized policy gradient.
793 The Dr.GRPO reduction has the linear reward vector $a\mathbf{1}_C$ with $a = (G - 1)/G \in (0, 1)$, which
794 satisfies their bounded-reward premise. This linear-reward result does not directly cover the nonlinear
795 finite-group MaxRL potential, the historical semantic estimator, a clipped PPO surrogate, or AdamW.

796 The KL directions explain a difference between certificates: $\text{KL}(u\|q)$ diverges if any protected
797 coordinate vanishes, whereas $\text{KL}(q\|u)$ stays finite. In particular, losing one mode and allocating
798 mass uniformly to the remaining modes leaves $H(q) = \log(m - 1)$. Section E.12 gives the sharp
799 entropy threshold and stronger replay-based probability bounds. This comparison concerns objective
800 level sets; it does not assert that exact entropy dynamics lose a mode.

801 E.7 When replay supplies a signal that fresh groups lack

802 High accuracy can remove the reward contrasts that fresh-group learning needs. A stored target can
803 still supply a learning signal in that regime.

804 The collapse theorem concerns an expected direction. A separate question is how often a finite group
805 supplies any task signal, and whether a stored mode must reappear in that group before receiving a
806 nonzero update from the replay objective.

807 **Lemma E.11** (Fresh-group starvation and replay signal). *Fix a categorical policy with correctness*
808 *probability P , and draw $G \geq 2$ fresh responses independently. For GRPO, Dr.GRPO, and the*
809 *implemented centered MaxRL estimator, the probability of a mixed-reward group is*

$$h_G(P) = 1 - P^G - (1 - P)^G \leq G \min\{P, 1 - P\}.$$

810 *Every all-equal group has zero fresh task gradient. Thus h_G is the probability of an opportunity for a*
811 *nonzero task gradient, rather than a lower bound on the gradient magnitude.*

812 *For a fixed bank with normalized positive weights w_b , let $R_w = -\sum_b w_b \log p_b$ and $\rho_w > 0$. Extend*
813 *w by zero outside the bank to obtain $\bar{w}$. The deterministic replay ascent vector is*

$$v^{\text{replay}} = -\rho_w \nabla R_w = \rho_w (\bar{w} - p).$$

814 *It vanishes only at $p = \bar{w}$. In particular, if $p_b \leq w_b/2$ for a banked mode, then $v_b^{\text{replay}} \geq \rho_w w_b/2$,*
815 *even when b is absent from the fresh group. For a uniform bank of $k \geq 2$ modes, as $p \rightarrow \mathbf{e}_b$ at*
816 *one banked correct mode, $h_G(P) \rightarrow 0$ whereas $\|v^{\text{replay}}\|_2 \rightarrow \rho_w \sqrt{1 - 1/k} > 0$. The replay norm*
817 *therefore stays positive as fresh task-signal opportunities vanish.*

818 *Proof.* The all-correct and all-incorrect events are disjoint and have probabilities P^G and $(1 - P)^G$.
819 A mixed group requires both a correct and an incorrect sample; the union bound gives the two bounds
820 GP and $G(1 - P)$. For $R_w = -\sum_b w_b \log p_b$, the softmax score identity gives $\partial R_w / \partial z_a = p_a - \bar{w}_a$,
821 proving the replay formula and coordinate bound. Finally,

$$\|\bar{w} - \mathbf{e}_b\|_2^2 = (1 - 1/k)^2 + (k - 1)/k^2 = 1 - 1/k,$$

822 which gives the strictly positive limiting norm for $k \geq 2$. □

823 **Worked example: a rare key need not reappear to receive replay.** Consider the hypothetical
824 policy $p = (0.600, 0.389, 0.001, 0.010)$, with the first three categories correct, and take $G = 16$. Its
825 $P = 0.99$ gives

$$h_{16}(0.99) = 1 - 0.99^{16} - 0.01^{16} \approx 0.14854.$$

Only about 14.85% of groups offer a fresh task-gradient opportunity. The third key is absent with probability $0.999^{16} \approx 0.98412$. Nevertheless, uniform replay over the three correct keys with $\rho_w = 0.1$ has third logit coordinate $v_3^{\text{replay}} = 0.1(1/3 - 0.001) \approx 0.03323$, including on groups that miss that key. The two sampling probabilities describe different events; the replay value is the deterministic full-bank vector from the lemma.

Here the replay vector averages the fixed bank; a stochastic draw of one exemplar has this vector only in expectation. Its nonzero logit coordinate does not guarantee a nonzero neural-parameter gradient after multiplication by a possibly degenerate Jacobian. Nor does replay necessarily increase the next mixed-group probability: h_G is largest at $P = 1/2$, so increasing an already high P decreases it. The precise benefit is a separate verified learning signal when the fresh task gradient is unavailable.

E.8 Why uniform verified replay prevents extinction

A persistent replay signal becomes a retention guarantee when vanishing probability would force a bounded loss to diverge. The next argument makes that loss barrier explicit, then shows what changes with unequal weights and incomplete coverage.

Let $\mathcal{B} \subseteq \mathcal{C}$ be a nonempty fixed bank of k modes that the policy has produced and the validator has accepted. Uniform verified replay uses one exemplar per banked mode and, in the categorical reduction, minimizes

$$R_{\mathcal{B}}(z) = -\frac{1}{k} \sum_{b \in \mathcal{B}} \log p_b. \quad (19)$$

This is ordinary cross-entropy against the distribution that assigns mass $1/k$ to each banked mode and zero elsewhere. It is not conditioned on the bank: decreasing (19) can raise total bank mass as well as redistribute mass within the bank.

In an averaged continuous-time model, absorb a fixed round-robin replay frequency and loss scale into an effective dose $\rho > 0$. This assumes a fixed eligible-bank set and constant relative weighting; it does not identify finite alternating updates with simultaneous gradient flow. The prompt-isolated categorical mean flow for any of the three fresh binary-reward objectives above plus verified replay is

$$\dot{z} = c_G(P) \nabla_z P - \rho \nabla_z R_{\mathcal{B}}(z). \quad (20)$$

Let $\Psi(P) = \int_0^P c_G(s) ds$. For the finite-group estimators in Lemmas E.1 and E.2, c_G has a finite positive extension to $[0, 1]$; in particular, Dr.GRPO has $c_G = (G - 1)/G$ and MaxRL has $c_G = [1 - (1 - P)^{G-1}]/P$. Define

$$F_{\mathcal{B}}(z) = \rho R_{\mathcal{B}}(z) - \Psi(P(z)). \quad (21)$$

Then (20) is exactly gradient flow $\dot{z} = -\nabla_z F_{\mathcal{B}}$.

Theorem E.12 (Verified replay prevents banked-mode extinction). *Suppose that after time T the nonempty verified bank $\mathcal{B}$ is fixed, fits within replay capacity, and receives a fixed positive effective replay dose ρ . Under (20), define*

$$C_T = R_{\mathcal{B}}(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho}.$$

Then for every $b \in \mathcal{B}$ and every $t \geq T$,

$$p_b(t) \geq \exp(-kC_T) > 0. \quad (22)$$

Thus each banked probability is uniformly bounded away from zero for all $t \geq T$, including asymptotically, even when the bank does not cover every correct mode. This is stronger than finite-time positivity, which already follows from finite softmax logits without replay.

Proof. Along the flow,

$$\frac{d}{dt} F_{\mathcal{B}}(z(t)) = -\|\nabla_z F_{\mathcal{B}}(z(t))\|_2^2 \leq 0.$$

Since $P(t) \leq 1$ and Ψ is increasing,

$$R_{\mathcal{B}}(z(t)) \leq R_{\mathcal{B}}(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho} = C_T.$$

Every $-\log p_b$ is nonnegative and their sum is $kR_{\mathcal{B}}$. Therefore $-\log p_b(t) \leq kC_T$ for every banked mode, which is exactly Equation (22) for every time after T . $\square$

Worked example: a rare mode cannot hide inside a bounded loss. In the running example, bank the first and third correct modes, whose probabilities are .30 and .05. Their uniform replay loss is $R_B = -\frac{1}{2} \log(.30 \cdot .05) \approx 2.09985$. For the illustrative Dr.GRPO choices $G = 4$ and $\rho = .30$, $C_T = R_B + (.75)(1 - .50)/.30 \approx 3.34985$, giving a floor $p_b \geq e^{-2C_T} \approx 0.00123127$. The point is the constraint on the loss: if either banked probability fell to 10^{-4} , its term alone would contribute $-\frac{1}{2} \log(10^{-4}) \approx 4.60517 > C_T$. The other term is nonnegative and cannot cancel this increase. These are hypothetical categorical values; the argument explains the barrier without making its conservative floor an experimental prediction.

This is a qualitative positivity statement, not a useful numerical guarantee. If a prompt receives fresh updates with frequency $\nu_x > 0$ and replay with frequency $\mu_x > 0$, normalizing time by its fresh-update frequency gives $\rho = \rho_{\text{raw}} \mu_x / \nu_x$. Equal fresh and replay frequencies cancel; round-robin scheduling alone does not imply dilution of the relative dose. As an illustrative equal-frequency categorical calculation, the registered $G = 16$ coefficient and per-rollout scale give $\rho = .10(15/256) = 5.86 \times 10^{-3}$. With $k = 16$, the worst-case Dr.GRPO potential term alone permits a bound on the order of $\exp(-2560)$ (and $R_B(z(T))$ can only weaken it). The bound is therefore vacuous at machine and experimental scales; it establishes only that an idealized continuous-time, fixed-bank coordinate cannot vanish asymptotically.

The retention bound also holds under smooth parameterization. The same argument applies to $\dot{\theta} = -\nabla_{\theta} F_B(z(\theta))$:

$$\frac{d}{dt} F_B(z(\theta(t))) = -\|J_{\theta}^{\top} \nabla_z F_B\|_2^2 \leq 0.$$

Thus Equation (22) holds regardless of parameter count for this fixed-bank, prompt-isolated parameter gradient flow. It does not require the shared-correctness geometry in Theorem E.5. This extends the qualitative retention bound, not convergence to a uniform distribution: a restricted or degenerate parameterization can stall. The next corollary retains independent logits.

Corollary E.13 (Conditional full-coverage categorical limit). *Assume the conditions of Theorem E.12 and that by time T the bank covers the full correct support, $\mathcal{B} = \mathcal{C}$. Let $u_c = 1/m$ on $\mathcal{C}$. Then the categorical mean flow converges to*

$$P(t) \longrightarrow 1, \quad q(t) \longrightarrow u.$$

Consequently

$$\min_{c \in \mathcal{C}} q_c(t) \longrightarrow \frac{1}{m}, \quad \mathcal{H}(q(t)) \longrightarrow \log m,$$

and conditional correct-mode collapse is impossible.

Proof. When $\mathcal{B} = \mathcal{C}$,

$$R_C = \log m - \log P + \text{KL}(u\|q). \quad (23)$$

Thus one uniform verified-likelihood loss already contains both a verified-mass term $-\log P$ and a within-support uniformity term $\text{KL}(u\|q)$. The potential (21) becomes

$$F_C = \rho \log m + \rho \text{KL}(u\|q) - \rho \log P - \Psi(P).$$

For fixed P , its unique minimizer is $q = u$. Its derivative in P is $-\rho/P - c_G(P) < 0$, so its unique minimizing distribution on the closed simplex is $P = 1, q = u$, placing all limiting probability uniformly on the correct support and zero mass on every incorrect category in the categorical limit.

For completeness, Theorem E.12 keeps every correct coordinate away from zero. The decreasing potential gives $\int_T^{\infty} \|\nabla F_C\|_2^2 dt < \infty$. The finite-group c_G is smooth on $[0, 1]$; softmax derivatives and the cross-entropy gradient are bounded. Thus the Hessian of F_C and the velocity $\dot{z} = -\nabla F_C$ are bounded, making $\nabla F_C(z(t))$ uniformly continuous. Together with square integrability, this implies $\nabla F_C \rightarrow 0$: recurring gradients of fixed positive size would persist over intervals of a uniform minimum length, contradicting the finite energy integral. At any limiting distribution the correct-coordinate stationarity equations are

$$0 = \rho(p_c - 1/m) - c_G(P)p_c(1 - P),$$

so all p_c are equal. Substituting $p_c = P/m$ gives $0 = -(1 - P)(\rho + c_G(P)P)/m$, hence $P = 1$. Each incorrect coordinate satisfies $0 = p_a(\rho + c_G(P)P)$ and therefore has $p_a = 0$. The limiting distribution is unique, proving convergence from every initialization covered by the corollary. $\square$

909 **Positive replay weights and exemplar length.** Uniformity of the target is separate from protection
 910 against extinction. For fixed weights $w_b > 0$ with $\sum_{b \in \mathcal{B}} w_b = 1$, define $R_w = -\sum_b w_b \log p_b$ and
 911 use replay coefficient $\rho_w > 0$. The same potential argument gives, for all $t \geq T$,

$$p_b(t) \geq \exp(-C_{T,w}/w_b), \quad C_{T,w} = R_w(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho_w}.$$

912 If $\mathcal{B} = \mathcal{C}$, then $R_w = H(w) - \log P + \text{KL}(w||q)$, and the independent-logit limit is $P \rightarrow 1$, $q \rightarrow w$.
 913 The preceding stationarity proof applies with $1/m$ replaced by w_c : summing the correct-coordinate
 914 equations forces $P = 1$, then each equation forces $p_c = w_c$.

915 This distinction matters for the implemented mean-token exemplar loss. Even if each key were
 916 represented by exactly one categorical exemplar of fixed length L_b , its loss would be AR_w , where

$$A = \frac{1}{k} \sum_{b \in \mathcal{B}} \frac{1}{L_b}, \quad w_b = \frac{L_b^{-1}}{\sum_{j \in \mathcal{B}} L_j^{-1}}.$$

917 Its effective categorical replay coefficient would be $\rho_w = \rho_{\text{LM}} A$. The full-coverage limit would
 918 therefore weight shorter exemplars more, and would be uniform only for equal lengths. For an actual
 919 language model, $\pi_\theta(e_b | x)$ is only one contribution to the total key probability. Equal averaging
 920 of per-token exemplar scores does not imply uniform learned key probabilities; the categorical
 921 uniformity corollary is not that claim.

922 **Worked example: equal key sampling need not imply a uniform target.** For three singleton
 923 categorical exemplars of lengths 10, 20, 40, equal averaging of mean-token losses gives $A = 7/120$
 924 and $w = (4/7, 2/7, 1/7)$. All three target weights are positive, so each mode is protected, but
 925 the full-coverage categorical limit assigns the shortest exemplar four times the probability of the
 926 longest. Equal lengths would instead give $(1/3, 1/3, 1/3)$. This example concerns the loss’s induced
 927 categorical target, not an asserted distribution over actual LM keys.

928 The premise $\mathcal{B} = \mathcal{C}$ is not a description of the released Python-factors experiments. Their prompts
 929 have 16–3,600 valid execution modes (mean 229.44), while the replay bank has capacity 16. Thus
 930 the corollary is inapplicable to Python factors as a domain-wide guarantee. More generally it applies
 931 only to prompts whose entire correct support fits in the bank and has actually been discovered; neither
 932 condition follows from replay.

933 **Lemma E.14** (Fixed-bank limits and deterministic finite steps). *Retain finite independent Euclidean*
 934 *logits, a nonempty fixed bank $\mathcal{B} \subseteq \mathcal{C}$, fixed positive normalized weights w_b , and $\rho_w > 0$. Let*
 935 *$F = \rho_w R_w - \Psi(P)$, with $\Psi \in C^2([0, 1])$ and $\Psi' = c_G \geq 0$. Set*

$$M_1 = \sup_{[0,1]} |\Psi'|, \quad M_2 = \sup_{[0,1]} |\Psi''|, \quad L = \frac{\rho_w}{2} + \frac{M_1}{2} + \frac{M_2}{8}.$$

936 *For constant $0 < \eta < 2/L$, deterministic updates $z_{n+1} = z_n - \eta \nabla F(z_n)$ preserve, for every $n \geq 0$,*

$$p_b(z_n) \geq \exp(-C_{0,w}/w_b), \quad C_{0,w} = R_w(z_0) + \frac{\Psi(1) - \Psi(P(z_0))}{\rho_w}.$$

937 *Moreover, $p(z_n) \rightarrow \bar{w}$, where $\bar{w}_b = w_b$ on the bank and $\bar{w}_a = 0$ elsewhere. The same probability*
 938 *limit holds for the corresponding continuous mean flow. The logits need not converge: zero limiting*
 939 *probabilities require divergent logit differences relative to banked coordinates.*

940 *Proof.* For a logit direction u , let $U = u_a$ and $Y = \mathbf{1}[a \in \mathcal{C}]$ under $a \sim p$. Differentiating along that
 941 direction gives

$$u^\top \nabla^2 P u = \mathbb{E}[(Y - P)(U - \mathbb{E}U)^2], \quad u^\top \nabla^2 R_w u = \text{Var}(U).$$

942 Since $|Y - P| \leq 1$, the first quadratic form is bounded in absolute value by $\text{Var}(U)$. The bounded-
 943 range variance inequality gives

$$\text{Var}(U) \leq \frac{(\max_a u_a - \min_a u_a)^2}{4} \leq \frac{\|u\|_2^2}{2}.$$

Both Hessians therefore have operator norm at most $1/2$. Also $\|\nabla P\|_2^2 \leq 2P^2(1-P)^2 \leq 1/8$. The chain rule therefore bounds $\|\nabla^2 F\|_{\text{op}} \leq L$ globally. The descent lemma yields

$$F(z_{n+1}) \leq F(z_n) - \eta(1 - L\eta/2)\|\nabla F(z_n)\|_2^2.$$

Energy monotonicity gives the preceding replay-loss bound and hence the stated retention floor. Since $F \geq -\Psi(1)$, the squared gradients are summable and $\nabla F(z_n) \rightarrow 0$. Its coordinate expression,

$$[\nabla F]_a = \rho_w(p_a - \bar{w}_a) - c_G(P)p_a(\mathbf{1}[a \in \mathcal{C}] - P),$$

extends continuously to the closed probability simplex. Every subsequential probability limit is therefore stationary. On an incorrect coordinate, stationarity gives $(\rho_w + c_G(P)P)p_a = 0$, forcing all incorrect mass to zero and $P = 1$. The correct-coordinate equations then give $p_a = \bar{w}_a$. Compactness and this unique limit prove convergence of the whole probability sequence. For the mean flow, bounded Hessian and velocity together with energy descent give vanishing gradient as in the preceding corollary; the same stationarity argument applies. $\square$

For Dr.GRPO, $M_1 = (G-1)/G$ and $M_2 = 0$. For the implemented finite-group MaxRL estimator, $M_1 = G-1$ and $M_2 = (G-1)(G-2)/2$, so one sufficient smoothness constant is $L = \rho_w/2 + (G-1)(G+6)/16$. These bounds concern categorical logit steps, not the neural optimizer's learning rate. With an incomplete frozen bank, the categorical limit assigns zero mass even to correct modes outside the bank: fixed-bank retention is not a guarantee of discovering or preserving all correct modes. The fixed-bank limit also gives

$$\lim_{t \rightarrow \infty} \text{distinct@}K = \sum_{b \in \mathcal{B}} [1 - (1 - w_b)^K] \leq k,$$

with the same expression along deterministic iterates. Thus the bank itself sets an asymptotic support ceiling in this model. The later stochastic and changing-bank results give conditional retention bounds under additional energy assumptions; they do not extend this fixed-bank convergence limit to arbitrary neural training.

Worked example: a safe logit step and an incomplete bank. Return to the two-mode bank above with $\rho_w = .30$, $G = 4$, and Dr.GRPO. Here $L = .525$, so $\eta = 1$ satisfies $0 < \eta < 2/L \approx 3.81$. One deterministic logit-gradient step changes the running distribution approximately as

$$(.30, .15, .05, .50) \mapsto (.3861, .1643, .0632, .3864),$$

while the potential falls from about .25496 to .09689. Both banked modes gain probability. The unbanked second correct mode also gains on this first step, but the lemma's eventual limit is $(.50, 0, .50, 0)$: early improvement of an unbanked mode does not make it protected. The step size here belongs to the independent-logit example; it is not a recommendation for the neural optimizer.

E.9 From categorical modes to complete replayed responses

The categorical analysis protects an execution key, whereas the learner scores a particular response string. The bridge is event containment: one complete verified response is one way to produce its key. A bound on its sequence surprisal therefore protects the key even when many other strings realize the same behavior. Length normalization changes the numerical loss scale, so its factor must remain explicit in that bound.

Lemma E.15 (Retention from a joint exemplar potential). *Fix finitely many verified complete-response exemplars (x_j, e_j) , each realizing key b_j , with lengths $L_j > 0$ and fixed weights $\alpha_j > 0$. Let*

$$F(\theta) = \sum_j \alpha_j \frac{-\log \pi_\theta(e_j | x_j)}{L_j} - J(\theta), \quad J(\theta) \leq J_{\max} < \infty.$$

Suppose $F(\theta(T))$ is finite and $F(\theta(t)) \leq F(\theta(T))$ for every $t \geq T$. With $D = F(\theta(T)) + J_{\max}$, every protected exemplar satisfies

$$p_\theta(b_j | x_j) \geq \pi_\theta(e_j | x_j) \geq \exp\left(-\frac{L_j D}{\alpha_j}\right) > 0 \quad (t \geq T).$$

Proof. The weighted exemplar loss equals $F(\theta(t)) + J(\theta(t)) \leq D$. Each summand is nonnegative, so $-\log \pi_\theta(e_j | x_j) \leq L_j D / \alpha_j$. The complete-response event is a subset of the event producing key b_j , which proves the stated lower bound on every protected key's probability. $\square$

Example: a token average is not a response probability. Suppose a complete response has four scored tokens, including termination, and each token has conditional probability $1/2$ along that response. Its mean log likelihood is $-\log 2$, but its complete-response probability is $(1/2)^4 = 1/16$; exponentiating the mean gives $1/2$, the geometric mean of token probabilities. For a single protected exemplar with $L = 4$, $\alpha = 1$, and joint energy ceiling $D = \log 2$, the lemma gives exactly the $1/16$ floor. If a different complete response with the same key has probability $1/32$, the two disjoint response events together give key probability at least $1/16 + 1/32 = 3/32$. The exemplar bound can thus be conservative even when its sequence probability is known exactly.

Exact gradient flow of a smooth joint potential is one sufficient condition for this monotonicity; the lemma requires only the stated energy inequality. Shared parameters and multiple prompts therefore need not invalidate retention. For example, the exact categorical mean model admits $J(\theta) = \sum_x \nu_x \Psi_x(P_x(\theta))$ with fixed nonnegative prompt weights and $J_{\max} = \sum_x \nu_x \Psi_x(1)$. The bound does not require uniform key weights or imply convergence to a uniform policy. It does not establish joint energy descent for the implemented stochastic, clipped, alternating optimizer. The likelihood must describe the complete response under the sampling law being bounded, including its termination event or a fixed deterministic horizon. A prefix likelihood alone, or a score under a different temperature or token-truncation rule, is insufficient.

Retention and finite evaluation visibility. At a fixed checkpoint, suppose k distinct banked keys for one prompt have probability floors $p_b \geq \delta_b > 0$. For K independent draws from that same policy, the banked distinct count $D_{\mathcal{B},K}$ satisfies

$$\mathbb{E}[D_{\mathcal{B},K}] \geq \sum_{b \in \mathcal{B}} [1 - (1 - \delta_b)^K],$$

$$\Pr(\text{some banked key is unseen}) \leq \sum_{b \in \mathcal{B}} (1 - \delta_b)^K \leq k e^{-K \delta_{\min}},$$

where $\delta_{\min} = \min_b \delta_b$. The first statement follows from indicator expectations and the second from the union bound. Thus $K \geq \log(k/\varepsilon)/\delta_{\min}$ suffices to observe all banked keys with probability at least $1 - \varepsilon$, for $0 < \varepsilon < 1$. The energy floors can make this budget impractically large; asymptotic retention does not imply visibility in an eight-draw evaluation.

Example: protection can still be hard to observe. A key with probability exactly $1/16$ appears in eight independent draws with probability $1 - (15/16)^8 \approx 0.403$. If four keys each have a $1/16$ floor, the sufficient budget above is $\lceil 16 \log(4/0.05) \rceil = 71$ draws for at least 95% probability of seeing all four. These hypothetical probabilities distinguish retaining a mode from reliably observing it in a short evaluation.

E.10 Retention under discrete stochastic optimizer updates

For discrete noisy updates, the useful quantity is the total upward drift allowed in the joint loss. Controlling that budget permits individual steps to increase the loss while keeping a high-probability ceiling on the loss over a stated update horizon.

Lemma E.15 converts a deterministic energy ceiling into complete-exemplar protection. We next replace pathwise descent with a controlled conditional drift, allowing shared parameters, noise, bias, and predictable preconditioning. The descent estimate follows the variable-metric analysis of Wang et al. [2017, Section 2]; the trajectory bound uses Ville’s nonnegative-supermartingale inequality, as stated by Howard et al. [2020, Lemma 1]. These are standard ingredients; the result below applies them to the verified-response potential.

Theorem E.16 (Conditional stochastic exemplar retention). *Fix finitely many complete verified responses (x_j, e_j) realizing keys b_j , with lengths $L_j > 0$ and weights $\alpha_j > 0$. Define*

$$R(\theta) = \sum_j \frac{\alpha_j}{L_j} [-\log \pi_\theta(e_j | x_j)], \quad F(\theta) = R(\theta) - J(\theta), \quad J(\theta) \leq J_{\max} < \infty.$$

Assume F is differentiable with globally L_F -Lipschitz gradient and θ_0 is deterministic with finite $F(\theta_0)$. Relative to the history filtration $(\mathcal{F}_t)$, consider

$$\theta_{t+1} = \theta_t - \eta_t H_t(g_t + b_t + \xi_t), \quad g_t = \nabla F(\theta_t).$$

1029 The variables $\theta_t, \eta_t, H_t, b_t$ are $\mathcal{F}_t$ -measurable, $0 \preceq H_t \preceq MI$, $\eta_t > 0$, and $\eta_t L_F M \leq 1$. The noise
 1030 satisfies $\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0$ and has finite conditional second moment. Suppose deterministic $d_t \geq 0$
 1031 satisfy

$$\frac{\eta_t}{2} \|b_t\|_{H_t}^2 + \frac{L_F \eta_t^2}{2} \mathbb{E}[\|H_t \xi_t\|_2^2 \mid \mathcal{F}_t] \leq d_t, \quad \|u\|_H^2 = u^\top H u. \quad (24)$$

1032 For $N \in \mathbb{N} \cup \{\infty\}$ with $B_N = \sum_{t < N} d_t < \infty$, write $D_0 = F(\theta_0) + J_{\max} \geq 0$. For every $0 < \delta < 1$,
 1033 with probability at least $1 - \delta$, the following holds simultaneously for all integer $t \leq N$ and all
 1034 protected exemplars:

$$p_{\theta_t}(b_j \mid x_j) \geq \pi_{\theta_t}(e_j \mid x_j) \geq \exp \left[-\frac{L_j(D_0 + B_N)}{\alpha_j \delta} \right]. \quad (25)$$

1035 For $N = \infty$, each protected probability also has an almost surely positive, potentially random,
 1036 infimum over all iterations. The likelihood must include the complete response and its termination
 1037 event, or a fixed deterministic horizon, under the sampling law whose key probability is bounded.

1038 *Proof.* Smoothness, conditional centering of ξ_t , and $H_t^2 \preceq M H_t$ imply

$$\begin{aligned} \mathbb{E}[F(\theta_{t+1}) \mid \mathcal{F}_t] &\leq F(\theta_t) - \eta_t g_t^\top H_t (g_t + b_t) + \frac{\eta_t}{2} \|g_t + b_t\|_{H_t}^2 \\ &\quad + \frac{L_F \eta_t^2}{2} \mathbb{E}[\|H_t \xi_t\|_2^2 \mid \mathcal{F}_t] \\ &\leq F(\theta_t) - \frac{\eta_t}{2} \|g_t\|_{H_t}^2 + d_t. \end{aligned}$$

1039 The bias cross term cancels when the square is expanded. Since $F + J_{\max} \geq R \geq 0$, the process

$$X_t = F(\theta_t) + J_{\max} + \sum_{s=t}^{N-1} d_s$$

1040 is a nonnegative supermartingale, using the convergent infinite tail when $N = \infty$. Its initial value
 1041 is $D_0 + B_N$. For $D_0 + B_N > 0$, Ville's inequality gives $\Pr(\sup_{t \leq N} X_t \geq (D_0 + B_N)/\delta) \leq \delta$. If
 1042 $D_0 + B_N = 0$, nonnegativity gives $X_t = 0$ almost surely instead. Every nonnegative summand of
 1043 $R(\theta_t) \leq X_t$ obeys the same ceiling. Exponentiation therefore yields (25) without a union factor
 1044 over exemplars. A complete-exemplar event is contained in its verified-key event. For $N = \infty$, the
 1045 same maximal bound gives $\Pr(\sup_t X_t \geq u) \leq (D_0 + B_\infty)/u$ for $u > 0$. Letting $u \rightarrow \infty$ proves
 1046 $\sup_t X_t < \infty$ almost surely and hence random positive all-time floors. $\square$

1047 **Finite and infinite horizons.** With zero bias and uniformly bounded noise variance, the cost in (24)
 1048 is $O(\eta_t^2)$, so square-summable steps suffice for an infinite-horizon budget. Bias instead contributes
 1049 $O(\eta_t \|b_t\|^2)$. Constant steps with persistent variance yield the finite-horizon result, with a budget that
 1050 can grow with N ; this argument does not give infinite-horizon protection in that regime. The classical
 1051 almost-supermartingale framework of Robbins and Siegmund [1971] supplies related convergence
 1052 tools, but the proof above needs only the displayed drift and maximal inequality. No positive lower
 1053 eigenvalue of H_t is needed for retention alone, and the theorem does not assert stationarity, parameter
 1054 convergence, or a useful evaluation-scale probability floor.

1055 **Example: a noise budget for one protected response.** Consider a hypothetical fixed one-token
 1056 horizon with two possible complete responses and $\pi_\theta(e) = (1 + e^{-\theta})^{-1}$, $L = \alpha = 1$, and $F =$
 1057 $R = -\log \pi_\theta(e)$, with $J = J_{\max} = 0$. Start at $\theta_0 = 0$, so $D_0 = \log 2$. This scalar objective has
 1058 $L_F = 1/4$ and $|g_t| \leq 1$. Take $H_t = 1$, zero bias, and independent noise $\xi_t \in \{-0.1, 0.1\}$ with equal
 1059 probabilities. At constant step $\eta_t = 0.1$, the theorem permits $d_t = 1/80000$, giving $B_{100} = 0.00125$
 1060 but $B_\infty = \infty$. With $\eta_t = 0.1/(t+1)$ instead, $B_\infty = \pi^2/480000 \approx 2.06 \times 10^{-5}$, which supplies
 1061 the infinite-horizon condition. For the constant-step, 100-update case, the 95% probability floor
 1062 from (25) is only $\exp[-(\log 2 + 0.00125)/0.05] \approx 9.30 \times 10^{-7}$. The next result uses bounded noise
 1063 to improve this confidence dependence.

1064 **Corollary E.17** (Sharper confidence dependence under bounded noise). *Under the assumptions*
 1065 *of Theorem E.16, suppose $b_t = 0$, $N < \infty$, and η_t is deterministic. Let deterministic bounds*
 1066 *$G_t, s_t < \infty$ satisfy $\|g_t\|_2 \leq G_t$ and $\|\xi_t\|_2 \leq s_t$ almost surely. Define*

$$c_t = \eta_t M G_t s_t, \quad V_N = \sum_{t < N} c_t^2, \quad Q_N = \frac{L_F M^2}{2} \sum_{t < N} \eta_t^2 s_t^2.$$

1067 For every $0 < \delta < 1$, with probability at least $1 - \delta$, simultaneously for all $t \leq N$ and j ,

$$p_{\theta_t}(b_j | x_j) \geq \pi_{\theta_t}(e_j | x_j) \geq \exp \left[-\frac{L_j}{\alpha_j} \left(D_0 + Q_N + \sqrt{2V_N \log(1/\delta)} \right) \right]. \quad (26)$$

1068 *Proof.* The pathwise smoothness estimate gives

$$F(\theta_{t+1}) - F(\theta_t) \leq -\frac{\eta_t}{2} \|g_t\|_{H_t}^2 + Z_{t+1} + \frac{L_F \eta_t^2 M^2 s_t^2}{2},$$

1069 where

$$Z_{t+1} = -\eta_t g_t^\top H_t (I - L_F \eta_t H_t) \xi_t.$$

1070 Predictability makes Z_{t+1} a martingale difference. The spectral bounds imply $0 \preceq I - L_F \eta_t H_t \preceq I$,
1071 so $|Z_{t+1}| \leq c_t$. Conditional Hoeffding's lemma therefore makes

$$\exp \left(\lambda \sum_{s < t} Z_{s+1} - \frac{\lambda^2}{2} \sum_{s < t} c_s^2 \right), \quad \lambda > 0,$$

1072 a nonnegative supermartingale starting at one. This standard exponential construction is reviewed by
1073 Howard et al. [2020]. Ville's inequality and optimization over λ yield

$$\Pr \left(\max_{t \leq N} \sum_{s < t} Z_{s+1} > \sqrt{2V_N \log(1/\delta)} \right) \leq \delta$$

1074 when $V_N > 0$; when $V_N = 0$, the sum is identically zero. Sum the drift inequalities, drop the
1075 nonpositive gradient terms, and apply the summand bound from Theorem E.16. $\square$

1076 In the same 100-update example, $G_t = M = 1$ and $s_t = 0.1$ give $V_{100} = 0.01$ and $Q_{100} = 0.00125$.
1077 The sharper 95% floor is therefore

$$\exp \left[-\log 2 - 0.00125 - \sqrt{0.02 \log 20} \right] \approx 0.391.$$

1078 Both bounds apply simultaneously to the toy trajectory's first 100 updates; the improvement comes
1079 from stronger noise assumptions. These toy constants are assigned for the calculation; they do not
1080 estimate noise or curvature in the language-model runs.

1081 **Relation to stochastic policy-gradient guarantees.** Zhang et al. [2021] analyze stochastic tabular
1082 REINFORCE with a state-action log barrier, phased step and regularization schedules, and policy
1083 mixing. Their regret guarantees provide a related stochastic policy-gradient precedent; they do not
1084 provide a fixed per-response floor for our replay bank. For the implemented optimizer, a current-
1085 gradient Adam second moment is not predictable relative to that gradient's noise. Momentum,
1086 clipping, prompt scheduling, and changing banks likewise require separate bias or energy control.
1087 The counterexamples of Reddi et al. [2018] explain why an unrestricted Adam guarantee cannot be
1088 assumed; they do not establish failure of these runs. The hypotheses above have not been established
1089 for the experiments' AdamW/PPO updates. In particular, positive replay weight and a small nominal
1090 learning rate do not verify the smoothness, predictability, or accumulated-error conditions.

1091 E.11 Discovery, admission, and retention through bank changes

1092 Discovery and retention require two separate budgets: enough opportunities to admit a missing key,
1093 and enough persistent loss weight to protect it after admission. Bank growth connects the two because
1094 admitting a key can also change the objective used to protect earlier keys.

1095 Retention begins only after a verified key enters active memory. Archive exploration separates
1096 discovery from reliable return to a discovered state [Ecoffet et al., 2021]; replay with behavioral
1097 cloning supplies a related empirical mechanism for reducing forgetting [Rolnick et al., 2019]. Neither
1098 result is a theorem that every execution key enters a finite bank or maintains a positive probability
1099 thereafter. We state the additional admission and energy conditions explicitly.

1100 **Actual insertion is the relevant event.** Fix a prompt and count opportunities according to a speci-
 1101 fied schedule. An opportunity can be a prompt-local query or a global update; the resulting bounds
 1102 concern that opportunity count. The admission probability includes verification, capacity, scheduling,
 1103 and candidate selection. For example, G conditionally independent samples with unconditional
 1104 accepted-key probability p_b contain key b with probability $1 - (1 - p_b)^G$. This is its admission
 1105 probability only if the key remains eligible and insertion is guaranteed whenever it appears. The
 1106 following argument permits adaptive histories; its probability foundation is conditional Borel–Cantelli
 1107 [Durrett, 2019, Theorem 4.3.4].

1108 **Lemma E.18** (Conditional admission hazards). *Fix a key b not initially banked. Let $(\mathcal{F}_n)_{n \geq 0}$ contain*
 1109 *the complete history after each opportunity, and let τ_b be the first opportunity at which a verified*
 1110 *representative of b is inserted into active replay memory, with $\tau_b = \infty$ if this never happens. Define*

$$h_{b,n} = \Pr(\tau_b = n \mid \mathcal{F}_{n-1}), \quad H_{b,N} = \sum_{n=1}^N h_{b,n}.$$

1111 *The stopped hazard $h_{b,n}$ is zero after an earlier admission. For $a \geq 0$,*

$$\Pr(\tau_b > N, H_{b,N} \geq a) \leq e^{-a}, \quad \Pr(\tau_b = \infty, H_{b,\infty} = \infty) = 0. \quad (27)$$

1112 *If deterministic $\underline{h}_{b,n} \in [0, 1]$ satisfy $h_{b,n} \geq \underline{h}_{b,n}$ on $\{\tau_b \geq n\}$, then*

$$\Pr(\tau_b > N) \leq \prod_{n=1}^N (1 - \underline{h}_{b,n}) \leq \exp\left(-\sum_{n=1}^N \underline{h}_{b,n}\right). \quad (28)$$

1113 *For finitely many targets, summing over initially missing keys bounds non-admission by N .*

1114 *Proof.* Set $Z_N = \mathbf{1}\{\tau_b > N\} \exp(H_{b,N})$ and $Z_0 = 1$. Earlier admission is known at time $n - 1$,
 1115 and both $h_{b,n}$ and Z_{n-1} vanish there. On survival through $n - 1$,

$$\mathbb{E}[Z_n \mid \mathcal{F}_{n-1}] = Z_{n-1} e^{h_{b,n}} (1 - h_{b,n}) \leq Z_{n-1}.$$

1116 Thus Z_n is a nonnegative supermartingale. Markov’s inequality proves the finite-hazard bound. Since
 1117 $\mathbb{E}Z_N \leq 1$, Fatou’s lemma excludes a positive-probability event on which survival continues forever
 1118 and $H_{b,N} \rightarrow \infty$, which would force $Z_N \rightarrow \infty$. With deterministic hazard floors, conditioning gives

$$\Pr(\tau_b > n) \leq (1 - \underline{h}_{b,n}) \Pr(\tau_b > n - 1).$$

1119 Iteration and a union bound prove the claims without independence across opportunities. $\square$

1120 **Example: turn an admission floor into a budget.** Suppose a prompt has four possible cor-
 1121 rect keys and at least four bank slots. At each opportunity, every missing key is generated with
 1122 probability at least 0.04 and inserted, conditional on appearing, with probability at least 1/2. Its
 1123 actual admission hazard is therefore at least 0.02. The lemma gives $4e^{-0.02N}$ as a failure bound, so
 1124 $N = \lceil \log(4/0.05)/0.02 \rceil = 220$ opportunities suffice for at least 95% probability of full admission.
 1125 With only two non-evicting slots, full admission of those four keys is impossible; after the bank fills,
 1126 the missing keys’ admission hazards become zero despite continued generation.

1127 Positive probability at every opportunity is insufficient. Independent opportunities with admission
 1128 probabilities $(n + 1)^{-2} > 0$ give

$$\Pr(\tau = \infty) = \prod_{n=1}^{\infty} \left(1 - \frac{1}{(n + 1)^2}\right) = \frac{1}{2}.$$

1129 A non-evicting capacity- K_{cap} bank also gives every unbanked key zero admission hazard once full.
 1130 For a fixed sampling law with no such gates, discovery reduces to coupon collection, with incorrect
 1131 responses acting as null coupons. Anceaume et al. [2015, Theorems 2–3] show that, at fixed total
 1132 non-null mass, uniform independent sampling minimizes collection time in stochastic order. This
 1133 comparison concerns a fixed independent sampling law; the adaptive admission bounds above supply
 1134 the appropriate statement when the training history changes sampling or eligibility.

1135 **Charge changes in the replay objective.** A key’s admission must preserve a representative with
 1136 positive loss weight. The energy can change when bank membership or weights change, so descent
 1137 between admissions alone is insufficient. We account for those changes explicitly, following the
 1138 general switched-energy viewpoint of Branicky [1998]. The following elementary bound is proved
 1139 directly and does not invoke an equilibrium-stability theorem for neural training.

1140 **Lemma E.19** (Retention with a finite switching-energy budget). *Let $t_0 < t_1 < \dots$ be switching*
 1141 *times, with finitely many switches on every bounded time interval. On interval r , define*

$$F_r(\theta) = \sum_{j: d_{rj} > 0} d_{rj} \ell_j(\theta) - J(\theta),$$

$$\ell_j(\theta) = -\log \pi_\theta(e_j | x_j), \quad J(\theta) \leq J_{\max} < \infty.$$

1142 *Each interval uses a finite collection of complete verified exemplars with nonnegative coefficients;*
 1143 *zero coefficients denote absent exemplars. Assume finite initial energy, continuity of parameters at*
 1144 *switches, nonincrease of F_r within interval r , and the finite positive-jump budget*

$$A := \sum_{r \geq 1} [F_r(\theta(t_r)) - F_{r-1}(\theta(t_r))]_+ < \infty. \quad (29)$$

1145 *If exemplar j is protected from interval r_j onward and $d_{rj} \geq \underline{d}_j > 0$ for all $r \geq r_j$, let $D =$*
 1146 *$F_0(\theta(t_0)) + A + J_{\max}$. Then*

$$p_{\theta(t)}(b_j | x_j) \geq \pi_{\theta(t)}(e_j | x_j) \geq \exp(-D/\underline{d}_j) > 0 \quad (t \geq t_{r_j}). \quad (30)$$

1147 *Each likelihood uses the complete verified-response probability under the sampling law being bounded.*
 1148 *For mean-token loss, $d_{rj} = \alpha_{rj}/L_j$, with fixed length $L_j > 0$.*

1149 *Proof.* Induction over interval descents and switch jumps gives $F_r(\theta(t)) \leq F_0(\theta(t_0)) + A$. Con-
 1150 sequently $\sum_j d_{rj} \ell_j(\theta(t)) \leq D$. Each summand is nonnegative, so a protected exemplar satisfies
 1151 $\ell_j(\theta(t)) \leq D/\underline{d}_j$. The key event contains its exemplar event, so exponentiation proves the bound. $\square$

1152 If admission preserves old coefficients, the jump equals the weighted surprisal of the newly added
 1153 exemplars at their admission probabilities. Uniform categorical replay with fixed total dose ρ instead
 1154 dilutes old weights. For a size- k bank, write $R_k = -k^{-1} \sum_{b \in \mathcal{B}_k} \log p_b$; adding one new key gives

$$\Delta F = \frac{\rho}{k+1} [-\log p_{\text{new}} - R_k].$$

1155 Old coefficients decrease from ρ/k to $\rho/(k+1)$ but stay at least ρ/K_{cap} under capacity K_{cap} .
 1156 Finitely many admissions with positive admission-time probabilities and no eviction therefore have
 1157 finite total switching cost in the exact descent model. Changing a representative must also be charged
 1158 to the objective jump. A key-level version of Lemma E.19 follows by requiring at least one complete
 1159 representative of that key with coefficient bounded below on every subsequent interval; the same
 1160 nonnegative-summand proof applies even if the representative changes.

1161 **Example: adding a rare key has a calculable energy cost.** Take $\rho = 0.3$ and a two-key categorical
 1162 bank whose keys each have probability $1/4$, so $R_2 = \log 4$. Add a third key with probability $1/16$,
 1163 without changing the policy at that instant. Then

$$\Delta F = \frac{0.3}{3} (\log 16 - \log 4) = 0.1 \log 4 \approx 0.1386.$$

1164 The earlier keys’ coefficients decrease from 0.15 to 0.10. A capacity-three bank still supplies a 0.10
 1165 coefficient floor, but the positive jump 0.1386 must enter the energy budget. This is why retention
 1166 through admission needs both the coefficient floor and the switch cost.

1167 **Corollary E.20** (Conditional discovery followed by retention). *Suppose one prompt’s entire correct*
 1168 *support $\mathcal{C}$ is finite, of size m no larger than bank capacity. Admissions do not evict keys, and every*
 1169 *missing key b has conditional admission hazard at least ϵr_b at every opportunity, for constants $\epsilon > 0$,*
 1170 *$r_b > 0$, and $\epsilon r_b \leq 1$. Assume the switching-energy conditions of Lemma E.19, including a finite*
 1171 *positive-jump budget and persistent positive coefficients for admitted complete exemplars. Then full*

1172 correct coverage is reached in finitely many opportunities almost surely. Thereafter each correct key
 1173 has a uniform-in-time positive probability floor. At horizon N ,

$$\Pr(\text{full coverage by } N) \geq 1 - \sum_{b \in \mathcal{C}} e^{-N\epsilon r_b}. \quad (31)$$

1174 If every $r_b \geq \gamma > 0$, this is at least $1 - me^{-N\epsilon\gamma}$.

1175 *Proof.* Apply Lemma E.18 to the initially missing keys and take a union bound. The failure bound
 1176 tends to zero. Finite support and non-eviction imply a finite almost-sure opportunity index of
 1177 simultaneous full coverage. Lemma E.19 protects each admitted complete exemplar and hence its
 1178 key; after full coverage this applies to all correct keys. $\square$

1179 A fixed full-support proposal mixture supplies the assumed hazard floor if it is queried at every
 1180 counted opportunity and insertion is guaranteed whenever a missing key appears. Capacity must
 1181 accommodate the entire admissible correct support, including keys competing for the same slots.
 1182 Finite-temperature sampling, bounded search attempts, or recurrent replay scheduling alone do not
 1183 establish this admission floor or the energy budget. The probability bound (31) controls admission by
 1184 a finite opportunity count; the subsequent retention floors can depend on the realized training history.
 1185 The corollary establishes neither uniform key probabilities nor convergence of the neural optimizer.

1186 E.12 Sharp per-mode certificates and finite-sample interpretation

1187 The energy bounds above can be sharpened by using probability normalization. We give self-contained
 1188 applications of relative-entropy data processing and Pinsker's inequality [van Erven and Harremoës,
 1189 2014, Theorems 9 and 31]; binary-KL inversion is also a standard statistical-learning tool [Garivier
 1190 and Cappé, 2011]. These certificates concern a specified probability distribution. Applying one at a
 1191 checkpoint requires a valid loss bound there; applying it throughout training additionally requires a
 1192 uniform energy bound. Neither application establishes descent for the implemented neural optimizer.
 1193 All logarithms are natural.

1194 E.12.1 A sharp certificate from normalized replay cross entropy

1195 Write $H(w) = -\sum_b w_b \log w_b$, $h_2(a) = H((a, 1-a))$, and

$$\text{kl}(a, r) = a \log \frac{a}{r} + (1-a) \log \frac{1-a}{1-r},$$

1196 with the usual endpoint conventions. For $0 < a < 1$ and $D \geq 0$, let $\ell_-(a, D)$ be the unique $r \in (0, a]$
 1197 satisfying $\text{kl}(a, r) = D$; set $\ell_-(1, D) = e^{-D}$.

1198 **Theorem E.21** (Sharp normalized replay certificate). *Let $\mathcal{B}$ be a finite nonempty bank, let $w_b > 0$
 1199 with $\sum_{b \in \mathcal{B}} w_b = 1$, and let p be a probability distribution on an alphabet containing $\mathcal{B}$. Suppose*

$$R_w(p) = -\sum_{b \in \mathcal{B}} w_b \log p_b \leq C < \infty, \quad D = C - H(w).$$

1200 Then $D \geq 0$ and, for every $b \in \mathcal{B}$,

$$p_b \geq \ell_-(w_b, D) > 0. \quad (32)$$

1201 This coordinate bound is sharp on the closed probability simplex whenever an outcome other than b
 1202 is available. Moreover,

$$p(\mathcal{B}) \geq e^{-D}, \quad p_b \geq \max\{0, w_b - \sqrt{D/2}\}.$$

1203 For $w_b < 1$, the positive bound $p_b \geq \exp[-(D + h_2(w_b))/w_b]$ is also valid.

1204 *Proof.* Extend w by zero off the bank, obtaining $\bar{w}$. Then

$$0 \leq \text{KL}(\bar{w} \| p) = R_w(p) - H(w) \leq D.$$

1205 For $a = w_b$ and $r = p_b$, the log-sum inequality on the complementary outcomes gives

$$\sum_{j \neq b} \bar{w}_j \log \frac{\bar{w}_j}{p_j} \geq (1-a) \log \frac{1-a}{1-r},$$

so $\text{kl}(a, r) \leq D$. For $0 < a < 1$, its derivative in r is $(r - a)/(r(1 - r)) < 0$ on $(0, a)$; it diverges at zero and vanishes at $r = a$. Thus $r < \ell_-(a, D)$ is impossible. Equality is attained by choosing $r = \ell_-(a, D)$, $p_j = (1 - r)w_j/(1 - a)$ on the remaining banked outcomes, and zero mass off the bank. For $a = 1$, the inequality is $-\log p_b \leq D$, with equality obtained by placing the remaining mass on another outcome. If the entire alphabet has one outcome, necessarily $p_b = 1$. For $D > 0$, equality can be approached with positive off-bank probabilities by increasing r slightly and sending off-bank mass to zero. At $D = 0$, feasibility instead forces $p = \bar{w}$ exactly.

Let $s = p(\mathcal{B})$ and $\tilde{p}_b = p_b/s$. The decomposition

$$\text{KL}(\bar{w} \| p) = -\log s + \text{KL}(w \| \tilde{p}) \geq -\log s$$

proves the total-mass bound. For fixed $r \in (0, 1)$, $\partial_a^2 \text{kl}(a, r) = 1/[a(1 - a)] \geq 4$, whereas its value and first derivative vanish at $a = r$. Hence $\text{kl}(a, r) \geq 2(a - r)^2$, including endpoints by continuity, which proves the Pinsker bound. Finally,

$$\text{kl}(a, r) = -h_2(a) - a \log r - (1 - a) \log(1 - r) \geq -h_2(a) - a \log r$$

Combining this inequality with $\text{kl}(a, r) \leq D$ yields the positive exponential coordinate bound. $\square$

For the categorical potential argument behind Eq. (22) and its positive-weight extension, one may substitute

$$C = R_w(p(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho_w}.$$

Normalization improves the termwise certificate $p_b \geq e^{-C/w_b}$.

Worked example: normalization turns a tiny floor into a useful one. Suppose hypothetically that a uniform bank of 16 has the certified loss bound $C = \log 16 + 0.01$: only .01 nat above the target's minimum cross entropy. The earlier termwise argument gives approximately 4.62×10^{-20} per mode. Accounting for the other fifteen probabilities instead gives $p_b \geq 0.0339712$, about 3.4% for every mode, against a uniform target of 6.25%. It also certifies total bank mass at least $e^{-.01} \approx 99.00\%$. The improvement uses normalization: one probability cannot become very small while the others absorb arbitrary mass and keep this loss almost minimal. These are mathematical inputs, not measured replay losses. A large excess cross entropy can still make even the sharp bound too small to be useful.

E.12.2 What a scalar entropy value can certify

Theorem E.22 (Sharp entropy threshold for all-mode retention). *Let q be a distribution on a known alphabet of $m \geq 2$ possible modes, allowing zero coordinates. A scalar bound $H(q) \geq h_0$, with $0 \leq h_0 \leq \log m$, implies a strictly positive uniform coordinate floor if and only if $h_0 > \log(m - 1)$. In that case the sharp floor is $q_b \geq r_m(h_0)$ for every b , where $r_m(h_0) \in (0, 1/m]$ uniquely solves*

$$h_2(r) + (1 - r) \log(m - 1) = h_0,$$

where m counts all possible modes, including those absent from the evaluation sample.

Proof. Fix $r = q_b < 1$. Splitting off that coordinate gives

$$H(q) = h_2(r) + (1 - r)H\left(\frac{q_{-b}}{1 - r}\right) \leq h_2(r) + (1 - r) \log(m - 1) =: f_m(r).$$

Uniform remaining coordinates attain equality. On $0 < r < 1/m$, $f'_m(r) = \log[(1 - r)/(r(m - 1))] > 0$, with $f_m(0) = \log(m - 1)$ and $f_m(1/m) = \log m$. Thus a coordinate below the stated root contradicts $H(q) \geq h_0$; a coordinate equal to one already exceeds the root. The distribution with one coordinate at the root and the others uniform proves sharpness. Conversely, if $h_0 \leq \log(m - 1)$, one zero coordinate with the rest uniform satisfies the entropy bound. Even restricting to strictly positive coordinates permits arbitrarily small values by approximation. $\square$

1242 **Worked example: a reported entropy of 97.7% can hide a missing mode.** On a known sixteen-
 1243 mode alphabet, let one mode have probability zero and the other fifteen share mass equally. Then

$$H(q) = \log 15, \quad \frac{H(q)}{\log 16} \approx 0.97672.$$

1244 Thus entropy rounds to 97.7% of its maximum although one mode is entirely absent. Its forward-KL
 1245 gap to uniform is only $\log(16/15) \approx 0.06454$ nat. The theorem requires a strict entropy bound above
 1246 $\log 15$ to exclude this example; a nearly maximal scalar entropy by itself does not certify all sixteen
 1247 coordinates.

1248 For uniform u , $H(q) = \log m - \text{KL}(q||u)$, and data processing gives $\text{kl}(q_b, 1/m) \leq \text{KL}(q||u)$.
 1249 A forward-KL budget therefore excludes every missing coordinate precisely when it is below
 1250 $\log[m/(m-1)]$. At the threshold, one missing coordinate is possible, whereas $\text{KL}(u||q) = \infty$. This
 1251 compares level sets; it does not assert that an exact-entropy gradient flow loses modes. Conditional
 1252 entropy alone also leaves total correct mass unconstrained, and entropy estimated from a finite sample
 1253 is not the exact $H(q)$ of a known alphabet.

1254 E.12.3 From complete-response likelihood to an execution key

1255 **Corollary E.23** (Response-to-key certificate). *Fix one prompt x , one checkpoint, and distinct*
 1256 *complete responses e_j with verified keys b_j . Give them weights $w_j > 0$ summing to one, and put*
 1257 *$W_b = \sum_{j:b_j=b} w_j$. Under a fixed decoding law, suppose*

$$-\sum_j w_j \log \pi(e_j | x) \leq C < \infty.$$

1258 *For every represented key b ,*

$$\text{kl}(W_b, p(b | x)) \leq C - H(w), \quad p(b | x) \geq \ell_-(W_b, C - H(w)) > 0.$$

1259 *Individually, $p(b_j | x) \geq \pi(e_j | x)$.*

1260 *Proof.* The distinct complete responses are disjoint events. Extend their target weights by zero to
 1261 the full response alphabet; include invalid outcomes and, if necessary, nontermination as additional
 1262 outcomes so that the policy law is normalized. The resulting response-level relative entropy is
 1263 $-\sum_j w_j \log \pi(e_j | x) - H(w) \leq C - H(w)$. Apply the log-sum inequality within each execution-
 1264 key event and then within its binary partition into key b and all other outcomes. The target assigns mass
 1265 W_b to the first event and the policy assigns $p(b | x)$, so the resulting divergence is $\text{kl}(W_b, p(b | x))$.
 1266 The inverse argument of Theorem E.21 proves the floor. Finally, each complete-response event is
 1267 contained in its verified-key event, proving the individual bound used in Lemma E.15. $\square$

1268 **Worked example: two responses can protect the same key.** At a hypothetical checkpoint, two
 1269 distinct complete responses for key A each have probability .20, a response for key B has probability
 1270 .40, and the remaining .20 is invalid. Give these three responses target weights $w = (1/4, 1/4, 1/2)$.
 1271 Their cross entropy satisfies $C - H(w) = \log(1.25) \approx .22314$. For key A , combine the first two target
 1272 weights to obtain $W_A = 1/2$; the certificate then gives $p(A) \geq .20$, since $\text{kl}(.50, .20) = \log(1.25)$.
 1273 Certifying each response separately and adding the two floors would give only about .1009. The
 1274 actual key probability in this constructed example is $.20 + .20 = .40$. Grouping complete responses
 1275 by their shared execution key therefore strengthens the certificate without treating either response as
 1276 the whole key.

1277 The probability being scored must include EOS or another specified termination event, or a fixed
 1278 deterministic horizon. It must use the same prompt formatting, temperature, vocabulary mask,
 1279 top- p /top- k rule, response-truncation rule, and verifier as the probability being bounded. A prefix
 1280 likelihood is insufficient because its continuations can produce other keys or invalid outputs. All
 1281 likelihoods entering one cross entropy must use the same checkpoint. For a complete sequence of
 1282 length L_j and mean-token log score ℓ_j , its probability is $\exp(L_j \ell_j)$, not $\exp(\bar{\ell}_j)$.

1283 Length-normalized loss requires normalization before the theorem applies: with $a_j = \alpha_j/L_j$,
 1284 $A = \sum_j a_j$, and $w_j = a_j/A$,

$$\sum_j \alpha_j \frac{-\log \pi(e_j | x)}{L_j} = AR_w.$$

Thus an upper bound D_0 on this loss supplies $C = D_0/A$; uniform weights on mean-token scores generally give inverse-length target weights on complete responses. Different prompts require separate conditional calculations or an explicitly normalized joint prompt mixture. A change in one exemplar’s likelihood need not equal the change in its whole key’s probability. Approximate log scores need controlled numerical error to supply rigorous numerical certificates.

E.12.4 Finite observations and repeated checkpoint measurement

At a frozen checkpoint, for a key specified before sampling, N independent draws under the same decoding law give $X \sim \text{Binomial}(N, p_b)$. Exact binomial-tail inversion gives one-sided confidence bounds [Clopper and Pearson, 1934]. If $X = 0$, the one-sided $(1 - \alpha)$ upper bound and lower bound are respectively

$$U_0 = 1 - \alpha^{1/N}, \quad 0.$$

Indeed, $\Pr_{p_b}(X = 0) = (1 - p_b)^N < \alpha$ whenever $p_b > U_0$; for nonzero counts, the full confidence procedure uses the corresponding binomial-tail inversion.

Worked example: unseen is not the same as absent. For one key selected before sampling, zero sightings in 32 draws give a one-sided 95% upper confidence bound of about 8.94%, with lower bound zero. The sample therefore does not establish exact extinction. If the claim must cover 16 prespecified keys simultaneously, Bonferroni allocates .05/16 error to each key; the zero-count upper bound becomes about 16.50%. Requiring simultaneous coverage makes the certificate weaker. Conversely, two sightings in 32 draws have empirical frequency 6.25%, but their one-sided 95% lower confidence bound is only about 1.12%. These hypothetical upper and lower examples are separate one-sided limits; a joint interval requires allocation to both tails.

For repeated inspection of draws from a fixed policy, a confidence sequence can provide validity under optional stopping [Howard et al., 2021]. For fixed beta-mixture parameters $a_0, b_0 > 0$, let S_t be the number of successes in t independent Bernoulli draws and let B denote the beta function. Define

$$M_t(p) = \frac{B(S_t + a_0, t - S_t + b_0)}{B(a_0, b_0)p^{S_t}(1 - p)^{t - S_t}}, \quad \mathcal{I}_t = \{p \in (0, 1) : M_t(p) < 1/\alpha\}.$$

For each true $p \in (0, 1)$, the numerator integrates the likelihood under a fixed beta distribution. Hence $M_t(p)$ is a mixture of likelihood-ratio martingales, is nonnegative, and has initial value one. The nonnegative martingale maximal inequality therefore gives $\Pr_p(\exists t : p \notin \mathcal{I}_t) \leq \alpha$. Error allocations across a fixed roster of keys make coverage simultaneous by a union bound; independence across keys is unnecessary.

This construction requires a fixed Bernoulli probability. Counts from different prompts, seeds, or changing checkpoints cannot be pooled to certify the current checkpoint’s probability. Separate frozen-checkpoint measurements can instead allocate error across prespecified prompt–key–checkpoint statements. Repeated deterministic teacher-forced scores are not independent Bernoulli observations, and choosing identities after inspecting the same samples also needs selection-aware inference. A probability bound at observed checkpoints alone makes no claim about unobserved intervening or subsequent training states.

Worked example: repeated checkpoints spend an error budget. Suppose the plan specifies 16 keys at each of ten frozen checkpoints, with 32 new draws per checkpoint. There are 160 key–checkpoint statements. Allocating .05/160 to each gives simultaneous 95% coverage by a union bound, without requiring the key counts to be independent. A zero count then has an upper bound of about 22.29%. Pooling all 320 draws would not fix this weakness: the ten checkpoints can have different key probabilities. The confidence sequence above instead permits repeated inspection and a data-dependent stopping time while sampling one fixed checkpoint.

For an unknown catalogue, missing mass is the total probability of categories not yet observed under a fixed iid law. Concentration results address this quantity [McAllester and Ortiz, 2003, Berend and Kontorovich, 2013]; it is neither the number of missing modes nor proof of their extinction. Such bounds do not certify each named key, and stationary missing-mass formulas do not directly apply to a history generated by changing policies. Named-key binomial counts provide the direct measurement of per-mode visibility. For example, a hypothetical missing mass of .10 could consist

1334 of one unseen mode with probability .10 or one hundred unseen modes of probability .001 each; the
1335 same aggregate mass does not identify how many keys are missing.

1336 *Remark E.24* (Exact scope of the ReplayMaxRL guarantee). The categorical replay theorem protects
1337 represented bank modes after discovery; it is not an unconditional end-to-end guarantee for a language
1338 model. Without full coverage it protects only $\mathcal{B}$, and Lemma E.14 shows that a frozen incomplete
1339 bank can exclude unbanked correct modes in the categorical limit. Capacity 16 can therefore truncate
1340 protected support. The admission results above identify sufficient coverage and switching conditions;
1341 they do not establish those conditions for the implemented proposal and insertion machinery.

1342 The categorical proof uses $\log p_b$, whereas the implementation uses length-normalized teacher-forced
1343 exemplar scores. The complete-exemplar bridge and stochastic extension permit shared parameters
1344 under explicit joint-energy and update-error assumptions. They do not prove uniform neural key
1345 probabilities or verify those assumptions for the current clipped, alternating AdamW updates. The
1346 sharper certificates require correctly normalized complete likelihoods under the decoding law being
1347 bounded. Finally, exact entropy and exact categorical natural gradient have their own protection
1348 guarantees; collapse is a property of the stated update model, not of every procedure that optimizes
1349 correctness or entropy. These results motivate direct per-mode survival measurements.

1350 **F Supporting Measurements and Direct Comparators**

1351 This appendix collects the comparator design, complete and partial UCPO and sparse RLEP-Dr
1352 blocks, and measurements re-derived from registered run logs. In the primary analyses, exact prefixes
1353 are not promoted to five-seed estimands and receive no block-level uncertainty intervals. Here and in
1354 the figure labels, *correctness-adjusted breadth* denotes extra verified modes, `distinct@8-pass@8`,
1355 which remains coupled to correctness.

1356 **F.1 Source integrity and admissible comparisons**

1357 The Falcon3-1B Countdown ReplayDr.GRPO source for seed 59 (job 30269051) contains
1358 two complete four-draw terminal evaluation sequences with matching request identities but
1359 conflicting metrics. The recorded E112-R1 integrity amendment excludes this source; we
1360 apply that exclusion to every efficacy endpoint, checkpoint, paired contrast, and aggregate
1361 that reuses the run (paper/preregistration/e112r1_two_scale_49pair_terminal_
1362 integrity_amendment_20260828.md). We select neither the first sequence, the last, nor an
1363 average. The source failure concerns ReplayDr.GRPO only: all 150 verifier-only Dr.GRPO/GRPO
1364 precheck runs remain admissible.

1365 The core replay contrast therefore contains 74 admissible pairs: 14 complete five-seed model-domain
1366 blocks and Falcon Countdown at $n = 4$ (seeds 55–58). All 74 raw `distinct@8` differences are
1367 positive. The primary replay and factorial analyses report four-seed means descriptively, without
1368 confidence intervals. The Falcon Dr.GRPO replay macro-average uses those same four seeds across
1369 all five domains, yielding $+ .441699$ on `pass@8` and $+ .773926$ on `distinct@8`. The ten complete
1370 MaxRL–ReplayMaxRL blocks retain all five seeds; the full four-arm factorial has nine complete
1371 blocks and one four-seed block. Interactions require the common admissible intersection. The
1372 separately reported historical E112-R1 comparison follows its exploratory amendment, including
1373 a Student- t interval with three degrees of freedom for its four-seed Falcon Countdown block; that
1374 interval is not primary evidence.

1375 All 14 complete core blocks have positive mean effects on each endpoint. Assuming independent
1376 block signs, the two-sided exact sign test gives $p = 2/2^{14} = 0.0001220703125$ per endpoint;
1377 Holm adjustment across the two endpoints gives $p_{\text{Holm}} = 0.000244140625$ for each. These are
1378 nominal, exploratory summaries, not confirmatory evidence or a correction for dependence among
1379 model-domain blocks.

1380 Trajectory readers use registered source jobs and documented replacements, excluding superseded
1381 attempts. A separate ambiguity affects the initial `mean@8` of Qwen2.5-3B MathIR ReplayDr.GRPO
1382 seed 71. We omit that descriptive checkpoint without selecting a retry; its terminal endpoint remains
1383 valid. All 17 registered checkpoints for the 50 Qwen2.5-0.5B core runs are unambiguous, so the
1384 reported trajectory AUC is unchanged. Source identities, row hashes, and recovery authorities are
1385 recorded in paper/audits/initial_auc_source_integrity_20260904.json.

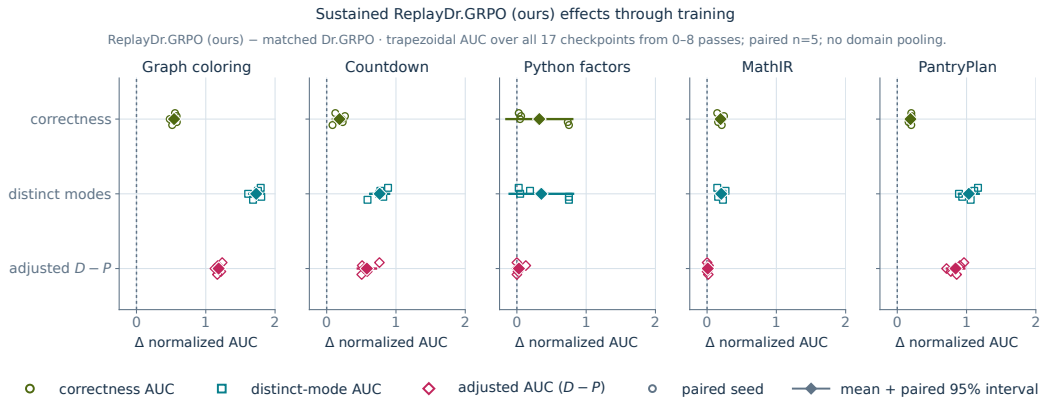


Figure 7: Sustained ReplayDr.GRPO effects through training. Qwen2.5-0.5B ReplayDr.GRPO is compared with matched Dr.GRPO separately on each static domain. Effects are trapezoidal trajectory AUCs over all 17 registered half-pass checkpoints from 0 through 8 passes, divided by the eight-pass horizon. Open markers are the five paired seeds 43–47, diamonds are paired means, and bars are unadjusted descriptive two-sided 95% Student- t estimation intervals. The adjusted effect subtracts correctness AUC from distinct-mode AUC per seed. No significance decision is made for an individual model–domain comparison, and no checkpoint, domain, or metric magnitude is pooled. An early-collapsed control can remain flat for most of the horizon, so AUC partly measures time spent in that state and is a sensitivity analysis, not independent mechanism evidence.

Table 1: Direct alternatives to canonical mode-balanced replay. Each method inherits the same eight-pass evaluation contract. Coverage counts source-admissible terminal runs; incomplete or excluded sources do not enter complete-block inference (Appendix F.1).

Method	Active distinction	Question isolated	Coverage
Dr.GRPO	Group-centered binary reward; no auxiliary derivative	verifier-only reference	75/75
GRPO	Reward-standardized group-relative update	estimator normalization	75/75
ReplayDr.GRPO	One retained exemplar per verified key, sampled uniformly over keys	persistent canonical memory	74/75 admissible
UCPO	Reallocates positive advantage within the current rollout group	on-policy diversity incentive without memory	50/75
Sparse RLEP-Dr	Replays verified successes in observed frequency, without canonical balancing	success memory without key balancing	47/75
MaxRL	Optimizes best- k jointly across sampling budgets, without replay	sampling-aware fresh search without persistent memory	10/15 blocks
Fixed semantic entropy	Adds detached surprisal over verified canonical outcomes to Dr.GRPO	explicit within-rollout diversity incentive	10/15 blocks

1388 Figure 8 gives the final-training comparison used to assess whether generic alternatives recover the
1389 retention pattern; the paragraphs below give exact domain-level values and the companion trajectories.

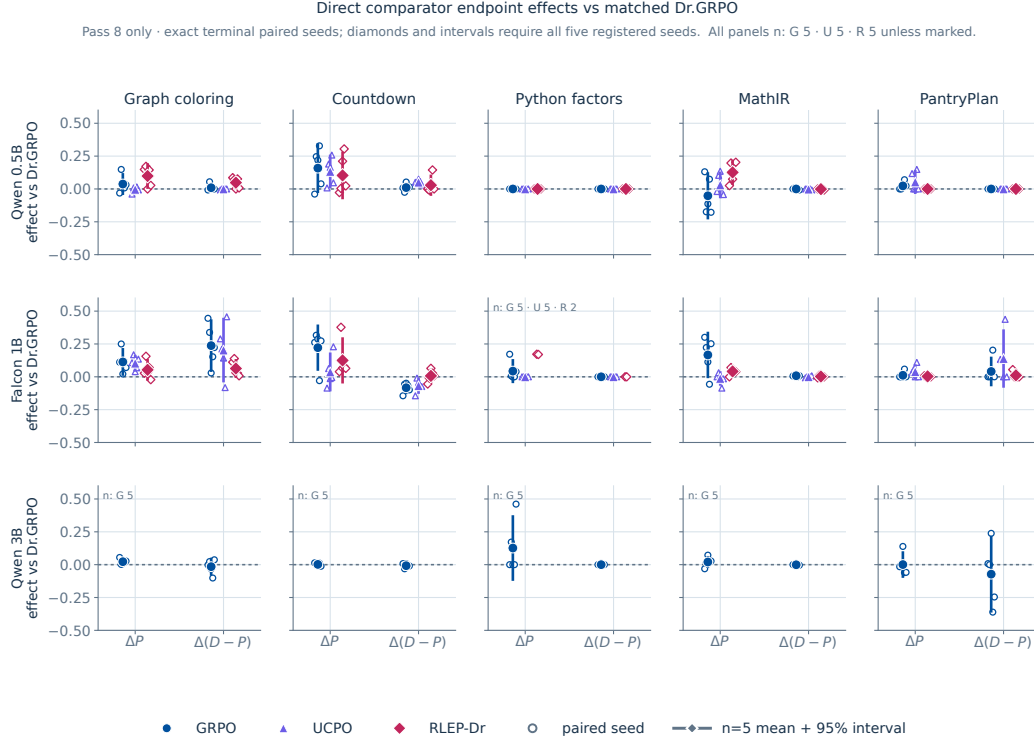


Figure 8: **Direct alternatives produce local rather than uniform final gains.** Each effect subtracts the same-seed Dr.GRPO control from the comparator after eight passes. Effects are shown for pass@8 and correctness-adjusted breadth. Circles are paired seeds; diamonds and 95% Student-*t* intervals appear only for complete five-seed blocks. Exact *n* is printed, and unsupported comparisons remain blank rather than being extrapolated or imputed at any seed.

1390 All five Qwen2.5-3B plain-GRPO blocks are now complete. Their pass@8 effects on Graph, Count-
 1391 down, Python, MathIR, and PantryPlan are $+.023 [-.002, +.048]$, $+.002 [-.012, +.016]$, $+.127$
 1392 $[-.123, +.376]$, $+.020 [-.026, +.066]$, and $+.001 [-.100, +.102]$. The corresponding correctness-
 1393 adjusted-breadth effects are $-.015 [-.084, +.053]$, $-.008 [-.028, +.013]$, exactly zero, $-.001$
 1394 $[-.003, +.001]$, and $-.072 [-.365, +.220]$. All five correctness means are positive, but none of
 1395 their individual 95% intervals excludes zero; the breadth intervals likewise include zero.

1396 Complete Qwen Countdown UCPO has a mean pass@8 effect of $+.133 [+.005, +.262]$ and adjusted-
 1397 breadth effect of $+.051 [+.031, +.071]$. The completed Qwen MathIR UCPO block remains neu-
 1398 tral: $+.031 [-.071, +.133]$ on pass@8 and $-.002 [-.006, +.003]$ on adjusted breadth. The com-
 1399 pleted Falcon Pantry RLEP-Dr block is inconclusive: $+.003 [-.002, +.008]$ on pass@8 and $+.011$
 1400 $[-.019, +.041]$ on adjusted breadth. Qwen Countdown RLEP-Dr is likewise inconclusive: $+.103$
 1401 $[-.078, +.285]$ and $+.029 [-.052, +.109]$. In contrast, Qwen MathIR RLEP-Dr has a positive cor-
 1402 rectness effect, $+.126 [+.030, +.222]$, with neutral adjusted breadth, $-.002 [-.006, +.003]$. All
 1403 complete blocks use five paired seeds and are not pooled across domains. The RLEP offline pool
 1404 used temperature .7 and top-*p* = .95 only during its frozen experience-collection job. Every reported
 1405 endpoint for RLEP and the other arms uses the shared temperature-1, top-*p* = 1 evaluation. Thus
 1406 there is no evaluation-decoding advantage for RLEP; its frozen offline pool versus our online bank
 1407 remains a substantive algorithmic difference.

Direct baselines across model scales — terminal evidence

Exact terminal prefixes are shown; cells without a terminal comparator are blank.

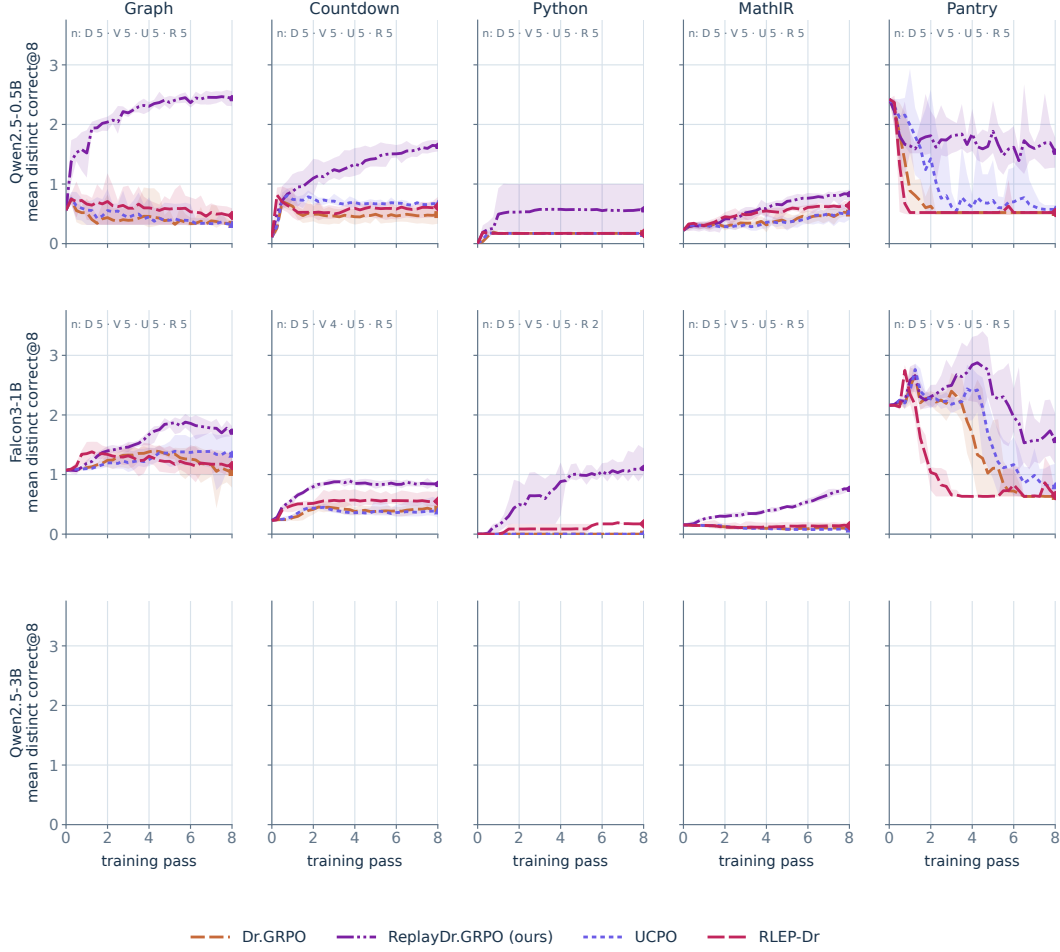


Figure 9: **Terminal direct-baseline learning trajectories in the invariant model grid.** Physical rows are Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B; columns are the five static domains. Each panel shows every terminal UCPO or sparse RLEP-Dr series together with matched Dr.GRPO and ReplayDr.GRPO, using its exact seed set and n . UCPO fills all ten smaller-model blocks at $n = 5$. RLEP-Dr fills all five Qwen domains and Falcon Graph/Countdown/MathIR/Pantry at $n = 5$, plus Falcon Python at $n = 2$. Models and domains without a final comparator remain blank; lines are descriptive means and bands are seed ranges. The core replay contrast appears in Figure 4.

1408 F.4 Fresh-frequency versus uniform replay: E120-R1 progress

1409 E120-R1 changes only the within-bank target weights: the treatment samples keys in proportion to
 1410 their fresh discovery frequency, while the matched ReplayDr.GRPO comparator weights retained
 1411 keys uniformly. The paper fixes this comparison to the snapshot generated on September 4, 2026
 1412 at 15:40:26 UTC (11:40:26 EDT). In that snapshot, 25/45 treatment cells have exact step-3,072
 1413 endpoints: all five Qwen2.5-0.5B domain blocks are complete, while the four registered Falcon3-1B
 1414 and Qwen2.5-3B blocks remain in progress. Table 2 reports the five terminal Qwen2.5-0.5B contrasts.

Table 2: **E120-R1 terminal Qwen2.5-0.5B blocks.** Each effect is fresh-frequency replay minus matched uniform key replay on the displayed five registered seeds. ΔP is the pass@8 difference and ΔD the raw distinct@8 difference. Means are descriptive; the adjacent machine-readable artifact retains paired, unadjusted 95% Student- t estimation intervals.

Domain	n	Seeds	ΔP	ΔD
Graph coloring	5	43,44,45,46,47	-.066	-.662
Countdown	5	43,44,45,46,47	-.022	-.661
Python factors	5	43,44,45,46,47	+.077	+.040
MathIR	5	43,44,45,46,47	+.005	-.014
PantryPlan	5	43,44,45,46,47	-.043	-.341

Fresh-frequency replay reduces raw distinct@8 relative to uniform key replay on Graph by $-.663$ ($[-.808, -.517]$), Countdown by $-.661$ ($[-.838, -.483]$), and PantryPlan by $-.341$ ($[-.564, -.117]$). MathIR is near zero, $-.014$ ($[-.160, +.132]$), while Python is heterogeneous across seeds, $+.040$ ($[-.926, +1.006]$). Correctness differences are smaller outside Python; Graph is $-.066$ ($[-.075, -.057]$), and the Countdown, MathIR, and PantryPlan intervals cross zero. Thus uniform weighting has a clear raw-support advantage in three of five complete blocks, but not universally. This supports uniform replay as an equal-retention design choice rather than as the optimizer of every downstream utility. Falcon3-1B and Qwen2.5-3B have no terminal E120-R1 cells in this frozen snapshot.

1424 F.5 When fresh Dr.GRPO gradients vanish

1425 Dr.GRPO centres advantages within a rollout group, so a group whose members all receive the same
 1426 reward contributes *exactly zero* fresh task gradient. Only a *mixed* group — at least one correct and at
 1427 least one incorrect member — can supply a nonzero fresh task gradient. The mixed-group fraction
 1428 therefore measures how often a group can supply fresh reward information. A zero current task
 1429 gradient does not imply an unchanged policy: AdamW momentum and weight decay can still move
 1430 parameters.

1431 Measured over all 3,072 updates of the Qwen2.5-0.5B cohort, five seeds for each domain:

Domain	matched Dr.GRPO	ReplayDr.GRPO
Graph coloring	.144	.954
Countdown	.140	.456
Python factors	.015	.185
MathIR	.364	.453
PantryPlan	.123	.789

1433 On Python factors, 98.5% of control updates therefore have zero fresh task gradient. The effect
 1434 sharpens with scale: the Qwen2.5-3B control on the same domain logged 7 mixed groups in 3,073
 1435 updates, versus 843 for its paired ReplayDr.GRPO run. Under the registered $G = 16$, $\beta_{KL} = 0$,
 1436 no-token-entropy-bonus protocol, the fresh task gradient is nearly always zero in this cell. Replay
 1437 supplies a verified-likelihood gradient directly and, after changing the policy, can also make later
 1438 fresh groups mixed again. This is evidence that replay is robust to this sparse binary-reward regime,
 1439 not that it dominates methods that keep the fresh RL signal active.

1440 **Direct output-collapse audit.** Exact endpoint equality is a useful screening signature: in the control
 1441 arm, mean@8, pass@8, and distinct@8 are identical in 14/15 Python seeds and 12/15 PantryPlan
 1442 seeds across the three scales. It is not our collapse criterion. We instead audited the stored raw
 1443 responses in all five E78 Qwen2.5-0.5B Python control seeds. At pass 0, a sampled $K = 8$ draw
 1444 contains 7.99 unique raw completions per prompt on average. By step 288–480 (0.75–1.25 passes),
 1445 every seed shows identical observed completions: for each of 128 prompts, all 32 temperature-1
 1446 samples are byte-identical to the separately logged temperature-0 greedy completion. This persists
 1447 through step 3,072, so the last 6.75–7.25 passes record persistence of this sampled-output pattern.
 1448 At the terminal checkpoint, a control draw contains exactly 1.00 unique raw completion per prompt,
 1449 versus 1.97 for ReplayDr.GRPO, averaged over the five seeds. When the singleton completion is

Table 3: **Absolute references for raw distinct@8.** *Uniform* samples the exposed action representation uniformly and is defined only for Graph (27 color strings) and PantryPlan (64 bit masks). Frozen is the matched pass-0 model; the final two columns are pass-8 means. Dashes avoid pretending that an open-ended program or derivation space has a canonical uniform distribution.

Model	Domain	Uniform	Frozen	Dr.GRPO	ReplayDr.GRPO
Qwen2.5-0.5B	Graph coloring	1.633	.561	.325	2.438
	PantryPlan	2.153	2.426	.522	1.539
Falcon3-1B	Graph coloring	1.633	1.070	1.034	1.717
	PantryPlan	2.153	2.162	.633	1.576
Qwen2.5-3B	Graph coloring	1.633	1.012	1.060	1.637
	PantryPlan	2.153	1.445	.841	1.514

correct it supplies one verified key; otherwise it supplies none. This is observed collapse to one completion per prompt, not one correct key on every prompt.

All three aggregate endpoint draw standard deviations are zero throughout the flat portion of these control traces. Zero draw spread is only corroborating: it is also zero at pass 0, when the varied completions are all invalid and raw within-draw uniqueness is 7.99. Raw per-prompt completion equality, rather than `draw_std` or equality of aggregate endpoints, establishes this observed collapse; it does not rule out unobserved low-probability outputs.

The logs also record masked-mean policy token entropy on sampled response tokens. The following descriptive values first average the 384 updates within each pass and then the five matched seeds:

Method	Training pass							
	1	2	3	4	5	6	7	8
Dr.GRPO	.679	.005	.003	.002	.002	.002	.002	.002
ReplayDr.GRPO	.325	.082	.088	.098	.087	.080	.072	.062

The first-pass mean includes the control’s rapid transition; after pass 1 its token entropy is near zero, whereas replay retains a larger, nonzero value. Token entropy is response-length dependent and is not canonical-key entropy, so we treat it as mechanism corroboration rather than a task-support endpoint.

Dynamic sampling that resamples until a group is mixed, larger G , reference KL, and token-entropy regularization directly target the same zero-gradient or collapse pathology. They are not present in the registered factorial and remain necessary controls for deciding whether replay adds value once fresh on-policy signal is kept alive. Likewise, the current arms have no temperature/top- p decoding sweep; that inexpensive robustness check would distinguish recoverable low-probability tails from collapse that persists under wider decoding.

F.6 Collapse is regression, not a smaller gain

Aligned methods share a pass-0 reference wherever they inherit the same initial model and the same fixed evaluation requests. That holds for every Qwen2.5-0.5B and Falcon3-1B domain and for three of the five Qwen2.5-3B domains. On Qwen2.5-3B MathIR and PantryPlan, E80-R1 seeds 71–74 carry marginally different pass-0 requests from the plain-GRPO method (.154 against .162, and .832 against .828 pass@8), so Figure 10 measures pass 0 within each method rather than drawing one shared origin. Table 4 gives the macro-average final values the main text quotes.

At Qwen2.5-0.5B, PantryPlan’s bit-uniform distinct@8 reference is 2.153 (2.714 on instruction-following support), above every final model; Graph’s reference is 1.633, which ReplayDr.GRPO reaches or exceeds. We therefore use PantryPlan as an enumerable stress test of support concentration, not as a claim of complete coverage. The table also identifies .108 as Countdown’s frozen pass-0 distinct@8; its terminal Dr.GRPO control is .487.

At Qwen2.5-0.5B, both verifier-only estimators regress on the two domains with substantial initial breadth: Dr.GRPO moves from .561 $\rightarrow$.325 on Graph and 2.426 $\rightarrow$.522 on PantryPlan; ordinary GRPO moves from .561 $\rightarrow$.372 and 2.426 $\rightarrow$.545, respectively. The other three domains begin with at most .011 extra verified modes per prompt on average, leaving little initial breadth to lose.

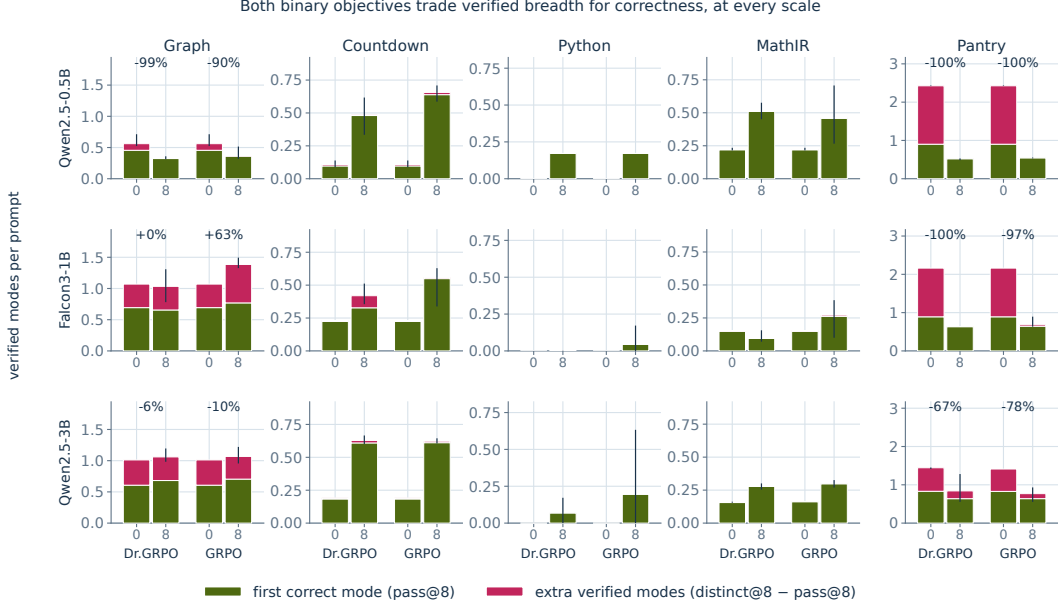


Figure 10: **Both verifier-only objectives reduce verified solution breadth, at every scale.** Bars split distinct@8 into the first correct mode beyond it, at training pass 0 and pass 8, for unmodified Dr.GRPO then GRPO in each panel. All 150 evaluations are complete at five seeds and four fixed $K = 8$ draws; whiskers are the seed range and pass 0 is measured per method. Badges give the change in extra verified modes, printed only where pass-0 breadth exceeds .05 of them; the other three domains start at .011 or less.

Table 4: **Problem precheck, five-domain macro-averages.** Both verifier-only objectives at pass 0 and pass 8, all three scales, five seeds and four fixed $K = 8$ draws per model–domain–method comparison. “Extra modes” is distinct@8–pass@8; “removed” is the fraction of that quantity lost between pass 0 and pass 8. Breadth loss is present at every scale and is far from uniform across scales, and at Falcon3-1B Dr.GRPO macro pass@8 falls as well.

Model	Dr.GRPO			GRPO		
	pass@8	extra modes	removed	pass@8	extra modes	removed
Qwen2.5-0.5B	0.335 → 0.402	0.329 → 0.002	99.4%	0.335 → 0.435	0.329 → 0.006	98.2%
Falcon3-1B	0.391 → 0.342	0.332 → 0.094	71.6%	0.391 → 0.454	0.332 → 0.135	59.4%
Qwen2.5-3B	0.356 → 0.455	0.205 → 0.120	41.5%	0.356 → 0.490	0.199 → 0.101	49.4%

Across the 15 evaluable Dr.GRPO panels, six finish below their pass-0 distinct@8: the two Qwen2.5-0.5B comparisons above; Falcon3-1B Graph 1.070 → 1.034, MathIR .154 → .097, and PantryPlan 2.162 → .633; and Qwen2.5-3B PantryPlan 1.445 → .841. Training on verified reward alone can therefore destroy correct-mode breadth already present in the base model. ReplayDr.GRPO ends below its pass-0 reference in only two of the 15 available domain summaries, both PantryPlan; the Falcon Countdown summary uses four admissible seeds (Appendix F.1).

The main-text precheck reads the matched Dr.GRPO controls (E78, E79, E80-R1) against the plain-GRPO cohort (E95 at all three scales, with the E114 seed extension supplying Qwen2.5-3B seeds 71–74), requiring complete four-draw evaluations at steps 0 and 3,072 for all five domains and all five registered seeds in each of the six method-by-scale blocks. All 150 evaluations are present, so no panel of Figure 10 is partial. A retained-breadth percentage is reported only where pass-0 extra modes exceed .05 per prompt: Graph and PantryPlan clear that floor at every scale, and Countdown, MathIR, and Python factors never reach .011, where one additional verified mode in one draw of one prompt already moves the four-draw mean by .002.

1499 G Semantic MaxEnt Baseline

1500 Semantic MaxEnt is included as a direct benchmark comparison, not as a component of ReplayMaxRL.
 1501 We state the implemented estimator here because its canonical-outcome input is closely related to
 1502 ModeBench measurement, while its training mechanism is different: it redistributes probability
 1503 among successful on-policy outcomes but does not maintain a persistent key-balanced memory.

1504 G.1 Score-function motivation

1505 For prompt x , let $q_\theta(a) = p_\theta^+(a | x)$ denote the policy conditioned on validator-positive canonical
 1506 outcomes. If this distribution were known, its entropy gradient would satisfy

$$\nabla_\theta H(q_\theta) = \mathbb{E}_{a \sim q_\theta} [(-\log q_\theta(a)) \nabla_\theta \log q_\theta(a)]. \quad (33)$$

1507 The implementation cannot observe q_θ and instead uses a detached, smoothed empirical predictor.
 1508 Equation 33 therefore motivates the redistribution direction; it is not a claim that the implemented
 1509 estimator is an unbiased entropy gradient.

1510 G.2 Implemented historical baseline

1511 For each prompt, the estimator counts validator-positive canonical keys in prior policy history and
 1512 successful leave-one-out peers. If K observed keys have total count N , a unit pseudocount and one
 1513 reserved unseen bucket define

$$\hat{p}(a) = \frac{n(a) + 1}{N + K + 1}, \quad \hat{p}(\text{UNSEEN}) = \frac{1}{N + K + 1}. \quad (34)$$

1514 For an eligible successful row, the added detached advantage is

$$A_{i,t}^{\text{sem}} = \eta \frac{\min\{-\log \hat{p}(a_i), C\} - \mathbb{E}_{\hat{p}}[\min\{-\log \hat{p}(a), C\}]}{C}, \quad \eta = .10, C = 5. \quad (35)$$

1515 Invalid and unsuccessful rows receive zero. The centered score is bounded by η in magnitude. The
 1516 unseen bucket assigns finite score to a newly observed key; it neither identifies an undiscovered
 1517 solution nor supplies a replay gradient for a remembered key after it stops appearing in fresh rollouts.
 1518 This is the conceptual distinction from ReplayMaxRL.

1519 The paper reports this fixed estimator only as a benchmark method. Its effects are heterogeneous
 1520 across domains and scales, so we make no general entropy or support-retention claim from it.

1521 **Predictor centering can change correctness as well as diversity.** Let $q = p(\cdot | C)$ and freeze a
 1522 predictor score s_c and baseline b independently of the scored action (for example, conditional on its
 1523 leave-one-out peers). Then

$$\mathbb{E}_{A \sim p}[\mathbf{1}\{A \in C\}(s_A - b)\nabla_z \log p_A] = P\mathbb{E}_{c \sim q}[s_c \nabla_z \log q_c] + (\mathbb{E}_q[s] - b)\nabla_z P.$$

1524 Indeed, on correct actions $\nabla \log p_c = \nabla \log q_c + \nabla \log P$, and $\mathbb{E}_q[\nabla \log q_c] = 0$. The second term
 1525 vanishes when the baseline equals the current conditional mean score, which a historical predictor
 1526 need not satisfy. Even with unclipped $s_c = -\log q_c$ and $b = H(q)$, averaging over all unconditional
 1527 rollout rows gives $P\nabla H(q)$, rather than $\nabla H(q)$.

1528 For an exact code-matched counterexample, suppose that there is one correct key and at least one
 1529 incorrect category. Conditional verified-key entropy is identically zero. With an explicit historical-
 1530 plus-peer count $N \geq 1$, the implemented open-set predictor has seen mass $u = (N + 1)/(N + 2)$
 1531 and unseen mass $v = 1/(N + 2)$. At the fixed $C = 5$ its advantage on the seen key is

$$-\frac{\eta v}{C}(\min\{\log(N + 2), C\} + \log u) < 0.$$

1532 Thus an all-success same-key group receives nonzero negative semantic pressure despite a zero
 1533 conditional-entropy gradient. This establishes estimator mismatch without determining the long-run
 1534 behavior of the resulting stochastic policy updates.

1535 G.3 Verified-support discovery comparator

1536 The discovery comparison in Figure 11 augments the row-local support with already verified replay
 1537 keys and keys found by an isolated proposal sampler. For row i , let H_i be successful policy history,
 1538 P_{-i} its successful leave-one-out peers, and R_i the verified replay/proposal support. Then

$$S_i = \text{supp}(H_i) \cup \text{supp}(P_{-i}) \cup R_i \cup \{a_i\}, \quad n_i(z) = H_i(z) + P_{-i}(z),$$

1539 and, with $N_i = \sum_z n_i(z)$,

$$q_i(z) = \frac{n_i(z) + 1}{N_i + |S_i|}, \quad A_i^{\text{sem}} = .10 \left[\frac{\min\{-\log q_i(a_i), 5\}}{5} - \sum_{z \in S_i} q_i(z) \frac{\min\{-\log q_i(z), 5\}}{5} \right]. \quad (36)$$

1540 Proposal-only keys contribute support but not counts; proposal rows never enter PPO or policy history.
 1541 A response enters replay only after the ordinary task verifier accepts it. This bundled comparison
 1542 adds both proposal sampling and semantic regularization to replay. It does not isolate their effects or
 1543 directly measure discovery or survival of individual modes. Adaptive controllers, group-centering
 1544 diagnostics, and open-bank development branches are excluded.

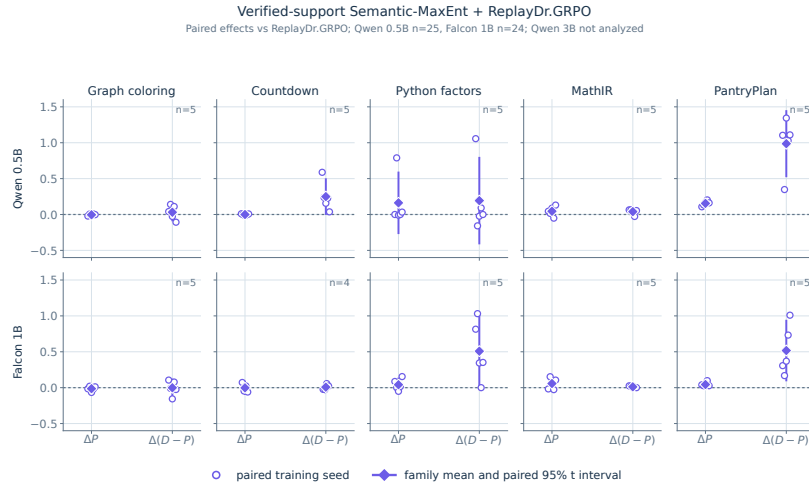


Figure 11: **Bundled verified-support comparison.** Effects compare ReplayDr.GRPO augmented with semantic regularization and proposal sampling against ReplayDr.GRPO after eight passes. Qwen uses $n = 5$; Falcon uses $n = 5$ except Countdown ($n = 4$). The historical exploratory amendment retains its $n = 4$ interval policy, separately from the primary analyses (Appendix F.1). This bundled historical comparison is supporting evidence, not a component of the primary ReplayMaxRL intervention evaluated here.

1545 Across 49 integrity-valid pairs, the mean correctness/adjusted-breadth gains are $+ .072 / + .300$ for
 1546 Qwen2.5-0.5B and $+ .026 / + .217$ for Falcon3-1B. PantryPlan is the only domain whose intervals
 1547 exclude zero on both measures at both scales; the remaining effects are heterogeneous.

1548 H External Python Verification Contract

1549 *Main-body link.* This appendix expands the Python Factors contract in Section 2 and its limitation in
 1550 Section 5, where observed behavior rather than source-code identity defines a Python mode.

1551 H.1 Accepted language and static bounds

1552 Every candidate is stripped, limited to 240 characters, and parsed by `ast.parse` in expression mode.
 1553 It must be exactly one `lambda n: EXPR`: positional-only arguments, defaults, keyword arguments,
 1554 variadic arguments, additional parameters, and any parameter name other than `n` are rejected. The

complete AST is capped at 64 nodes. Task specifications contain two to eight distinct integer inputs in $[4, 1000]$ and are rejected unless their exact behavior space contains at least two valid output vectors. The expression grammar admits integer literals with absolute value at most 10,000; the input variable n ; $+$, $-$, $*$, floor division, and remainder; integer comparisons; Boolean `and/or`; unary plus, minus, and `not`; and conditional expressions. Boolean and non-integer literals are rejected. Calls, imports, attributes, comprehensions, containers, subscripts, assignment, loops, and every identifier except n have no accepted AST node and therefore cannot reach execution. The grammar is defined once in parent-importable contract code, and the external worker applies it immediately before compilation.

1563 H.2 External process and fail-closed semantics

The trainer never evaluates model-written code. A locked parent client starts the same Python executable with isolated mode `-I`, a fresh process session, and a persistent JSON-lines worker. Each request contains only the restricted source and trusted task JSON. The worker reparses the candidate, compiles it in expression mode, evaluates it with `__builtins__={}`, and runs every frozen input under a 0.25-second real-time alarm. The parent imposes a separate one-second response deadline.

The worker returns one JSON object containing either `valid=true`, the integer output vector, and its canonical key, or `valid=false` and an exception class. A worker timeout, parent timeout, empty or malformed output, broken pipe, non-integer output, or failed divisor check yields $\perp$ and reward zero. Broken pipes, operating-system errors, and malformed responses terminate the worker and permit one clean restart; a timed out request is failed immediately. Worker `stdout` is never interpreted as a candidate answer, and no exception path manufactures a key.

1575 H.3 Correctness and behavioral identity

For each frozen input n_j , the returned value d_j must be an integer but not a Boolean and must satisfy $1 < d_j < n_j$ and $n_j \bmod d_j = 0$. Correctness is conjunctive across the complete input list. The canonical mode key is the ordered vector

$$\text{python_factor}: d_1, \dots, d_m.$$

Consequently, syntactically different functions that execute to the same vector merge, while a difference on any frozen input creates a different mode. The exact support size used only for dataset validation is $\prod_j |\{d : 1 < d < n_j, d \mid n_j\}|$; neither that catalogue nor its size is given to the policy, bank, scheduler, or stopping rule.

For inputs $[6, 10, 15]$,

```
lambda n: 2 if n % 2 == 0 else 3
```

returns $(2, 2, 3)$, while

```
lambda n: 3 if n % 3 == 0 else 5
```

returns $(3, 5, 3)$. Both verify and receive different behavior keys.

1586 I Domain Prompts

1587 *Main-body link.* This appendix reproduces the exact prompt the policy receives in each of the five
1588 MODEBENCH domains of Section 2 and Table 5.

1589 Every domain is single-turn: one system message, one user message, and no few-shot examples. The
1590 user message is the dataset problem field; the wrapper is the chat template the launcher exported
1591 through OAT_ZERO_PROMPT_TEMPLATE, defined in `src/oat_drgrepo/templates.py`. Both meth-
1592 ods of every domain use the identical string, and the same template is applied at training and at evalua-
1593 tion, so the prompt is never a difference between methods. Each block below is the first row of that co-
1594 hort’s frozen evaluation split, rendered by that template; `ops/check_paper_domain_prompts.py`
1595 regenerates these blocks from the datasets and fails closed if the typeset text drifts from them. Lines
1596 longer than the column are wrapped here with a two-space continuation indent; joining each indented
1597 continuation to its predecessor with one space recovers the string byte for byte, and the generator
1598 asserts that round trip. The only real newlines are the ones inside the PantryPlan pantry listing.

1599 I.1 Graph coloring

1600 Template qwen_boxed; 384 training and 128 evaluation
1601 rows in `var/data/graph_coloring_modebench_v2`.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Color this graph with vertices 1 through 6. The undirected edges are: (1,2), (1,3), (3,5),
(3,6), (4,6), (5,6). Assign each vertex one color from {1, 2, 3} so that no edge has the
same color on both endpoints. A partial coloring is 2??1?2, where ? means uncolored.
Keep every shown digit fixed. There are exactly 3 question marks. Return exactly one
final answer and no explanation: exactly 3 digits, each one 1, 2, or 3, for the missing
positions from left to right, inside \boxed{}.<|im_end|>
<|im_start|>assistant
```

1602 I.2 Countdown

1603 Template qwen_boxed; 384 training and 128 evaluation
1604 rows in `var/data/exact_countdown_easy3_probe`.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Using the numbers [3, 6, 9], create an arithmetic expression that equals 18. Use each
given number exactly once. You may use +, -, *, /, and parentheses. Return exactly one
final answer and no explanation: the expression inside \boxed{}.<|im_end|>
<|im_start|>assistant
```

1605 I.3 Python factors

1606 Template qwen_boxed; 384 training and 128 evaluation
1607 rows in `var/data/python_factor_modebench_v1`.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Write one pure Python function with the exact form lambda n: EXPR. An external Python tool
will call it once for each n in [18, 82, 91, 93]. For every call, return an integer d
satisfying 1 < d < n and n % d == 0. Different valid return vectors are different
solution modes. You may use integer literals, n, +, -, *, /, //, %, comparisons, Boolean
operators, and conditional expressions; calls, imports, attributes, containers, and
other names are forbidden. Return exactly the one-line lambda inside \boxed{} and no
explanation.<|im_end|>
<|im_start|>assistant
```

1608 I.4 MathIR action menu

1609 Template qwen_boxed; 384 training and 128 evaluation
 1610 rows in var/data/mathir_action_menu_v1.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
MathIR linear-menu-v1. Solve for x by choosing an executable action sequence. Every
selected action is applied exactly to BOTH sides of the current equation, followed by
exact simplification. Bindings: a=2, b=-9, c=-1. Initial equation: x/a + b = c. Action
menu: A means div(a); B means add(b); C means sub(b); D means sub(c); E means
sub(mul(a,b)); F means mul(a). Choose 1-4 action IDs. Illegal operations, repeated
states, and paths that do not finish with x isolated are rejected. Return only the IDs
separated by semicolons inside the answer box; for example formatting only: \boxed{A;C}.
Do not return x's numeric value.<|im_end|>
<|im_start|>assistant
```

1611 I.5 PantryPlan

1612 Template qwen_pantry_support_mask; 384 training and 128 evaluation
 1613 rows in var/data/pantry_plan_modebench_v2.

```
<|im_start|>system
Return only six binary digits and no other text. Each digit maps to the corresponding
Pantry row in printed order. A trusted environment will choose exact quantities on
precisely the selected rows.<|im_end|>
<|im_start|>user
Create one feasible high fiber snack from this frozen pantry.
Nutrition values are per 100 g. Quantities are exact integer grams.

Pantry:
- almonds: available=100g, step=25g, minimum-if-used=50g; energy=584kcal, protein=21.5g,
  fiber=10.8g, sodium=0.0mg; tags=tree_nut
- banana: available=125g, step=25g, minimum-if-used=50g; energy=97.0kcal, protein=0.74g,
  fiber=4.62g, sodium=0.0mg; tags=none
- carrots: available=150g, step=25g, minimum-if-used=50g; energy=45.0kcal, protein=0.941g,
  fiber=3.1g, sodium=86.6mg; tags=none
- grape_tomatoes: available=100g, step=25g, minimum-if-used=50g; energy=27.0kcal,
  protein=0.83g, fiber=2.1g, sodium=6.0mg; tags=none
- navel_orange: available=150g, step=25g, minimum-if-used=50g; energy=47.0kcal,
  protein=0.91g, fiber=2.0g, sodium=9.0mg; tags=none
- sunflower_seeds: available=125g, step=25g, minimum-if-used=50g; energy=571kcal,
  protein=18.9g, fiber=7.22g, sodium=0.0mg; tags=none

Use 2 to 4 ingredients.
Forbidden tags: none
Targets:
- mass_g: minimum 125, maximum 200
- energy_kcal: minimum 239.4, maximum 432.25
- protein_g: minimum 6.423
- fiber_g: minimum 3.478
- sodium_mg: maximum 37.6

Choose the ingredient support as a six-bit mask in the exact Pantry row order. Bit 1
includes that row and bit 0 excludes it.<|im_end|>
<|im_start|>assistant
```

1614 J Dataset and Identity Validation

1615 *Main-body link.* This appendix documents the materialization and identity claims in Section 2 and
 1616 Table 5, including the checks that bind correctness and identity to one execution.

1617 J.1 Executable correctness and identity inventory

Table 5: **MODEBENCH**. Correctness and identity come from one executable path; training starts without the mode catalogue. *Valid modes* is the mean number of correct execution modes per evaluation prompt, exact over the released multi-answer split. Every domain trains on 384 prompts and evaluates on a fixed 128-prompt split; the exact prompt each policy receives is reproduced verbatim in Appendix I for a direct, byte-level audit of the evaluation interface.

Domain	Generated answer / check	Executed key	Aliases that merge	valid modes
Graph coloring	Complete assignment; check fixed colors and edges	Color vector	Spacing only	6.27
Countdown	Parse operands; execute exactly	Normalized AST	Commutative order	4.52
Python factors	Execute restricted lambda externally	Return vector	Source programs with same behavior	229.44
MathIR	Execute bounded action program	State trajectory	Programs with the same states	5.00
PantryPlan	Exact ingredient allocation; check all bounds	Ingredient support	Quantity and ordering aliases	18.19

1618 J.2 Deterministic split materialization

Table 6: Deterministic materialization identities. For every Python row, two independently executed functions must verify with different keys. Every MathIR certified route is executed by the same interpreter used for model outputs. Together these checks bind both acceptance and identity to deterministic execution for every row in the released split.

	Python factors	MathIR action menu
Training rows	384	384
Evaluation rows	128	128
Exact modes per prompt	16–3,600	5
Training-row SHA-256	bcfa9acc...37cf7	bfc98362...5d3835
Evaluation-row SHA-256	be0f621c...94d183	df5783dd...328b96
Gold catalogue supplied to policy	no	no
Training-bank initialization	empty	empty

1619 J.3 Matched Level-2 construction

1620 ModeBench currently releases two levels for each domain. Level 2 changes the problem structure
 1621 while preserving the Level-1 verifier, canonicalizer, split sizes, and exact per-split histogram of
 1622 valid-mode counts. Its 384/128/128 train/development/evaluation identities are disjoint from every
 1623 Level-1 and Level-2 split. Consequently, a level comparison changes difficulty without silently
 1624 changing the amount of valid support available to be found.

Domain	Level 1	Level 2 structural change
Graph coloring	Three hidden vertices	Four hidden vertices
Countdown	Three operands	Four operands, with values up to 14
Python factors	Four executed cases, all at most 96	Four cases, including a value above 96
MathIR	One-sided linear families	Variable-on-both-sides families
PantryPlan	Base dietary/composition constraints	More active dietary exclusions and tighter composition

1626 Before treatment training, frozen Qwen2.5-0.5B evaluation admitted a Level-2 domain only if its
1627 `pass@8` was in $[.10, .90]$ and strictly below the paired Level-1 value under identical decoding. The
1628 released manifest `var/data/modebench_harder_v2_matched_r5/identity.json` records the
1629 support histograms, row hashes, construction seeds, disjointness checks, and admission receipts.
1630 The same generator-and-admission recipe can create further levels; the benchmark definition is not
1631 restricted to the two instantiated here.

1632 **J.4 PantryPlan split validation**

1633 The deterministic v2 release contains 384 train, 64 development, and 128 evaluation prompts
1634 across four formulation families; exact exhaustive support counts range from 8 to 45. The re-
1635 ported comparison uses the full 384-prompt training split and the fixed 128-row evaluation split,
1636 whose 2,328 feasible supports average 18.1875 per prompt. The train/evaluation row hashes are
1637 `a4883c5e...a757f` and `5d81ba8d...5923`. The terminal validation status is `pass` with decision
1638 `pantry_terminal_eligible`. Correctness checks exact inventory, serving steps, mass, nutrient
1639 bounds, and dietary tags; identity is ingredient support. The benchmark does not claim that a verified
1640 allocation is a recipe, palatable, cookable, or safe.

1641 K Algorithmic Details

1642 *Main-body link.* This appendix formalizes the replay state transition summarized in Section 3.

1643 For each prompt x , let $\mathcal{B}_x$ be the capacity- M verified-exemplar bank and let τ be the recurrent replay
1644 cursor, which makes the next bank selection exactly resumable after interruption.

1645 K.1 Binary MaxRL advantage

1646 For a fresh on-policy group of G responses, let $r_i \in \{0, 1\}$ be the verified reward of response i and
1647 let $m = \sum_{i=1}^G r_i$ count successes. The binary MaxRL task advantage is

$$A_i^{\text{MaxRL}} = \begin{cases} Gr_i/m - 1, & m > 0, \\ 0, & m = 0. \end{cases}$$

1648 Mixed groups assign failures advantage -1 and increase the weight of each success when successes
1649 are scarce. All-correct and all-failure groups have zero task advantage. This is the centered binary
1650 MaxRL estimator [Tajwar et al., 2026]. Lemma E.2 derives its on-policy mean objective: a harmonic
1651 mixture of $\text{pass}@k$ for $k = 1, \dots, G - 1$. It is applied only to fresh on-policy responses; replay
1652 exemplars enter through the separate verified-likelihood loss.

1653 K.2 State and optimizer update

Algorithm 1 One ReplayMaxRL optimizer update

Require: Policy π_θ ; prompt x and verifier specification s_x ; group size G ; fresh objective $o \in \{\text{MaxRL}, \text{Dr.GRPO}\}$; replay weight λ_{replay} ; states $(\mathcal{B}, \tau)$

Ensure: Atomically updated $(\theta', \mathcal{B}', \tau')$

- 1: Draw the on-policy group $\mathcal{G} = \{y_i \sim \pi_\theta(\cdot | x)\}_{i=1}^G$
 - 2: **for** $i = 1, \dots, G$ **do**
 - 3: Execute $\tilde{c}_i \leftarrow V(y_i, s_x)$ and observe task reward r_i
 - 4: Set $c_i \leftarrow \tilde{c}_i$ iff $r_i > 0 \wedge \tilde{c}_i \neq \perp$; otherwise set $c_i \leftarrow \perp$
 - 5: **end for**
 - 6: Compute fresh task advantages A_i^o from $\{r_j\}_{j=1}^G$; for ReplayMaxRL use the formula in Appendix K.1
 - 7: After the full group is validated, deterministically retain one policy exemplar for each new $c_i \neq \perp$ in $\mathcal{B}_x$, up to capacity M
 - 8: $(z, \tau') \leftarrow \text{NEXTBANK}(\{\mathcal{B}_u\}_u, \tau)$ by a fixed round-robin cursor
 - 9: **if** a replay bank $\mathcal{B}_z$ exists **then**
 - 10: For each retained exemplar e_c , compute $s_c = |e_c|^{-1} \log \pi_\theta(e_c | z)$
 - 11: $L_{\text{replay}} \leftarrow -|\mathcal{B}_z|^{-1} \sum_{c \in \mathcal{B}_z} s_c$
 - 12: **else**
 - 13: $L_{\text{replay}} \leftarrow 0$
 - 14: **end if**
 - 15: $w_{\text{rep}} \leftarrow (G - 1)/G^2$ for one replay pseudo-group per G on-policy rows
 - 16: $\theta' \leftarrow \text{UPDATE}(\theta, L_{\text{PPO}}(A^o) + w_{\text{rep}} \lambda_{\text{replay}} L_{\text{replay}})$
 - 17: Checkpoint $(\theta', \mathcal{B}', \tau')$ as one atomic state
-

1654 K.3 Bank admission, determinism, and capacity

1655 A bank is keyed by the SHA-256 digest of the exact prompt-token sequence. Only loss-active rows
1656 with positive task reward and a nonempty validator key are eligible. The full rollout group is validated
1657 before new keys are committed, so every candidate is considered against the same pre-group bank
1658 state.

1659 For each newly observed key, candidates are normalized to token tuples and the lexicographically
1660 smallest eligible response is stored. New keys are processed in sorted-key order, making admission
1661 invariant to rollout-row order. The bank holds at most one exemplar per canonical key and at most
1662 $M = 16$ exemplars per prompt. Existing exemplars are never evicted or overwritten by a later surface
1663 form; once capacity is full, later keys may remain in diagnostic counts but cannot enter the replay
1664 support. A prompt-hash collision or a changed token sequence under an existing prompt hash is a
1665 hard error.

1666 **K.4 Recurrent scheduling and verified-likelihood loss**

1667 At each optimizer update the global scheduler sorts the nonempty prompt-bank hashes and selects one
1668 by a round-robin cursor. Singleton banks are eligible: verified likelihood can preserve a discovered
1669 success before a second mode is found. The cursor advances independently of reward, evaluation,
1670 desired support, or bank entropy and is serialized with the bank, so an interruption cannot silently
1671 resample the schedule.

1672 For the selected bank, stored prompt and response tokens are materialized unchanged into a right-
1673 padded teacher-forcing batch. Prompt and padding tokens are masked; each response receives its
1674 mean token log probability s_c . Uniform verified likelihood averages $-s_c$ first across modes within
1675 the selected prompt, then applies the registered replay coefficient and the per-rollout scaling shown in
1676 Algorithm 1. Thus each stored mode has equal target weight even if its on-policy frequency was high
1677 when discovered. Exhaustive support and evaluation outputs never enter the loss.

1678 **K.5 Matched control, telemetry, and resume state**

1679 The matched control uses the same validation, bank-update, scheduling, replay-materialization, and
1680 score-pass code, but replaces the replay score-gradient vector by exact zeros. The treatment changes
1681 that derivative from zero to live. Policy-dependent samples, bank contents, selected banks, exemplar
1682 lengths, and realized computation can subsequently differ; sharing the code does not hold those
1683 quantities fixed. Both methods log available and replayed banks, retained modes, realized token
1684 counts, raw and applied replay losses, score-gradient sums and norms, scheduler phase, and whether
1685 the exact-zero switch was active.

1686 Model, optimizer, bank counts, exemplar tokens, replay cursor, data cursor, and request-stream state
1687 are checkpointed as one resume contract. The loader checks the schema, replay capacity, scheduler
1688 settings, estimator constants, token types, prompt hashes, exemplar/support consistency, and cursor
1689 bounds before accepting a checkpoint.

1690 **L Replay Mechanism Checks**

1691 The intervention changes whether the replay derivative is applied. Treatment and control share
1692 verifier, bank-admission, round-robin scheduling, replay-materialization, and score-pass code. The
1693 control zeros the replay derivative; the treatment applies it. The resulting policies can build different
1694 banks and incur different replay workloads. ReplayMaxRL and ReplayDr.GRPO differ only in the
1695 fresh-rollout advantage. No semantic-entropy term, proposal model, novelty reward, adaptive replay
1696 coefficient, reference KL, or gold support is active.

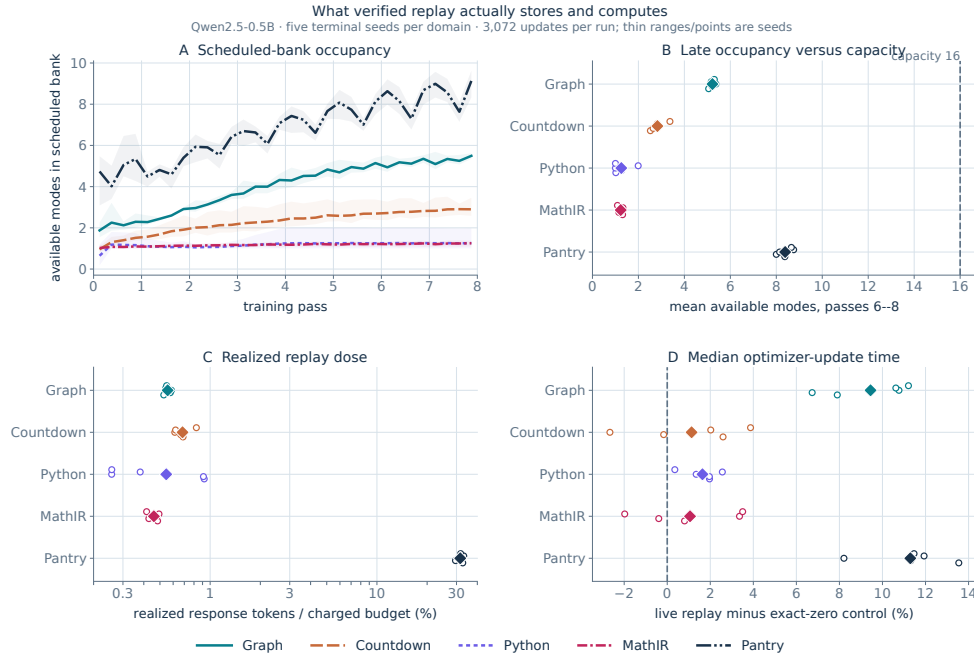


Figure 12: The verified replay path is active and bounded. Telemetry summarizes all 25 terminal Qwen2.5-0.5B ReplayDr.GRPO runs and their matched controls. It records bank availability and occupancy, capacity hits, realized replay-token dose, and paired optimizer time. These measurements validate the intervention but do not by themselves establish causal mediation through any individual stored mode across training checkpoints.

The bank stores at most one surface exemplar per verified canonical key and at most 16 keys per prompt. Identity-level survival histories were not retained, so occupancy cannot be interpreted as a mode-survival curve and Theorem E.12’s individual-mode prediction is not tested by the experiments in this paper. Existing fingerprints verify key/weight alignment but do not record each fixed exemplar’s longitudinal policy score; the disjoint evaluation prompts cannot supply that missing measurement. The registered E121 fixed-bank survival study addresses this gap by freezing a Graph bank after one pass and recording per-exemplar score trajectories (paper/preregistration/e121_fixed_bank_survival_telemetry_20260903.md). It is planned mechanism evidence; no E121 empirical result is reported here. The present telemetry is used only to verify that replay was scheduled, nonzero in treatment, exactly zero in controls, and within its registered resource envelope. Earlier bank-normalization, starvation-fallback, and open-bank experiments altered multiple mechanisms and are excluded from the appendix because they do not identify the ReplayMaxRL effect reported here.

1711 **M Aligned Scale Extensions**

1712 **M.1 Matched design**

The scale extensions preserve the five ModeBench domains, paired initialization and data identities, eight-pass horizon, evaluation grid, verifier, bank capacity, replay scheduling code, and recovery rules. Registered seeds are 55–59 for Falcon3-1B and 70–74 for Qwen2.5-3B. The Falcon Countdown ReplayDr.GRPO source at seed 59 is excluded from efficacy summaries (Appendix F.1). Hardware and optimizer settings are matched within a model–domain–seed comparison. Results are reported separately by model and domain; no missing result is imputed and no cross-model aggregate is interpreted as a scaling law.

1720 M.2 Terminal paired evidence

1721 Figure 4 in the main text gives the paired ReplayDr.GRPO effects. Table 7 records the corresponding
1722 exact matched final results.

Table 7: **Matched results after eight training passes.** Columns compare Dr.GRPO with ReplayDr.GRPO after exactly eight training passes. Entries are means over the displayed paired-seed intersection; $n@pass$ gives its size and training pass. P, M, and D denote pass@8, mean@8, and distinct@8. Every available source-admissible terminal pair is included, no shallower checkpoint is carried forward, and no missing result is imputed. Falcon Countdown uses four admissible seeds (Appendix F.1). Pale green marks a ReplayDr.GRPO mean that is strictly higher than its matched Dr.GRPO mean.

Model	Domain	$n@pass$	Dr.GRPO			ReplayDr.GRPO		
			P	M	D	P	M	D
Qwen2.5-0.5B	Graph coloring	5@8.0	.323	.323	.325	.969	.404	2.438
	Countdown	5@8.0	.480	.465	.487	.672	.597	1.637
	Python factors	5@8.0	.172	.172	.172	.528	.507	.571
	MathIR	5@8.0	.512	.464	.513	.796	.688	.829
	PantryPlan	5@8.0	.522	.522	.522	.729	.545	1.539
Falcon3-1B	Graph coloring	5@8.0	.656	.300	1.034	.809	.334	1.717
	Countdown	4@8.0	.325	.267	.415	.632	.343	.838
	Python factors	5@8.0	.000	.000	.000	.979	.722	1.107
	MathIR	5@8.0	.095	.068	.097	.736	.292	.756
	PantryPlan	5@8.0	.633	.633	.633	.751	.560	1.576
Qwen2.5-3B	Graph coloring	5@8.0	.682	.360	1.060	.836	.349	1.637
	Countdown	5@8.0	.609	.569	.628	.746	.505	.962
	Python factors	5@8.0	.069	.069	.069	.659	.603	.852
	MathIR	5@8.0	.278	.168	.279	.686	.580	.706
	PantryPlan	5@8.0	.638	.558	.841	.744	.546	1.514

1723 M.3 MaxRL factorial per-domain results

1724 For a common paired seed j and endpoint Y , the factorial interaction is

$$I_j = (Y_j^{\text{ReplayMaxRL}} - Y_j^{\text{MaxRL}}) - (Y_j^{\text{ReplayDr.GRPO}} - Y_j^{\text{Dr.GRPO}}).$$

1725 A positive replay effect under each fresh objective need not give $I_j > 0$. Here “complementary”
1726 means that replay adds value under both objectives; it does not assert a positive interaction. Negative
1727 interactions can coexist with those gains, including when a bounded endpoint approaches its ceiling.
1728 Interaction uncertainty must use these within-seed differences of differences, not differences between
1729 separately calculated confidence limits.

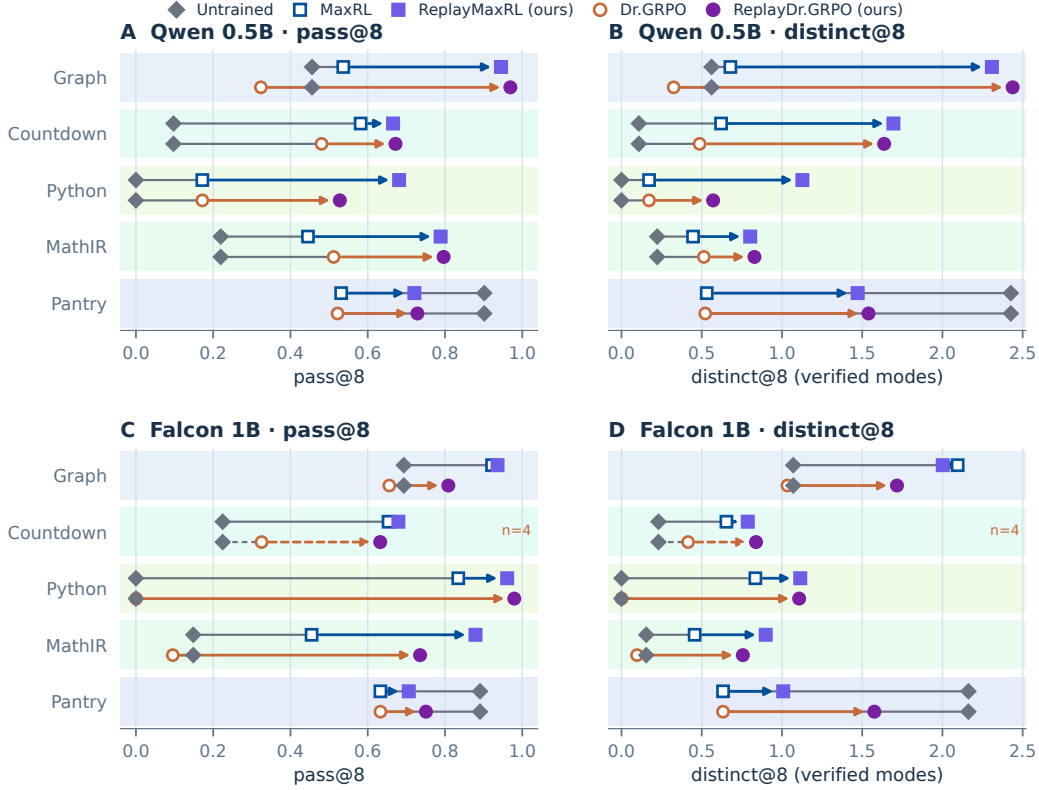


Figure 13: **Per-domain results behind the cross-domain averages in Figure 5.** Panels A–B show Qwen2.5-0.5B and panels C–D show Falcon3-1B. MaxRL–ReplayMaxRL pairs retain five terminal seeds in every domain. Dr.GRPO–ReplayDr.GRPO uses four admissible seeds for Falcon Countdown and five elsewhere. Domain tints match Figure 2. Replay arrows, trained-arm markers, and untrained references use the same encoding as the main figure.

1730 All five Qwen2.5-0.5B ReplayMaxRL–MaxRL blocks are complete. Both endpoint effects are posi-
 1731 tive in every domain: pass@8 effects range from $+0.084$ to $+0.509$, and raw distinct@8 effects range
 1732 from $+0.356$ to $+1.630$. All five Falcon MaxRL–ReplayMaxRL blocks are also complete. Replay-
 1733 MaxRL has a positive mean pass@8 effect in every Falcon domain and a positive raw distinct@8
 1734 effect in four of five. Graph is the exception on raw support, with -0.093 modes (unadjusted 95% inter-
 1735 val $[-0.229, +0.043]$) and a small correctness effect of $+0.014$ ($[-0.016, +0.044]$). MathIR has the largest
 1736 Falcon effects: $+0.424$ on pass@8 ($[+0.310, +0.539]$) and $+0.443$ on distinct@8 ($[+0.317, +0.569]$).
 1737 The within-seed cross-domain effects are $+0.133$ ($[+0.043, +0.223]$) and $+0.228$ ($[+0.094, +0.361]$), re-
 1738 spectively. Falcon Python remains inconclusive despite positive means: $+0.127$ ($[-0.372, +0.625]$) and
 1739 $+0.280$ ($[-0.209, +0.769]$).

1740 Qwen2.5-3B currently contributes only a two-seed Python terminal prefix to the MaxRL factorial.
 1741 It remains in the machine-readable progress record but is excluded from both figures until all five
 1742 domain blocks are complete; it receives no interval, cross-domain average, or efficacy interpretation.

1743 M.4 Interpretation boundary

1744 At terminal pass 8, the ReplayDr.GRPO contrast has 24 admissible Falcon3-1B pairs and 25 Qwen2.5-
 1745 3B pairs. Falcon Countdown uses four seeds; the other blocks use five. Raw distinct@8 is higher
 1746 in every admissible pair. Correctness-adjusted breadth is strongest on Graph coloring, Countdown,
 1747 and PantryPlan; Python factors and MathIR more often improve correctness without a consistent
 1748 breadth effect. These are per-domain, within-scale comparisons. They do not establish monotonic
 1749 improvement with parameter count. MaxRL–ReplayMaxRL pair results use their own admissible
 1750 seed sets; full factorial interactions use the four-arm intersection. Incomplete Level-2 blocks remain
 1751 descriptive and do not support cross-domain generality.

N Reproducibility and Claim Boundaries

N.1 Comparison registry

Table 8: **The three experiments and the evidence boundary at the September 4 freeze.** A complete block contains all five registered paired terminal seeds. Partial or source-excluded intersections are not counted as complete blocks. E120 uses its September 4, 15:40:26 UTC snapshot.

	Question	Registered unit	Complete	Still needed
Exp. 1	Does canonical replay retain verified support?	Level 1; 3 scales $\times$ 5 domains; Dr./ReplayDr.	14/15 blocks; one $n = 4$	retain Falcon Countdown as descriptive after source exclusion
Exp. 2	Does memory add value beyond MaxRL?	Level 1; 3 scales $\times$ 5 domains; MaxRL/ReplayMaxRL	10/15 blocks	five Qwen-3B blocks; full factorial has nine complete blocks and one $n = 4$
Exp. 3	Does the factorial transfer to a harder matched level?	Level 2; Qwen-0.5B; 5 domains $\times$ 4 methods	1/5 blocks; Graph complete; MathIR $n = 1$	finish the other four domains
Ablation	Does fresh-frequency replay differ from uniform key replay?	Level 1; 9 model-domain blocks; 45 cells	5/9 blocks; 25/45 cells	larger-model blocks absent from the frozen E120 snapshot

The primary registry is the 2×2 factorial over fresh objective (Dr.GRPO or MaxRL) and verified replay (off or on). Level 1 covers five domains at Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B; Level 2 repeats the full factorial on a separately generated, support-matched set at Qwen2.5-0.5B. A result enters the paper only at a registered checkpoint shared by the paired methods. Seed prefixes are labeled with exact n and are never promoted to five-seed estimates. GRPO, UCPO, sparse RLEP-Dr, and fixed Semantic MaxEnt remain direct comparators rather than components of the primary factorial. The ten complete MaxRL–ReplayMaxRL blocks remain valid at five seeds each. Requiring all four arms leaves nine complete five-seed blocks and one four-seed Falcon Countdown block. Cross-objective interactions require the common admissible seed intersection; they are not differences of aggregates with different seed sets.

N.2 Training and evaluation contract

Table 9: **Core matched run contract.** Scale-specific systems changes are applied to every method within a paired comparison, so systems changes cannot define a treatment arm.

Component	Frozen setting
Data	384 training and 128 disjoint evaluation prompts per domain and level; no evaluation row enters training or replay.
Rollouts	$G = 16$, temperature 1, $\text{top-}p = 1$; one prompt group per update.
Training	Eight passes (3,072 updates), one PPO epoch, $\beta_{\text{KL}} = 0$; paired methods share initialization, prompt stream, optimizer, and checkpoint cadence.
Evaluation	Passes 0, .5, . . . , 8; a separate temperature-0 greedy trace and four fixed temperature-1, $\text{top-}p = 1$, $K = 8$ draws per prompt for <code>pass@8</code> , <code>mean@8</code> , and <code>distinct@8</code> ; no best-checkpoint selection.
Replay	One capacity-16 bank per prompt; one bank selected by a serialized round-robin cursor per update; $\lambda_{\text{replay}} = .10$; controls use the same replay code with an exact-zero derivative; realized banks and workloads can differ.

The pinned revisions are listed separately to keep model identity explicit:

- Qwen2.5-0.5B-Instruct: 7ae557604adf67be50417f59c2c2f167def9a775;
- Falcon3-1B-Instruct: 28ba2251970a01dd1edc7ba7dad2eb71216ccfd;
- Qwen2.5-3B-Instruct: aa8e72537993ba99e69dfaafa59ed015b17504d1.

Training seeds are 43–47, 55–59, and 70–74, respectively. Qwen2.5-0.5B uses a constant 2×10^{-7} learning rate; Falcon3-1B uses the same rate with optimizer state offload; Qwen2.5-3B uses a 10^{-7}

1771 peak cosine schedule, 10^{-8} floor, and 10% warmup. Every paired comparison uses the same hardware
1772 class within its model–domain–seed block.

1773 Graph coloring and Countdown allow 192 response tokens; Python factors uses 192/512/192 tokens
1774 at 0.5B/1B/3B; MathIR uses 64/128/64; PantryPlan uses eight. Evaluation base seeds are 610100,
1775 610200, 610300, 610400, and 76299 in domain order. Sampled draw d uses base seed plus d , with
1776 independent responses within each draw. Metrics average over all 128 prompts and four draws; raw
1777 prompt responses, per-draw endpoints, and their aggregate spread are retained. Evaluation outputs
1778 never enter training, replay, checkpoint choice, or stopping.

1779 **N.3 Recovery and fail-closed rules**

1780 Model and optimizer state, bank contents, replay cursor, data cursor, and random request-stream
1781 state are checkpointed atomically. Resume checks schema, capacity, scheduler settings, objective
1782 constants, token types, prompt hashes, exemplar/key consistency, and cursor bounds. Non-finite
1783 values, state mismatch, missing completion markers, changed data identities, or aggregate drift are
1784 hard failures. Placement-only recovery amendments may change hardware or wall time but not the
1785 scientific update.

1786 **N.4 Artifacts and implementation map**

1787 Every reported figure has an adjacent machine-readable record containing its source paths, exact
1788 seed set, checkpoint rule, displayed values, and SHA-256 hashes. `paper/FIGURE_MANIFEST.md`
1789 maps figures to builders and versioned records. The ModeBench example renderer re-executes every
1790 displayed coloring, factorization, arithmetic transition, and pantry allocation. Result builders abort
1791 on missing matched comparisons or source-hash drift rather than carrying a shallower checkpoint
1792 forward.

1793 Under `src/oat_dgrpo/`, benchmark environments are implemented in `canonical_actions.`
1794 `py`, `python_modebench.py`, `mathir.py`, and `pantry_plan.py`; isolated Python execu-
1795 tion uses `python_modebench_process.py` and `python_modebench_worker.py`. Veri-
1796 fied replay and bank state are implemented in `canonical_replay.py` and `online_`
1797 `canonical_bank.py`; learner integration is in `learner/grpo.py` and `learner/run.py`.
1798 The post-hoc raw-collapse and token-entropy audit is regenerated by `ops/exp_scaling/`
1799 `build_paper_e78_python_collapse_diagnostics.py` into `paper/results/e78_python_`
1800 `collapse_diagnostics.json`. E119 all-domain terminal progress is stored beside Figure 6;
1801 E120-R1 is regenerated by `ops/exp_scaling/build_paper_e120_frequency_progress.py`
1802 into `paper/results/e120_frequency_progress.json` and its table body. Dataset manifests
1803 under `var/data/` bind every train/evaluation identity and level.

1804 **O Limitations and Disclosures**

1805 Appendix E gives categorical collapse and retention results, plus conditional extensions to stochastic
1806 updates and bank admission. These require the stated geometry, energy, and coverage assumptions;
1807 they do not certify the implemented neural optimizer. Convergence to uniform correct mass in the
1808 categorical uniform-replay model requires full verified coverage, which Python’s bank capacity
1809 precludes. ModeBench measures task-defined executable support on small synthetic tasks rather
1810 than free-form reasoning diversity or downstream utility. Experiments fix $G = 16$, zero reference
1811 KL, no token-entropy bonus, and a single decoding protocol; dynamic sampling, larger groups,
1812 regularization, and decoding sweeps remain open. The uniform-versus-frequency ablation has five
1813 of nine complete blocks, all at Qwen2.5-0.5B, in the frozen September 4, 15:40:26 UTC E120
1814 snapshot. The across-scale MaxRL pair has 10/15 complete blocks; requiring all four arms leaves
1815 nine complete blocks and one four-seed block. Core replay evidence contains 14 complete blocks
1816 and one four-seed block. Level-2 transfer remains complete only on Graph. Sign tests are nominal
1817 exploratory summaries assuming independent block signs; individual intervals quantify estimation
1818 uncertainty.

1819 **O.1 AI Use Statement**

1820 In this work, we used a general-purpose generative AI tool to organize and copy-edit the manuscript,
1821 help interpret recorded experimental results for reporting, develop and check mathematical arguments,
1822 write auxiliary result-reporting code, and check internal consistency, sources, formatting, and compi-
1823 lation. We did not use the tool to generate experimental observations, select reported checkpoints,
1824 alter registered stopping rules, or make final decisions about scientific claims. Every numerical
1825 statement is regenerated from the recorded run artifacts by the scripts identified in Appendix N. The
1826 authors reviewed all AI-assisted prose, code, and claims and take responsibility for the final content
1827 of this work.

1828 **O.2 Ethics Statement**

1829 The benchmark and experiment-specific data are synthetic; this study collects no human-subject
1830 or personal data. More verified modes are not inherently better: a distinct execution may be risky
1831 or costly even when it passes the validator, and `distinct@K` does not measure downstream value.
1832 Systems that evaluate or optimize breadth should report correctness, invalidity, and application-
1833 specific costs alongside mode counts.

1834 **O.3 Reproducibility Statement**

1835 Claims are computed from versioned result files; missing final results and aggregate drift cause
1836 the build to fail. Appendix N records protocols, hashes, validations, and regeneration commands;
1837 Appendix J records split identities. The AI Use Statement above discloses generative-AI assistance.